

Replication precedes individuality, and heritable variation follows it, in a digital primordial soup

Reading copy (figures placed in the text) assembled 2026-10-10; author list and affiliations to be added.

Internal review copy, 2026-10-10 (revision 3). Working title, to be replaced: the registered branching-transfer test (X1) showed that the first replicator carries no heritable variation, so the earlier title no longer fits. Nature Article format: summary 199 words, main text 4,568 words, six figures, Methods 3,186 words, ten Extended Data items, and Supplementary Information (Supplementary Methods, figures and tables, theorems and proofs). Every number is from a generated table of the project repository; every experiment reported is complete; the genealogies were settled on 17 worlds and replicated as registered on 20 more, except for one prediction (N6-4).

Organisational theories define a living individual as a system whose future depends on itself rather than on its surroundings^{1–4}; somewhere in the origin of life, spreading patterns must have become such individuals. We followed that passage through thousands of worlds of random machine code. On tapes of 16 to 64 bytes with harmless by-products, in the Z80 and an 8080-like subset, the first replicator whose copies copy in turn is almost always a two-byte word that writes itself when executed, runs on into its partner and makes copies that vary with it. Its lineages persist when they branch, but no site keeps a mutation: it replicates a pattern, not a genome. Closure, execution confined to the replicator’s own bytes, follows where a closing design is within reach, and seeded closers displace the open form; the evolved closers we tested rely on registers that the harness empties. Heritable variation arrives only with closure, and only in some closers: in a recurring block-copy core one byte decides whether a closer rewrites itself, erasing mutations, or copies its whole tape, keeping them. Whether by-products halt execution sets their balance. In a second machine, adding one self-writing instruction made the first replicator open.

Main

What distinguishes a living thing from the chemistry around it has two answers: a list of properties, metabolism, compartment and heredity, with the question of which came first^{5–10}, or a relation: a living system is one whose operations produce the organisation that produces them, so that its future depends on its own past rather than on its environment^{1–4,11,12}. Evolutionary theory places the start of Darwinian evolution proper at a related passage, where vertical descent comes to dominate communal exchange¹³ and replication outcompetes the chemistry that produces replicators¹⁴. Few experimental systems allow such a transition to be observed as it happens, repeated under controlled variation and taken apart.

Digital organisms usually descend from a designed ancestor^{15,16}; random-program soups come closer. When random byte strings for a simple machine execute one another, with no fitness function and no replacement, self-replicating programs arise and take over¹⁷, as in earlier program soups and artificial chemistries^{18–22}. For the Z80 processor the first replicators are load–push programs that give way to a block-copy family^{17,23}, a takeover attributed to mutational robustness²³, and a Z80 soup with an energy budget has been used to study cooperation²⁴; whether the successors descend from the first replicators, and how, has not been asked. We treated such a soup as a test of one hypothesis: that the smallest programs that write themselves run on into their surroundings, so that replication begins open (Fig. 1a), and that closure, a path of execution that returns into the organism and one step towards individuality, is a later invention; and we asked when heritable variation, the material of Darwinian evolution, appears. We ran more than 3,600 worlds of random Z80 code, varying the instruction set, program length, execution budget, mutation rate and memory geometry, with confirmatory stages pre-registered with kill criteria (Methods). Heredity is measured by intervention, as in serial-transfer tests of self-replication²⁵: a program is lifted out and cultured against fresh random partners. We repeated the central experiment on an 8080-like subset of the Z80 (the 8080 subset) and on a second, minimal machine whose instruction set we could edit, and recorded every encounter of 40 worlds to rebuild their genealogies.

Order before replicators

The soup holds 20,000 programs of L bytes on a 160×125 lattice (Fig. 2a). At each step 8,192 random cells are each paired with a lattice neighbour, the pair is executed as one program for 128 instructions and written back, and a background mutation replaces a fixed number of random bytes; nothing else happens (Methods). A tape is *heritable* if, lifted out and executed against random partners, it makes copies that copy in turn; the heritable fraction of random cells is the population measure.

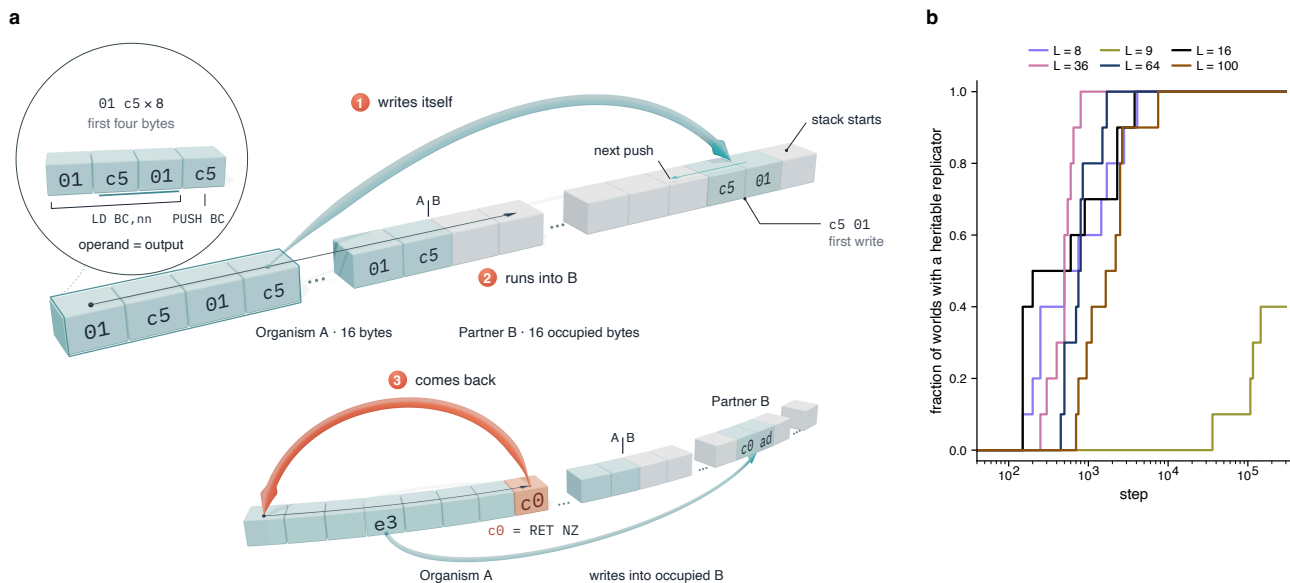


Fig. 1 | The first replicator and its successor. **a**, The first replicator and its encounter. Organism A (teal, 16 bytes) and partner B (grey, 16 bytes) share one 32-byte memory (tick A | B, where A ends); execution starts at A's first byte and moves right, the stack pointer starts at B's last byte and moves two bytes left with every push, and the bytes A writes into B are teal, A's own word. The first replicator is the word 01 c5 repeated: LD BC, nn loads its own next two bytes, c5 01, and PUSH BC writes exactly those two bytes at the stack pointer (operand = output), so the partner fills from its far end with the organism's own word (1) while execution runs into the partner (2). Later a closer appears (3): its conditional return c0 (RET NZ, the only vermilion byte) sends execution back into the organism's own code (vermilion arc), while e3 (EX (SP), HL) writes into B, which already holds bytes (occupied); the organism's bytes remain as exposed as before. **b**, Kaplan-Meier fraction of worlds with a heritable replicator by step for six lengths from 8 to 100 bytes (8, 9, 16, 36, 64 and 100; Stage E, unablated, 10 worlds per length, 300,000 steps).

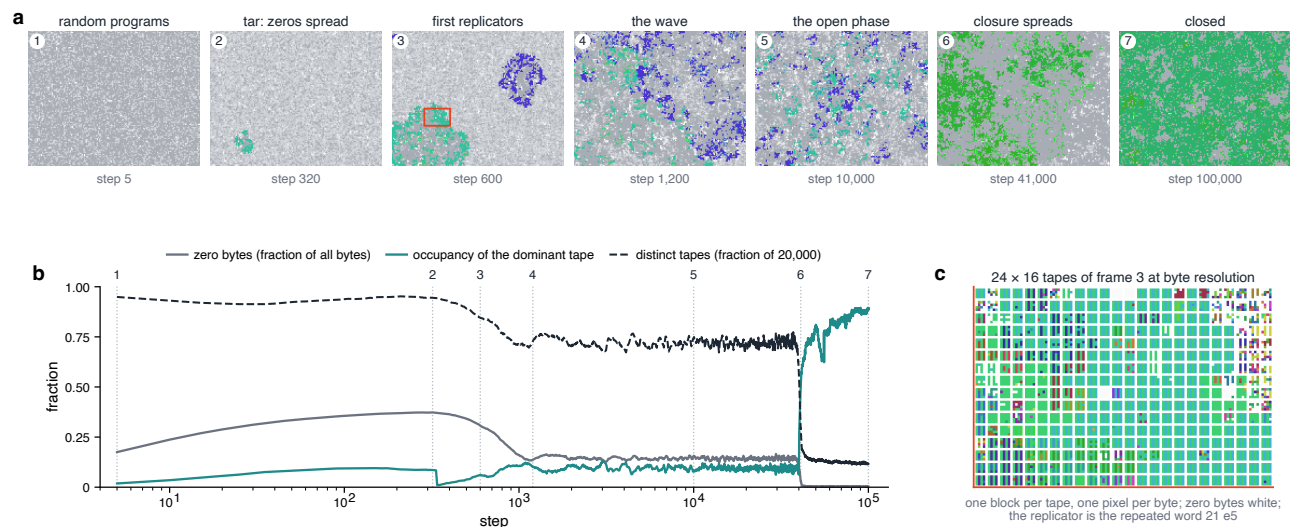


Fig. 2 | One world, watched. Seed 2002 at $L = 16$, run with dense snapshots for Supplementary Video 1. **a**, The lattice (160×125 tapes, one pixel per tape) at seven steps. The four most common tapes with at least 20 copies are coloured by the mean colour of their bytes (the colour code of c and of the video); every other tape is grey, lighter the more of its bytes are zero, so the tar shows as bleaching. Zeros spread first (frame 2: 37% of all bytes at step 320); the first replicators appear as patches of their own copies (3; the teal patch is the word 21 e5, the blue one 01 c5, both load-push words); they sweep the lattice (4) and compete for 40,000 steps among pockets of zeros (5); a class carrying the return closer's motif with the conditional return e0 (RET PO) spreads from about step 40,000 (6), and its variant with c0 (RET NZ) takes over from about step 55,000 while the zeros disappear (7): at step 100,000, 89% of cells lie within Hamming distance 4 of the dominant tape and 62.5% are identical to it, 2,290 distinct tapes remain of 20,000, and 0.4% of bytes are zero. **b**, The same world's zero-byte fraction, dominant-tape occupancy (cells within Hamming distance 4 of the most common tape) and number of distinct tapes against step; dotted lines, the frames of a. **c**, The 24×16 tapes inside the red rectangle of frame 3 at byte resolution (one 4×4 block per tape, one pixel per byte, zero bytes white): the replicator's tapes are the repeated word; the rest is random code and tar.

The first thing to form is order that does not inherit. When the most common pattern first occupies a tenth of the soup, a quarter to a third of all bytes are zero (means of 24–34% at each of seven lengths from 9 to 100 bytes, Stage B; Fig. 2 follows one world). Push, call and reset instructions executed with empty registers write zeros wherever the stack pointer points, and zero is itself an instruction that does nothing, so the pattern spreads by being written and changes nothing when executed. We call this flood *tar*, after the asphalt that prebiotic chemistry makes from the reactions that also make sugars²⁶. In the standard soup the tar is *benign*; in some soups we made a fetched zero halt the pair, making the tar *lethal*. The tar is what detectors see: the high-order entropy used to detect the transition¹⁷ is near chance or inverted in unablated worlds at six of eight lengths from 25 to 100 bytes, and an assembly measure of selection^{27,28}, which rises with any duplication, passes ten times its random-soup value before the first heritable replicator in 71 of 80 worlds (Extended Data Fig. 1). Only intervention tells the two apart.

The first replicator is open

With benign tar and the harness's standard conventions (Methods), the first heritable replicator of every confirmatory world of 16 to 64 bytes is the same program, which we call the *pusher*: the two-byte word `01 c5`, or one of its register variants, repeated to fill the tape (Fig. 1a). It comes first in 170 of 170 such worlds, across five lengths, the Z80 and its 8080 subset, and two pairing rules, at Kaplan–Meier medians of 450–750 steps in Stage G (Supplementary Table 1; Fig. 1b); on shorter tapes block copiers often come first (at 5, 6, 9 and 12 bytes; Extended Data Fig. 2b). It still comes first when every register is drawn at random before each encounter (20 of 20 worlds); with a random stack pointer the first replicator is open in 18 of 20, though in 13 it is the stack move `e1 e3`, which has no operand (Methods). Executed, `01 c5` loads its own next two bytes into a register pair and writes that pair where the stack pointer points, in the partner: its operand bytes pass verbatim into the output, a *literal channel* from code to copy, so it copies itself as a side effect of being run, with no loop, no counter and no test of its surroundings. It is nearly the only thing of its size that replicates: of all 65,536 two-byte words tiled to 16 bytes, exactly five, besides the zero word, copy themselves into an all-zero partner, each pairing a load with a push (Supplementary Information, Proposition 3). Removing the immediate loads or the stack writers delays the first replicator by orders of magnitude or prevents it, and no ablation that keeps them changed its time significantly with ten worlds each (Extended Data Fig. 2).

The first replicator is *open*. Its pointer fetches partner bytes in every one of 256 encounters, for every traced first replicator under benign tar (100 of 100). Against 256 random partners it copies into a median 0.66–0.73 of them and loses a quarter or more of its own bytes in a median 0.06–0.43 of encounters (Fig. 3a,b), because after writing its copy it executes the partner's bytes, which can poison the register pair before the next push. The offspring is an exact copy in a median 58% of encounters at $L = 16$ and almost never at $L = 64$, and the information that flows from the partner into it, its entropy over random partners (Supplementary Information, Lemma 7), runs from 3.9 bits at $L = 16$ to the 8-bit ceiling of the sample at $L = 64$ (Fig. 3c): a child depends in part on the neighbour it was made against. Against cells drawn from the whole soup at step 5,000 the offspring still carry 3.8 bits of the partner at $L = 16$ and the pointer enters in every encounter (80 of 80 first replicators; registered), although on the lattice a pusher's neighbour is its own kind in 57% of its encounters, and inflow among lattice neighbours was not measured (Supplementary Methods). The median heritable fraction of random cells stays between 0.09 and 0.19 from step 2,000 to step 20,000 at $L = 16$ (Fig. 3d). Copied serially into fresh random partners, one offspring per transfer, 96–100% of first-replicator lines end within eight transfers (Fig. 4a), about as fast as a copy success of 0.66 per encounter predicts. Allowed to branch, with two fresh partners per member and generation, 48 of 64 unmutated lineages survive eight generations, yet none of 512 single mutants (2,048 populations) still carries its mutation, with random partners or with cells of open-phase soups (registered; Supplementary Table 2): the pusher replicates a pattern, not a genome.

Closure, where a closer is within reach

The first replicator does not stay. At $L = 16$ the dominant tape carries a control-flow instruction in 20 of 20 worlds by 300,000 steps, 17 of them ending on the identical eight-byte sequence whose conditional return sends the pointer back into the organism's own body (Fig. 3e; Extended Data Table 1). Every such successor copies into all 256 random partners without self-damage (Fig. 3a,b,f), and in 63 of 67 loop-bearing final dominants (the most heritable of the three commonest tapes at the end) the offspring is one and the same string in all 256 encounters (Fig. 3c). At the last snapshot every sampled heritable cell is confined in all 20 worlds at $L = 16$, and most are in 19 of 20 worlds at $L = 20$, 12 of 20 at $L = 32$ and 8 of 20 at $L = 64$ (registered population sample; Supplementary Table 1); the heritable fraction of random cells rises to 0.75–0.94 at $L = 16$ (Fig. 3d).

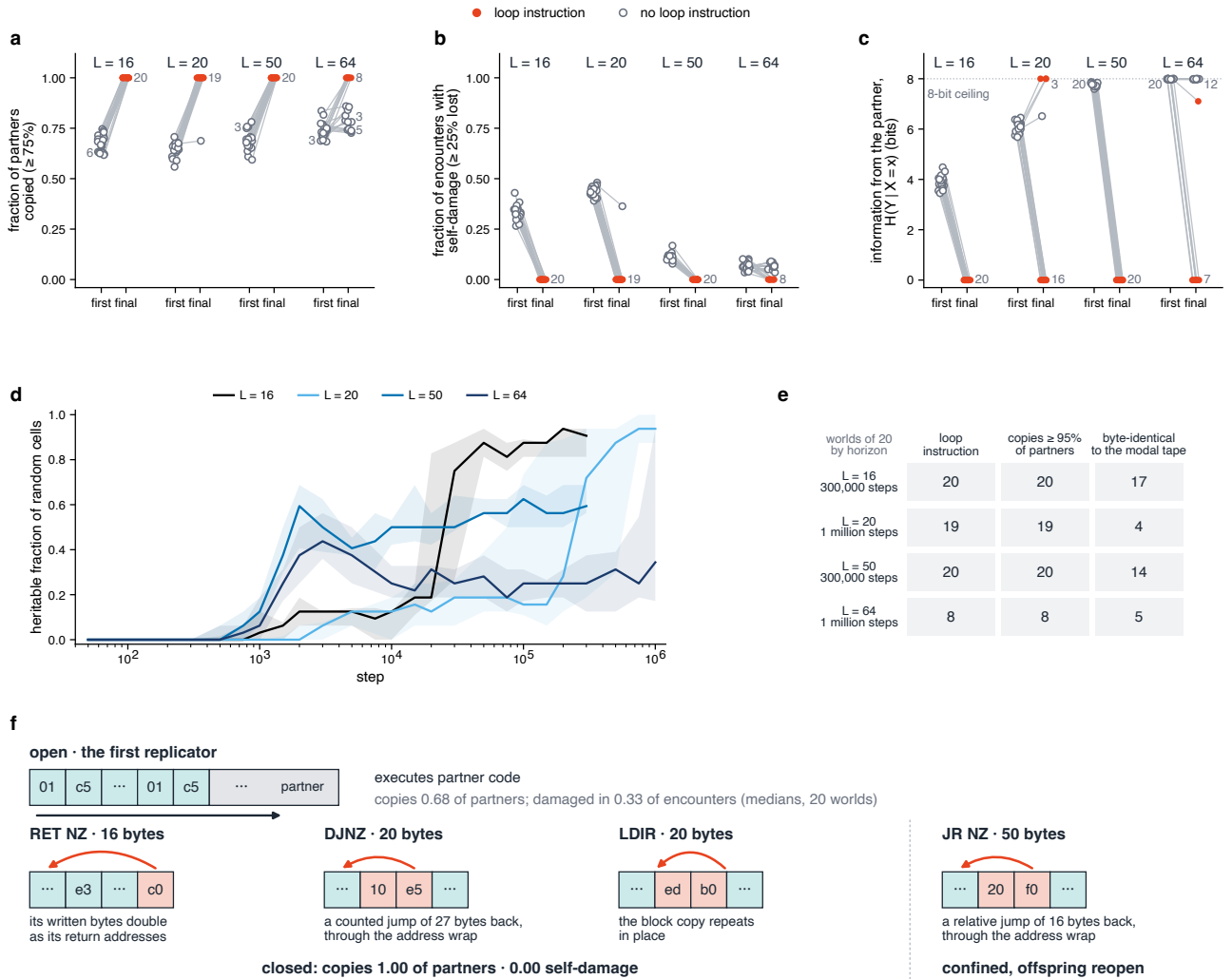


Fig. 3 | The first replicator is open, the successor closed. **a**, Fraction of 256 random partners that become a $\geq 75\%$ copy of the first heritable replicator (first) and of the final dominant (final) of every Stage G world, grouped by tape length (20 worlds per L; grey lines join the same world; points spread sideways within each column, heights exact; numbers, coincident points, although points can still overlap within dense groups; open circles, no loop instruction; filled, a jump, call, return or restart instruction or a block repeat). **b**, Fraction of encounters in which the organism loses $\geq 25\%$ of its bytes. **c**, Information the offspring carries about the partner, $H(Y | X = x)$ in bits over 256 random partners (dotted line, the 8-bit ceiling of 256 partners). **d**, Heritable fraction of 16 random cells against step (lines, medians over worlds; shaded bands, interquartile ranges). **e**, Worlds (of 20) whose final dominant carries a loop instruction, copies $\geq 95\%$ of partners, and equals the modal final tape byte for byte, by tape length and horizon, each column counted on its own; at L = 50 the final dominant is confined but its offspring is not (Supplementary Methods), and at L = 64 the five worlds that end on the modal tape end on the open pusher. **f**, Control flow of the first replicator, of three closed successors and, set apart, of the L = 50 final dominant, schematically (bytes abbreviated; the instruction that returns control into the organism is shaded and its return drawn as a vermillion arc): the pusher's pointer runs on into the partner, whereas RET NZ returns to addresses formed from the organism's own written bytes, DJNZ jumps back through the 16-bit address wrap, and LDIR repeats in place; at L = 50 JR NZ also jumps back through the wrap and the final dominant is confined, but its offspring reopen (e). Medians of the partner test over the Stage G worlds at L = 16: the first replicator copies 0.68 of partners and is damaged in 0.33 of encounters; every closed successor copies 1.00 with 0.00 self-damage. Exact traces, Extended Data Fig. 3.

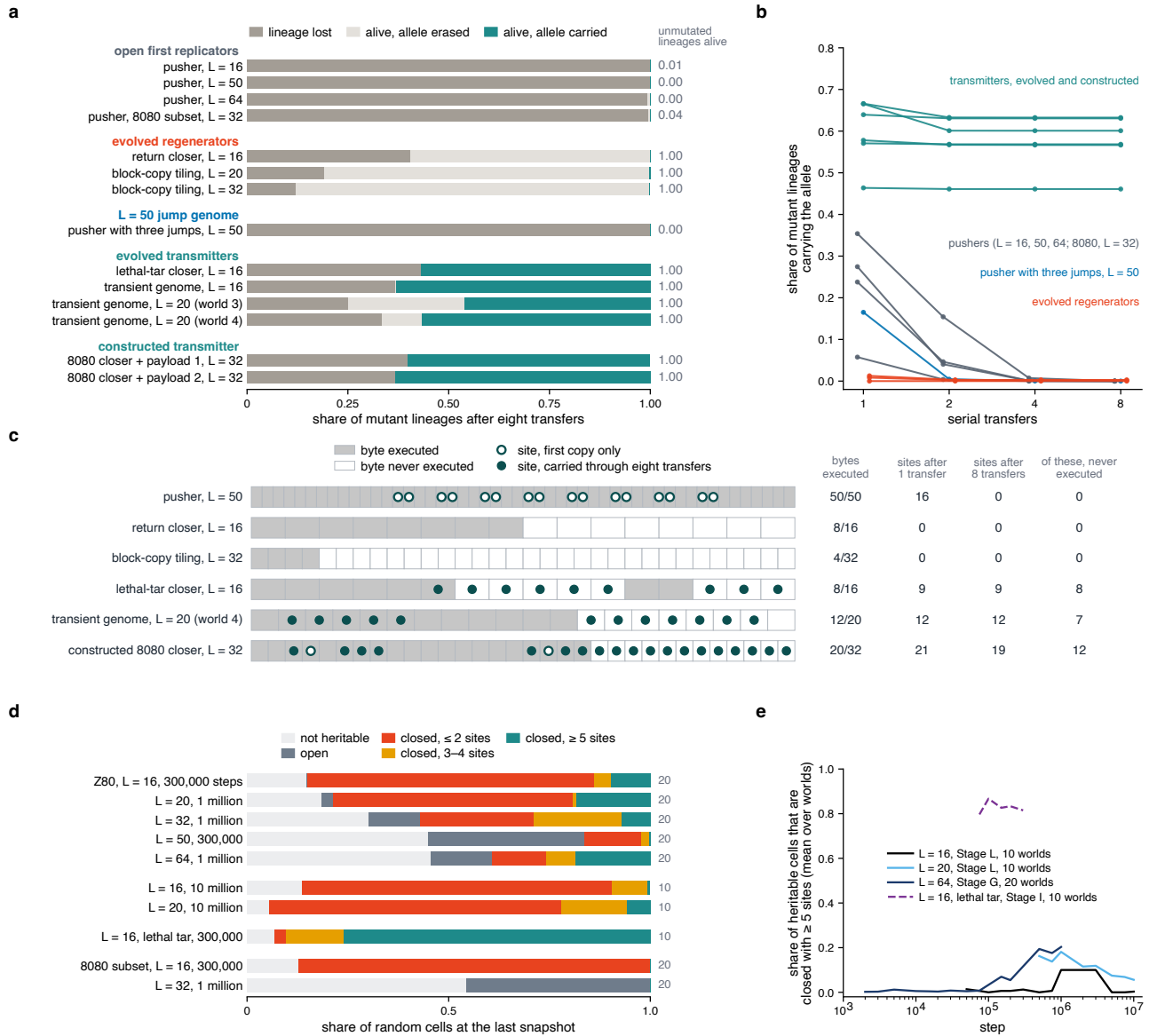


Fig. 4 | Closure regenerates or transmits. **a**, Fate of every single-byte mutant of 14 genomes after eight serial transfers into fresh random partners (16 lineages per mutant; all 255 alternative bytes per position at $L \leq 32$, 32 seeded values at $L = 50$ and 64): the lineage was lost (a transfer failed to make a $\geq 75\%$ copy), is alive with the mutant byte erased, or is alive carrying it (the mutant's five-byte window anywhere in the tape). The $L = 50$ jump genome forms its own group (blue, as in **b**). Regenerators (vermillion) keep almost no allele and transmitters (teal) keep alleles at several sites, the colours of **d**; a transient genome is the most common tape of a Stage L world at the first snapshot at which a genome-bearing block copier was most common (seed 6006 at $L = 16$; worlds 3 and 4, seeds 6003 and 6004, at $L = 20$). Right, the share of 256 unmutated lineages still copying after eight transfers. **b**, Share of all mutant lineages carrying the allele against the number of transfers, one line per genome (colours as in **a**; markers dodged sideways by group at each transfer, regenerators on top). **c**, Six of the genomes byte by byte: grey, fetched as instruction stream (opcode or operand) by the unmutated tape in any of 64 encounters; white, never fetched; open circle, a site whose alleles reach first-generation copies only; filled circle, a site whose alleles are still carried after eight transfers (for at least half of its values, the share of live lineages carrying the allele exceeds the chance background by at least 0.5); right, bytes executed, sites after one and after eight transfers, and how many of the latter are never executed. **d**, Classes of 64 random cells at the last snapshot, mean over worlds (number of worlds at right): not heritable; heritable and open (the pointer fetches a partner byte in at least one of 16 encounters); closed, by the number of transmissible sites in a single-mutant scan of up to 16 heritable cells per snapshot (at most two: regenerators; three or four: intermediate; five or more: transmitters). **e**, Share of heritable cells that are closed with five or more sites against step, mean over worlds, at the snapshots at which at least 80% of a series' worlds are recorded and at least a fifth of the cells are heritable. Lines: $L = 16$ and 20, Stage L (10 worlds each, to 10 million steps); $L = 64$, Stage G (20 worlds, 1 million steps); dashed, $L = 16$ with lethal tar, Stage I (10 worlds); Stage K ($L = 32$) and Stage G at $L = 50$ are not drawn: their means stay low and would lie on the other lines.

We call this property *closure*, and a replicator that has it a *closer*. A loop in the code, confinement of the pointer and independence of the offspring from the partner travel together here, and each was measured: every loop-bearing final dominant fetches no partner byte as an instruction in any traced encounter (89 of 89, including the 20 at $L = 50$ whose offspring reopen), and confinement and zero inflow agree in 106 of 110 finals (Extended Data Fig. 3). Closure in this sense is a property of execution, not the organisational closure by which theories of autonomy define living systems^{1,3,11}, though it is one component of individuality in the information-theoretic sense⁴. It is not a membrane, since the organism's bytes stay exposed, nor a property of place²⁹: it also arose with well-mixed pairing, though later (post hoc; Supplementary Methods). The longer the tape, the slower and rarer was closure (Cochran–Armitage test, $P = 6 \times 10^{-5}$, post hoc; the per-byte mutation rate also falls with length): 20 of 20 worlds by 300,000 steps at $L = 16$, 12 of 20 by one million at $L = 32$ and 8 of 20 at $L = 64$, and given ten million steps every world closed at $L = 16$ and 20 (Extended Data Fig. 4). At $L = 50$ a jump spreads but its carriers' offspring reopen, and only 18–39% of heritable cells are confined (registered population sample). Closers rely on conventions of the harness: under random registers the four evolved or evolved-core closers we tested lose their lineages and only a constructed self-initialising closer survives (Supplementary Methods, Parent $\times$ environment), and some at $L = 20$ and 50 use a 16-bit program counter in a memory addressed modulo 2L (Methods).

Closure is selected where it is found. Seeded at 1% into a world filled with pushers, the evolved return closer at $L = 16$ and the block-copy tiling at $L = 32$ held half the cells within 400–750 and 260–290 steps, in 5 of 5 soups each; the pusher seeded into a closer's world vanished in 5 of 5 (Extended Data Fig. 5). Where closure did not come, the closer was missing from what evolution had found, not from what the instruction set allows. In the 8080 subset no world closed at $L = 32$ in a million steps, yet a 20-byte closer of 8080 instructions, seeded at 1%, had its loop in half the cells within 300–320 steps in 5 of 5 soups. With registers drawn at random before each encounter, closure came in 8 of 40 worlds by three million steps, and a self-initialising closer, seeded, displaced the pusher in 5 of 5 soups (Extended Data Fig. 6g; Supplementary Methods). The barrier was discovery, not expressibility, and closers seeded at 1% were not held back by competition.

The first closers are not point mutants

How the first closer is made cannot be read from snapshots, and the soup cannot be replayed exactly, so we recorded every encounter and mutation of 20 worlds at $L = 16$ until closure, rebuilt the lines of descent of their final confined populations and checked the record by replaying encounters byte for byte (Methods); once the analysis had been settled on these worlds, we registered it and repeated it unchanged on 20 new ones (Supplementary Table 3). Seventeen of the first 20 worlds closed and all 20 of the second. On every sampled line of every closed world the first copier was open (registered; 37 of 37 worlds). On these lines the founder, the first confined copier after the newest open copier, appeared a median of 79 and 46.5 steps after the newest open copier in the two sets, through non-copyers only in 16 of 17 and 20 of 20 worlds (Extended Data Fig. 7a,b,e; in 2 and 3 worlds a few confined copyers, at most 0.05% of cells, were already present off these lines). No founder came from the tape before it by a point mutation: it differs from that tape in 2 to 11 bytes, and the 54 and 55 recorded mutations on these lines were expected to make 0.002 and 0.006 founders (Extended Data Fig. 7d). Each founder was written in one encounter (Extended Data Fig. 7c) by a tape that does not copy itself whole: in 23 of the 37 worlds by one that copies neither itself nor the code it runs, rarely or never with soup partners (median 0.2% and 0% of encounters, excluding three writers that make it with most partners; never in 12 of the 23), and in 14 by a confined tape that copies the code it runs but not the rest of itself and always makes the founder. Non-copyers fill 84–91% of these soups, so the writers' kind largely reflects their abundance ($P = 0.076$ and 0.062). In 34 of 37 worlds the founder carries the return closer's motif, built of one-byte variants of the pusher's word 21 e5 (Extended Data Fig. 7). Closers thus descend from the open lineage, as expected in soups the pushers dominate, but the first one is not a point mutant of the tape before it: it is written in one encounter by a tape that does not copy itself whole. The registered replication confirmed how the founder is made and that only non-copyers lie between it and the newest open copier; its prediction of the writer's kind by the code it copies failed (11 of 20 copy neither, against 12 required).

Closure regenerates or transmits

Heritable variation comes only with closure, and not with every closer. We followed every single-byte mutant of 14 genomes through eight serial transfers into fresh random partners (Fig. 4a–c; Methods). The pusher's mutations reach its first copies, at 16 of 50 and 30 of 64 positions, but they fade as its lines of transfer end, mutated or not (0 and 7 sites after four transfers, none after eight); our registered prediction that its sites persist failed, and they fade at least as fast when its lineages branch (Supplementary Table 2). The evolved closers, the return at $L = 16$ and the block-copy tilings at $L = 20$ and 32, lose no unmutated lineage, but they copy the code they run and rebuild the rest of the tape from it: 60–88% of their mutant lineages live on with the mutation erased, and almost none keeps it (at most 0.3%). A third design, a short copying

core followed by bytes it copies but never runs, keeps the allele in 46–63% of its mutant lineages, at 9 to 20 positions, 7 to 12 of them never executed (Fig. 4c). Such bytes are copied without being interpreted, half of what Langton's and von Neumann's descriptions do^{30,31}; and erasure is not canalisation³², since the variation never reaches the offspring.

One byte decides which. A four-byte block-copy core recurs across stages, `XX 5e ed b0`: one instruction, then `LD E, (HL)` and the repeating block move `LDIR`; with its variant `le NN ed b0` it is carried by 2,347 of the 15,655 confined heritable cells we scanned (Methods). Because the registers start at zero, the first byte, which ref. 23 annotates as the partner offset, is both an instruction and the copy offset: each byte is copied $d = XX \bmod 2L$ addresses ahead, and the block move tiles the first d bytes around the ring, the organism included. With $d < L$ the closer rewrites its own body from its first d bytes and erases every mutation beyond them; with $d = L$ it copies the tape verbatim and keeps them. A closed lineage can therefore transmit $d - 4$ sites, set by one byte of its own code, whenever d divides L and the first byte passes control to the core without writing below address d (Proposition 8). Of 2,343 confined block-copy cells with such an offset, sampled in Stages G, I, K and L (846 distinct tapes, 42 worlds), the single-mutant scan finds exactly $d - 4$ sites in 2,289 (98%).

A world's most common tape hides which setting it holds, since a regenerating population is nearly clonal and a transmitting one a cloud of distinct genomes, so we classified random cells: confined cells with at most two transmissible sites are *regenerators*, those with five or more *transmitters* (Fig. 4d,e; Methods). Under benign tar, regenerators are the majority of heritable cells at the last snapshot in 17 of 20 worlds at $L = 16$ and 14 of 20 at $L = 20$, and in 20 of 20 in the 8080 subset at $L = 16$; longer tapes hold mixtures (at $L = 32$ regenerators are the majority in 3 of 20 worlds and no class in 9; at $L = 64$ regenerators and transmitters each in 2). Under lethal tar, transmitters are the majority in 9 of 10 worlds, with a median of 11.5 sites they never execute, while the most common tape holds 0.2–1.5% of the soup.

The mode belongs to the parent, which keeps it in every environment where its lineages survive (Extended Data Fig. 8a), and the halting rule chooses among parents. A matched regenerator and transmitter at $L = 16$, differing only in the offset byte and in what follows the core, copy alike and survive alike as partners of cells drawn from the soups, to within 0.01 (Extended Data Fig. 6e), yet populations converge from either side to a share that the tar sets: at $L = 32$ transmitters hold 0.03–0.37 of the core-carrying cells after 300,000 steps with benign tar and 0.55–0.87 with lethal tar, whether they start at 1%, 50% or 99% (Extended Data Fig. 6a,b). Encounter censuses of 20 such soups under each rule show where the two kinds differ (post hoc; Methods). Copies between them balance to within 0.1%; what differs is how each fares against the cells that carry no core, a median 13% of the soup. A core-free executor whose pointer enters a regenerator runs the copies of the copying code that make up its body and destroys it in 0.75–0.89 of such entries; one that enters a transmitter wanders through bytes never run and, under the lethal rule, is often halted by a zero first. These terms account for the drift of the transmitters' share between rules, but not for why regenerators win under benign tar, which remains open (Methods).

Robustness³³ does not explain the succession: the first replicator is at least as robust to single mutations as its successor (0.74 against 0.59 of single mutants remain heritable at $L = 16$; Supplementary Fig. 1), and by the criterion of ref. 23, an exact self-copy after mutation, the matched transmitter scores 0.88 and the regenerator 0 (Extended Data Fig. 6f): the criterion measures the transmission of mutations, not survival. A transmitter tolerates a mutation and passes it on, a regenerator survives it by erasing it, and only tolerance builds variation³⁴.

What decides the beginning

If the order of events follows the literal channel, a machine without one should be born loop-bearing, and giving it one should start replication open. BFF, the Brainfuck family of ref. 17, has no literal channel: copying a 64-byte tape into a partner that differs from it everywhere takes at least 64 instructions that change a cell and 64 that move a head, so a copier confined to its own 64 cells must execute some cell twice (Theorem 1). In soups of 2^{17} programs run for 16,384 epochs (Fig. 5), BFF as published and with a wrapping pointer produced a heritable replicator in 28 of 48 worlds; every first replicator contains a loop (28 of 28), and none enters its partner in more than 22 of 64 encounters (with harmless brackets, 7 of 12 worlds replicated, every first replicator loop-bearing but only 5 confined (entering the partner in at most 5% of encounters; 4 in none): the missing channel explains a loop, not confinement). Adding one instruction that writes its own two operand bytes below the write head changes the beginning: in 12 of 12 worlds under either pointer rule the first replicator is the one-byte tiling of that instruction, loop-free, with its pointer entering the partner in every encounter (Fig. 5b). It is the `01 c5` of BFF, and without a wrapping pointer its offspring carry 8 bits of the partner, the ceiling of the measurement (Supplementary Fig. 2).

Then it dies. An unmatched bracket ends BFF execution, so the soup fills with tar that halts the pointer, and by epoch 1,024 the open replicator holds under 1% of the soup; no closed class evolves (0 of 12; Supplementary Methods). The open phase is a *window* whose length the tar sets. With harmless brackets the open replicator holds for all 16,384 epochs

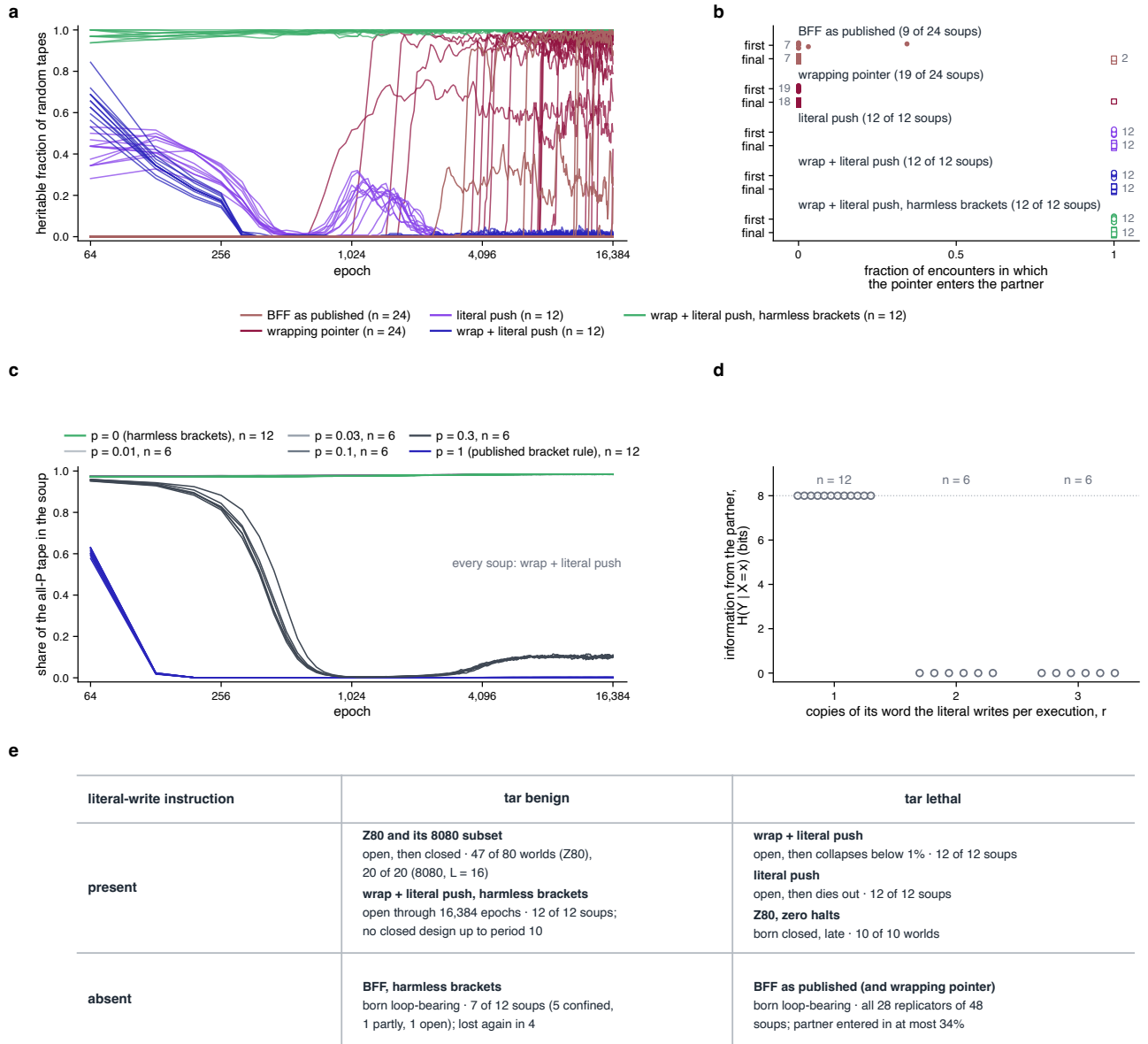


Fig. 5 | Two properties of the substrate decide the beginning. **a**, Heritable fraction of 32 random tapes against epoch for every BFF soup, by variant: as published ($n = 24$), with a wrapping pointer (24), with the literal push (12), with both (12), and with both and harmless brackets, which are no-ops instead of halting the pointer (12); samples every 64 epochs from epoch 64, one line per soup, each a trailing mean of four samples. **b**, Fraction of culture-test encounters in which the pointer enters the partner, for the first replicator (circles) and the final dominant (squares) of every soup with a replicator, by variant (open symbols, no loop instruction; filled, loop instruction; points spread vertically within each row, horizontal positions exact; numbers, points that share one horizontal position). **c**, Lethality as a dial: share of the all-P tape, the literal instruction tiled, against epoch under wrap + literal push when an unmatched bracket halts the pointer with probability p ($p = 0$, 12 soups; 0.01, 0.03, 0.1 and 0.3, six each; 1, the published bracket rule, 12); the curves for $0 < p \leq 0.1$ lie on top of one another near 1, with $p = 0$ drawn over them. **d**, The write ratio ($2r/3$): information inflow of the first replicator, $H(Y | X = x)$ over 256 random partners (dotted line, the 8-bit ceiling), when the literal push without a wrapping pointer writes $r = 1, 2$ or 3 copies of its word per execution; one point per soup, spread evenly sideways within each r (heights exact), every first replicator loop-free. **e**, The map: the observed order of events in each cell defined by a literal-write instruction (present or absent) and by the lethality of the tar to an open replicator, with world counts; variants named as in **a**, and BFF, harmless brackets, the published rules with brackets that are no-ops (12 soups); “collapses below 1%”, the share of the all-P tape (**c**); “dies out”, the heritable fraction of random tapes falls to zero by the last sample (12 of 12 literal-push soups without a wrapping pointer); Z80 closure is counted at the population level (most heritable random cells confined at the last snapshot; registered population sample, Methods).

in 12 of 12 worlds; with brackets that halt with probability p it holds 98% of the soup at $0 < p \leq 0.1$ (18 of 18 soups), and at $p = 0.3$ it falls below 1% within about 900 epochs in 6 of 6 soups but recovers to 10% or more in 5, against the registered prediction (Fig. 5c). When the literal writes two or three copies of its word per execution, raising the *write ratio*, bytes written per byte read, from 2/3 to 4/3 or 2, so that a copy completes within one pass, the open tiling's offspring carry 0 bits of the partner, information-closed from birth without a loop, and every random tape is heritable at 16,384 epochs in 12 of 12 soups (Fig. 5d).

Lethal tar also removes the open phase in the first machine. When a zero byte halts the Z80, its own tar becomes lethal and the open beginning disappears: in 10 of 10 pre-registered worlds no pusher establishes, contrary to the registered expectation of an open phase followed by extinction, and replication begins closed and late, as a block copier, at a Kaplan–Meier median of 38,000 steps against 450 with benign tar (Extended Data Fig. 9). Between the two rules lies a threshold. When a fetched zero halts the pair with probability p , the first replicator is an open load–push word in 10 of 10 worlds at every p up to 0.1, in 2 of 10 at $p = 0.3$ and in none at $p = 1$, and the median share of transmitters among heritable cells at 300,000 steps rises from 0 at $p \leq 0.03$ to 0.78 at $p = 1$ (Spearman $\rho = 0.50$, permutation $P < 5 \times 10^{-5}$; Extended Data Fig. 8b,d), though they are the majority in only 6 of 10 worlds at $p = 1$ (7 registered); closure timing is not monotone in p (Extended Data Fig. 8c).

The beginning does not need the Z80's own extensions. In the 8080 subset, made by turning every Z80-only instruction into a no-op, which keeps the literal word but removes the block copies and relative jumps (Methods), the load–push word comes first in every world, and at $L = 16$ the Z80's own return closer, pure 8080 code, dominates 20 of 20 worlds by 300,000 steps. Together these fill a map (Fig. 5e): a literal channel suffices for replication to start open, though it is not the only route; the lethality of the tar decides whether an open beginning lasts, or happens at all; and the closed designs within reach decide whether what follows is closure, a late closed beginning, or extinction.

Two counting statements bound this order of events without forcing it, and a third bounds heritable variation (Fig. 6; Methods). A straight-line replicator is open by construction, and a confined one must execute some address twice wherever execution cannot stop inside it before the budget is spent (Fig. 6a). Counting complete self-writers, the usual estimate of how likely a replicator is to arise by chance^{35,36}, does not explain which comes first: the five of period two are open, but 558 of the 641 of period four are closed (Fig. 6b). And a closer with the block-copy core transmits $d - 4$ sites (Fig. 6c).

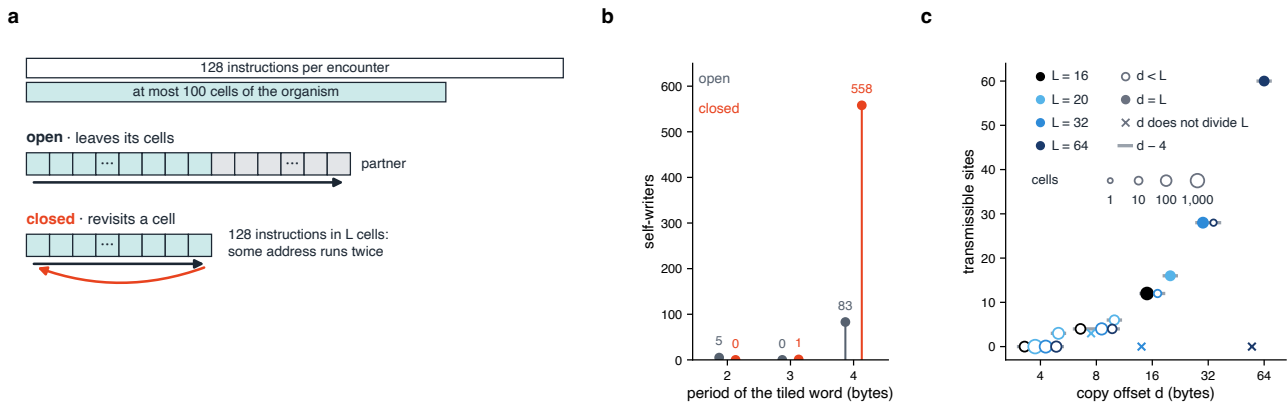


Fig. 6 | What is proved and what is counted. **a**, Theorem 2 for the Z80 soups with benign tar: an encounter executes 128 instructions, an organism has at most 100 cells (bars to scale) and no instruction stops execution inside it (a HALT counts, by convention, as executing its address again), so a program whose pointer never leaves its own cells (closed) must execute some address at least twice, whereas a program without such a return runs on into the partner (open). Under lethal tar a zero byte can stop execution inside the organism, and the argument does not apply. In BFF the same conclusion follows from the cost of a copy (Theorem 1). **b**, Proposition 3: every word of period two, three and four, tiled to 16 bytes and executed against the all-zero partner; self-writers by period, open (the pointer fetches a partner byte against the zero partner or one of 16 random partners) or closed. The five of period two each pair a load with a push; the one of period three, `b0 5e ed`, and the 558 closed ones of period four are block copies. **c**, Proposition 8: in the core `XX 5e ed b0` the first byte sets the copy offset $d = XX \bmod 2L$, the closer tiles its first d bytes over organism and partner, and the $d - 4$ bytes between the core and the offset are transmitted. Points, median number of transmissible sites (single-mutant scan) of confined heritable cells carrying the core or its variant `1e NN ed b0`, by tape length and offset, for all 2,346 cells with $d \leq L$; 2,289 of the 2,343 cells whose offset divides L lie exactly on $d - 4$, with zero interquartile range in every such group; crosses, the three cells whose offset does not divide L ; open circles, $d < L$; filled, $d = L$; grey bars, $d - 4$, one across the markers of each offset. Marker size grows with the number of cells (key, 1 to 1,000; largest group 1,121); the lengths present at an offset are drawn side by side in order of L , centred on the true d , so that a lone marker sits on its offset; the $d = 5$ point ($L = 20$) is drawn with its bar 2 sites higher so as not to touch the $d = 4$ group.

Discussion

The order of events is observed, and the counts bound it. The first replicator is open, and its children depend on the partner they were made against; closure, execution confined to the replicator, is found where a closing design is within reach and selected once found; and the first closer was written in one encounter, not made by a point mutation. Replication came first, heritable variation only after closure, and the first closer was written by other tapes before it copied itself, which recalls, at the level of single programs and in an analysis whose founder result was replicated as registered, the passage from communal exchange to vertical descent that Woese placed at the Darwinian threshold¹³. Closure is necessary for heritable variation here but not sufficient. In the terms of ref. 8, the open replicator replicates without heritable variation, and so does a closer that copies only the code it runs, regenerating itself and erasing its mutants, as autocatalytic networks that lack evolvability do³⁷; a closer that also copies bytes it never runs has a limited heredity at $d = 4$ sites. In a recurring block-copy core one byte decides which, and the halting rule shifts the balance between them by a mechanism our encounter censuses only partly account for.

For chemistry these are analogies, not results. Template copying is also a literal channel, a strand specifying its complement by physical fit with no interpreter, and short self-templating oligomers replicate without enzymes³⁸; by analogy, whether by-products are inert or inhibitory would decide whether an open phase lasts long enough for control to evolve, processivity, the chemical write ratio, now selectable in ribozyme polymerases³⁹, how far copies depend on their surroundings, and if the first self-sufficient replicators are written rather than refined, they would first appear as products of molecules that do not replicate, as replicating RNAs arise de novo in reactions of Q β replicase⁴⁰. In these soups detection required intervention, since low-complexity order precedes the first replicators and trips compression and assembly measures alike^{27,28}.

What limits the claims is plain: one processor, a restriction of it and one toy language; units of fixed size, start address and pairing imposed by the harness; a culture test that is our own ground truth, though the open beginning held under both register conventions, at every confirmatory length from 16 to 64 bytes and under heredity thresholds from 0.2 to 0.4; closure incomplete at the largest lengths within our horizons, and more than a hundred times slower when the harness does not empty the registers, on which the evolved closers depend as scaffolded reproducers⁴¹; and genealogies at one length, with definitions settled on 17 worlds and confirmed, in part, on 20 more. Within those limits the result stands: replication came first; execution confined to the replicator, with offspring independent of the partner, came later; and heritable variation came only with closure, in closers that one byte of a recurring block-copy core sets to transmit rather than regenerate.

Methods

Soup. The soup holds $N = 20,000$ tapes of L bytes (L from 3 to 100), initialised with uniformly random bytes, on a 160×125 square lattice. At every step 8,192 cells are drawn uniformly at random and each is paired with one of its four lattice neighbours (a step off the edge is reflected; the lattice does not wrap); a pair is active if neither member is already in another pair that step, which leaves $\approx 3,950$ active pairs. An active pair is concatenated, organism first, into a pair memory of P bytes ($P = 2L$ except in the padding experiment of Stage F), executed by a Z80 emulator for a fixed budget of 128 instructions (32–2,048 in the budget arms) from byte 0 with all registers zero and the stack pointer at the last byte of the partner, and both tapes are written back. After each step 8,192/2^k bytes chosen uniformly at random over the soup are replaced by random bytes ($k = 4$ unless stated; the size-axis control arms of Stage E also vary N and the mutation rate, Supplementary Methods); the replacements are applied in parallel without synchronisation, so two falling in the same four-byte word in one step can lose one, at most about 0.3% of mutations at $L = 16$ and fewer at longer tapes. There is no fitness function, no replacement and no death. Instruction-set ablations replace whole opcode families by no-ops through a prefix-aware fetch hook that sees the operation after DD/FD and ED/CB prefix resolution; the ablations and their opcode counts are listed in Supplementary Table 4. The executor was validated against an independent Z80 reference (z80-emulator 2.3.0 by L. Kesteloot), both mapping every 16-bit address modulo the pair size: at pair sizes of 32, 38, 40, 64, 100 and 128 bytes, 100,000 random and 9,000 targeted programs per size in each of two seeded runs (jumps, stack moves and block copies across address 0) left no unexplained mismatch, every difference tracing to a documented defect of the reference or to one of two undocumented details the executor does not model (the F3 and F5 flags after BIT n , (HL), and the refresh register after a wasted prefix); the odd and padded rings of Stage F were not differentially tested, and no claim depends on cycle-exact timing.

Culture test and partner tests. A tape runs as the first program against 64 uniformly random partners for the condition's budget; copy success is the fraction of partners that become $\geq 75\%$ copies at the best cyclic shift. The score of a tape x is the mean, over partners j whose best-shift match to x before the encounter, s_j , is below 0.75, of $(s'_j - s_j)/(1 - s_j)$, s'_j being the match after it. The second-generation score, gen2, is the same quantity one generation on: the offspring of

encounter j runs as organism against a fresh random partner, whose match to x is measured before and after; gen2 thus approximates the share of encounters whose offspring copies in turn (Supplementary Methods). Heritable means $\text{gen2} \geq 0.3$, faithful additionally $\geq 50\%$ of partners copied. Partner-independence tests use 256 random partners and record self-damage ($\geq 25\%$ of the organism's bytes changed in one encounter). In BFF the culture test also records whether the instruction pointer entered the partner in each encounter; closed means $\leq 5\%$ of encounters entered and $\geq 95\%$ of partners copied, open means $\geq 50\%$ entered. Information inflow from the partner, the conditional mutual information⁴² $I(E; Y | X = x) = H(Y | X = x) - H(Y | X = x, E)$ between partner E and offspring Y (the partner's bytes after the encounter) with the organism x fixed, equals $H(Y | X = x)$ because the executor is deterministic; it is estimated as the plug-in entropy in bits over 256 random partners, ceiling 8 bits.

Events and statistics. The event t_{rep} is the first sampled step at which one of the three most common tapes has a share $\geq 0.5\%$ and is heritable by the culture test (a clonality criterion by design: it dates the first *clone*, not the first heritable material); t_{faith} additionally requires faithfulness; the *final dominant* is the most heritable of the three most common tapes at the horizon. Because the most common tape can misrepresent a diverse population, population claims rest on random cells (Population sampling, below); $t_{q_{10}}$ (occupancy of the quasispecies⁴³ of the most common tape $\geq 10\%$) is the pre-registered occupancy event and fires on zero floods as well as on replicators. Emergence fractions are quoted as worlds reaching the event within the horizon with Wilson 95% intervals; times as Kaplan–Meier medians with censoring at the horizon. Ablation effects are ratios of Kaplan–Meier medians with seed-paired sign tests (the same seeds run under every ablation); confirmatory contrasts use one-sided Fisher exact tests and the Cochran–Mantel–Haenszel test pooled over budgets. All thresholds (0.75 copy, $\text{gen2} \geq 0.3$, 0.5% share, 10% occupancy, 95%/5% closure) were fixed before the confirmatory stages. Principal medians are given with 95% bootstrap intervals across worlds (Supplementary Table 1). Same-seed Z80 runs diverge because GPU scheduling makes the pairing order non-deterministic, so every Z80 claim is a population claim over seeds (the spread of ten same-seed repeats is 55–85% of that of ten seeds); the BFF soup is bitwise reproducible.

Harness conventions. Two derived soup and executor shaders, byte-identical to the Stage G ones except after the register reset of every encounter (`gen_conv_shader.py`): *random registers* draws A, F, B, C, D, E, H, L, their alternates, IX and IY uniformly (program counter 0, stack pointer as before); *random stack pointer* draws SP uniformly. Twenty worlds each at $L = 16$ under Stage G conditions for 300,000 steps (seeds 8001–8020), and 40 random-register worlds for 3,000,000 steps (seeds 8101–8120, and 8201–8220 as a registered replication after the first set closed only in odd seeds); every assay, including the culture test and tracing, runs under the world's own convention. Closed: most heritable cells among 64 random cells confined in 16 traced encounters. Registered predictions and outcomes, invasion designs and the constructed 8080 closer are in Supplementary Methods (Harness conventions; Invasions at aligned lengths; The 8080 subset; Invasions of a constructed closer).

First closure. For every world of Stages G, H, I, K, L and M (Stage G at $L = 50$, whose offspring reopen, excluded), the first sample at which one of the three most common tapes holding at least 0.5% of cells is heritable and confined in 16 of 16 traced encounters, censored at the horizon; Kaplan–Meier medians and a log-rank test of benign $L = 16$ (Stages G and L, 30 worlds) against lethal $L = 16$ (Stage I, 10 worlds), with the decision rule registered before the analysis (`first_closure.py`).

The copy-offset switch. In the block-copy core `XX 5e ed b0` with $HL = D = B = C = 0$, register E is loaded from the organism's first byte, so `LDIR` copies each byte $d = XX \bmod 2L$ addresses ahead; for `1e NN ed b0`, $d = NN \bmod 2L$. The working values of `XX` were found by executing the core with every byte value (`offset_census.py`). The copy-period law pools every confined heritable cell of the population scans whose first four bytes are either core and compares d with the number of transmissible sites of its single-mutant scan (`offset_population.py`). The closing-core counts (2,347 of 15,655 confined heritable cells carry `XX 5e ed b0` or `1e NN ed b0`) are from `core_families.py`.

Encounter censuses (exploratory). All encounters of 20 final matched-pair soups at $L = 32$, ten of each tar, were recorded for 2,000 steps under each rule, cells classed by their first four bytes as regenerator (core with $d < L$), transmitter ($d = L$) or core-free; copying, destruction, conversion and mutational terms were compared with the 2,000-step drift of the transmitters' share (post hoc). Registered hypotheses that regenerators survive through backup cores (N2) and that inherited zeros carry the lethal rule's protection of transmitters (N4) were rejected; payload zeros (N1) are inherited stack-damage scars, not a selected defence (Supplementary Methods).

Genealogies (exploratory, then replicated). Same-seed runs diverge (Events and statistics), so to inspect events 20 new benign-tar worlds at $L = 16$ (seeds 31001–31020, Stage G dynamics) were recorded encounter by encounter from step 0 (`lod.py`): the pairs drawn, each active pair's memories before and after execution, and the soup after mutation, until the confined share of copy events had exceeded 0.5 for 2,000 steps, then 5,000 steps more (cap 80,000). Each changed cell receives a birth record: a *copy* (within Hamming distance $L/4$ of the executor at some cyclic shift, closer to it than to its

own former tape, and with at least $L/4$ bytes newly matching it), *damage*, or partial overwrite (within $L/4$ of its former tape), *copy of the partner* (the executor's own half becomes such a copy of its partner), *rewrite* (none of these, a residual class that does not imply that the bytes of two parents were joined; its functional parent is the tape that supplied the bytes it executes) or *mutation*. Replay on the traced executor reproduced the recorded memories byte for byte in a validation world and all 40 recorded worlds (Supplementary Methods). On the lines of 64 sampled cells per closed world, every tape was classed against 64 random partners as a copier if it made a $\geq 75\%$ copy in at least half of the encounters, confined if its pointer entered the partner in fewer than half, and otherwise a non-copier; the founder is the first confined copier after the newest open copier older than the final stretch in which at least 90% of the line is confined, and a world's first founder is its earliest. On each line the first copier is the earliest record whose tape is an open or confined copier (registered, X3). The copy-gain rule, the founder, the functional parent and the separation of first founders from later conversions by existing closers were defined after inspecting data, as dated in the change log; the frozen pipeline was then registered and run on 20 new worlds (N6, seeds 32001–32020; Supplementary Table 3). Word shares, founding-encounter replays, the executed-bytes copier test, the production test and the reach and abundance nulls are in Supplementary Methods.

Traced execution. A derived copy of the culture-test executor records every address fetched as instruction stream (opcodes, prefixes, operands) in a 256-bit map per encounter, with byte-identical execution (checked). Pointer confinement is the fraction of 256 encounters with a seeded random partner in which no partner address is fetched; the executed set is the union of fetched organism positions; a transmissible position of the mutational scan is called unexecuted if it is never fetched. Every first replicator and final dominant of Stages G, K and I (110 of each) was traced. In a registered test of openness among the soup's own cells (X2), each Stage G first replicator was also run against 256 cells drawn uniformly from its world at step 5,000 ($L = 16$ and 20) or at the snapshot nearest twice its t_{rep} ($L = 50$ and 64) (`inflow_insoup.py`; kin encounters on the lattice, Supplementary Methods, Controls). Confinement concerns the instruction pointer only; a confined copier may read its partner's bytes as data.

Mutational scan. A mutant is heritable if $\text{gen2} \geq 0.3$; its mutation is transmitted if the mutant byte is present at the aligned position in at least half of the offspring that are $\geq 75\%$ copies; a position is a transmissible site if at least half of its tested values are both heritable and transmitted. Transmission is scored in first-generation copies only; serial retention follows alleles further.

Branching transfer (registered, X1). As serial retention, except that each generation every member of a population runs against two fresh partners, every offspring that is a $\geq 75\%$ copy of its parent joins the next generation, populations are capped at 32 by random subsampling, and each of 32 seeded values per position founds four populations, against 64 unmutated ones; a population carries the allele if at least half of its members contain the mutant's five-byte window, and a site is transmissible under the serial-retention rule (`branching_transfer.py`; partners uniform or drawn from the step-5,000 and 10,000 snapshots of the 20 Stage G worlds of the same length).

Serial allele retention. Fourteen genomes (Fig. 4a) were scanned with every single-byte mutant (all 255 values per position at $L \leq 32$, 32 seeded values at $L \geq 50$). Each mutant founded 16 lineages and the unmutated tape 256. In each of eight transfers the lineage tape ran as organism against a fresh uniformly random partner under the genome's own rule (8080 subset, lethal tar) and the partner's bytes after the encounter became the next lineage tape. A lineage is alive while every transfer has been a copy (offspring matching its parent at $\geq 75\%$ of positions at the best cyclic shift). The allele is carried if the mutant's five-byte window centred on the mutation occurs anywhere in the lineage tape. A position is serially transmissible after g transfers if, for at least half of its values, the share of alive lineages carrying the allele exceeds by at least 0.5 the share of alive unmutated lineages that contain the same window by chance (scoring details, Supplementary Methods).

Population sampling. At the final snapshot of every Stage G, I, K, L and M world, 256 random cells were culture-tested (32 partners) and traced against 16 random partners; a cell is confined if no partner address is fetched as instruction in any of the 16 (registered). At every snapshot of the same worlds, 64 random cells were culture-tested and traced, and up to 16 random heritable cells were given the single-mutant scan (16 values per position, 16 partners) and their executed set (defined before the run and exploratory, since the first sample motivated it). A heritable cell is open if not confined; a confined one is a regenerator (at most two transmissible sites), intermediate (three or four) or a transmitter (five or more); the classes of unscanned confined cells are inferred from the scanned ones. A world is closed at the population level if most of its heritable cells are confined. For confined cells the first-generation scan agrees with serial retention (Fig. 4a,c); for open cells it overstates it, so open cells are reported as open.

Address mapping and aligned lengths. Pair memory is a ring of $2L$ bytes: every memory access uses the 16-bit address modulo $2L$, and the stack pointer starts at the largest 16-bit value that aliases the last byte of the partner. The evolved closers tested rely on the empty registers (Extended Data Fig. 8a). The program counter and the stack pointer themselves are 16-bit, so a relative jump or a stack move that crosses address 0 reaches $(65,536 + \text{target}) \bmod 2L$, which equals the

true-ring target only when 2L divides 65,536. Stage K (L = 32, 20 worlds, one million steps, seeds 5001–5020) tested closure at an aligned intermediate length (outcomes, Supplementary Methods).

The 8080 subset. The `i8080` suppression makes every Z80-only instruction a no-op: the 256 CB-page opcodes, the 78 defined ED-page opcodes (among them the block copies and their repeats, 16-bit ADC and SBC, IN and OUT (C), NEG, RETN and RETI, IM, LD A,I and A,R, RLD and RRD) and EX AF,AF', DJNZ, the five JR forms and EXX. This is an 8080-like restriction of the same emulator, not an 8080. Stage M: Stage G conditions at L = 16 (300,000 steps) and L = 32 (one million), 20 worlds each, seeds 7001–7020, with the predictions registered before the run.

Ten-million-step runs. Stage L: the Stage G conditions at L = 16 and 20, ten worlds each (seeds 6001–6010), run for ten million steps.

Controls. Stage H, well-mixed pairing (the partner drawn uniformly from the soup) at L = 16, ten worlds, seeds 3001–3010, against the 20 Stage G worlds at L = 16. Stage I, lethal tar: a zero byte fetched as an opcode halts the pair for the rest of the encounter (a change to the machine's semantics, not only to the fate of its by-product), at L = 16, ten worlds, seeds 4001–4010, with the culture and partner tests under the same rule.

BFF. The default semantics of `cubff`, the code of ref. 17, verified from source: two 64-byte tapes concatenated, pointer and heads at byte 0, ten instructions, every other byte a no-op, unmatched brackets halt, 2^{13} steps per encounter, 2^{17} programs, random ordered pairs each epoch, mutation 2^{-12} per byte per epoch. Switches: `ip_wrap` (pointer wraps modulo 128); `literal` (byte 0x50 = P a b: write a at head1–1 and b at head1–2, then head1 += 2, skip the operands; designed so that its one-byte tiling writes itself, the analogue of the Z80's load–push word); `nohalt` (unmatched brackets are no-ops); `halt_p` (an unmatched bracket halts with probability p, decided by a hash of the pair, the step and the seed); `lit_rep` (the literal instruction writes r copies of its word per execution). Our GPU kernel reproduces `cubff`'s validation replicator and matches a CPU reference on randomised tests (`tests/test_bff.py`, 15 of 15 passed). Seed counts, registered predictions and their outcomes are in Supplementary Methods.

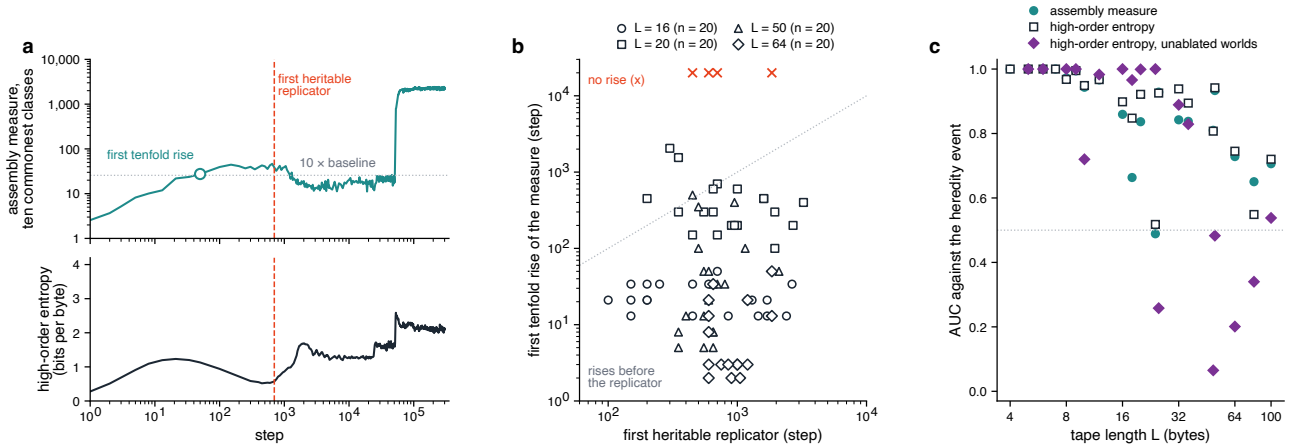
What is proved. Theorems 1 and 2 (Supplementary Information; Fig. 6a): a straight-line replicator runs on into its partner, open by construction; a confined one must re-execute an address (return control into code already run) wherever execution cannot stop inside it before the budget ends (benign-tar Z80: 128 executions, ≤ 100 cells; HALT counts as a repeat by convention) or a copy costs more executions than it has cells (BFF). Lethal tar breaks the first proviso (a zero can stop execution inside the organism); the second is unproven for the Z80. Counting complete self-writers, the usual chance-origin estimate, does not explain which comes first: all five of period two are open, 558 of 641 of period four closed (Fig. 6b; Proposition 3); the open word's head start (Kaplan–Meier medians at L = 16, Stage G: first replicator 450 steps, first closer 21,000) is measured, not counted. Proposition 8 bounds heritable variation: a closer with the block-copy core transmits $d - 4$ sites (Fig. 6c). Perfect fidelity requires zero information inflow (Lemma 7), reached by confinement in Z80 loop-bearing finals and, while entering the partner in every encounter, by the BFF literal writing two or three copies (Figs 3c and 5d). A window model (closure probability $\approx 1 - \exp(-q \int n dt)$; q, closed-variant rate; n, open population) fits the lethality dial's ends, not its middle (Supplementary Information, Model 5).

Pre-registration and stages. Supplementary Table 5 lists every run of the numbered stages. Exploratory Stages A, B, C and E came first (Supplementary Methods). Stages D, F, G, H and I, the BFF cells and every stage added in revision (K, L, M, the mutational scan, the pair invasion, the traced execution and the dials) were pre-registered with predictions and kill criteria before launch. Misses are reported in the text and listed, with the registered tests of the second and third revisions, in Supplementary Methods. Confirmatory stages: D (280: the L = 9 reversal, seeds 1001–1020), F (210: padded pair memories), G (80: closure, seeds 2001–2020, L = 16 and 50 for 300,000 steps, L = 20 and 64 for one million), K, L, M as above.

Data availability. Every condition file, raw run summary and sample record, every generated table from which a number in the text or a figure is drawn, and the pre-registration documents with their dated change log are in the project repository, to be archived with a DOI on acceptance. Supplementary Video 1 shows the world of Fig. 2.

Code availability. The simulator (a Z80 emulator in WGSF with the prefix-aware ablation hook, the traced executor and the BFF kernel with its CPU reference), the differential tests of the executor against the reference, the assays, the analysis scripts and the figure scripts are in the same repository; each figure is produced from the generated tables by one script.

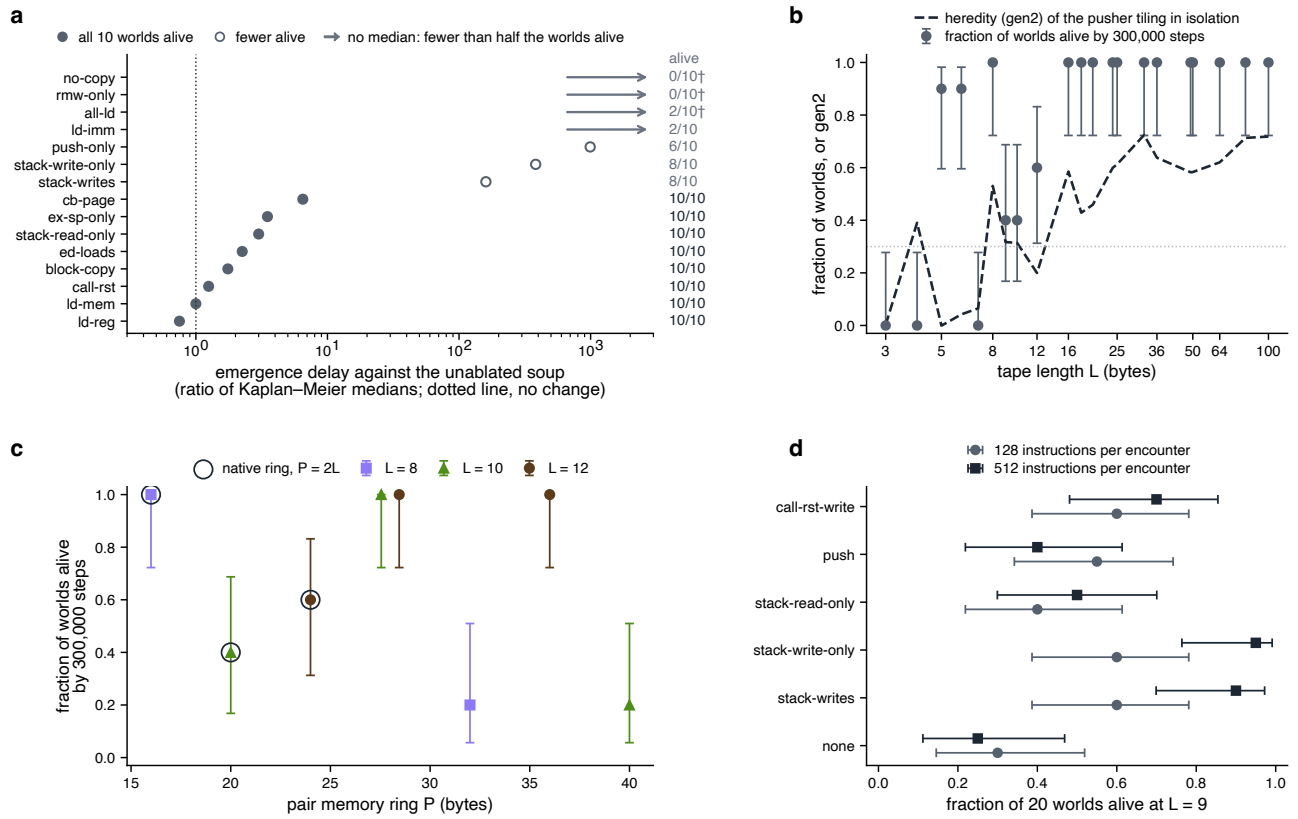
Extended Data



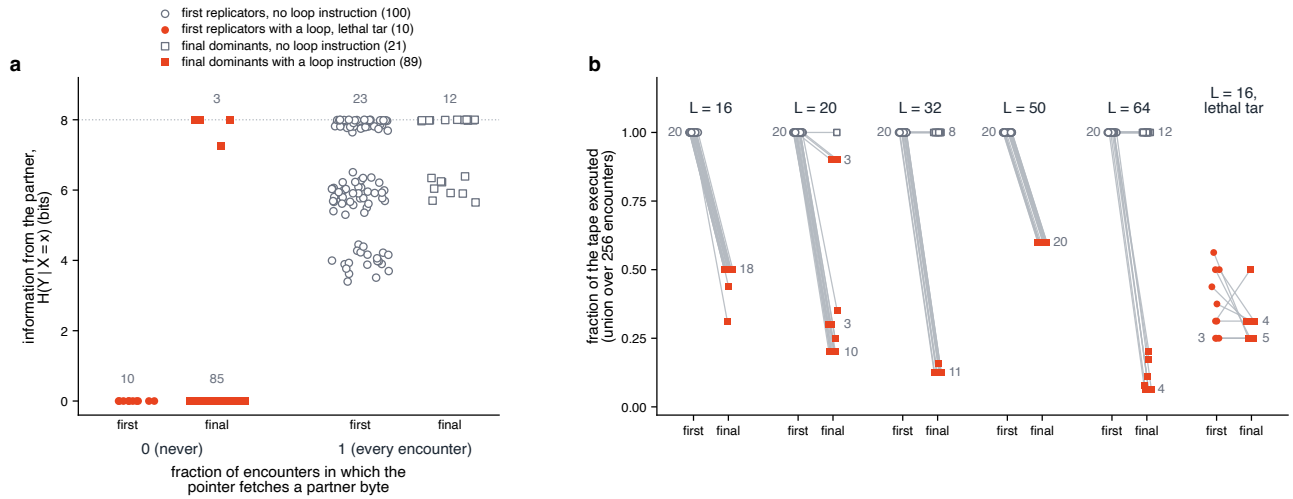
Extended Data Fig. 1 | Two pattern detectors against the culture test. **a**, The assembly measure over the ten most common classes (a Re-Pair⁴⁴ upper bound on the assembly index in the formula of ref. 27; top) and the high-order entropy of ref. 17 (bottom) against step for one world ($L = 16$, seed 2001): the measure rises tenfold above its random-soup baseline at step 50 (circle), 650 steps before the first heritable replicator (dashed line), while the soup is filling with tar. **b**, Step of the measure's first tenfold rise against the step of the first heritable replicator for the 80 Stage G worlds (dotted line, equality; crosses at the top edge, worlds without a rise). **c**, Area under the ROC curve against the heredity event by tape length: the assembly measure (filled circles) and the high-order entropy (open squares) over every arm of the size-axis stage (Stage E) at its snapshot steps, and the high-order entropy in the unablated worlds of that stage alone (diamonds), near chance or inverted at six of the eight lengths from 25 to 100 bytes; dotted line, chance. The Re-Pair index of every first replicator equals the minimum for any string of its length.

Extended Data Table 1 | Final dominants of the confirmatory closure stage, by the instruction that closes their execution. For each tape length, the 20 final dominants of Stage G (the most heritable of the three most common tapes; at $L = 50$ a stage of a lineage whose offspring reopens, Supplementary Methods) grouped by the control-flow or block-repeat instruction they carry (none for the open pusher), with the most common tape of each class (first eight bytes) and the number of worlds that end on it byte for byte, and the medians of the partner test over 256 random partners.

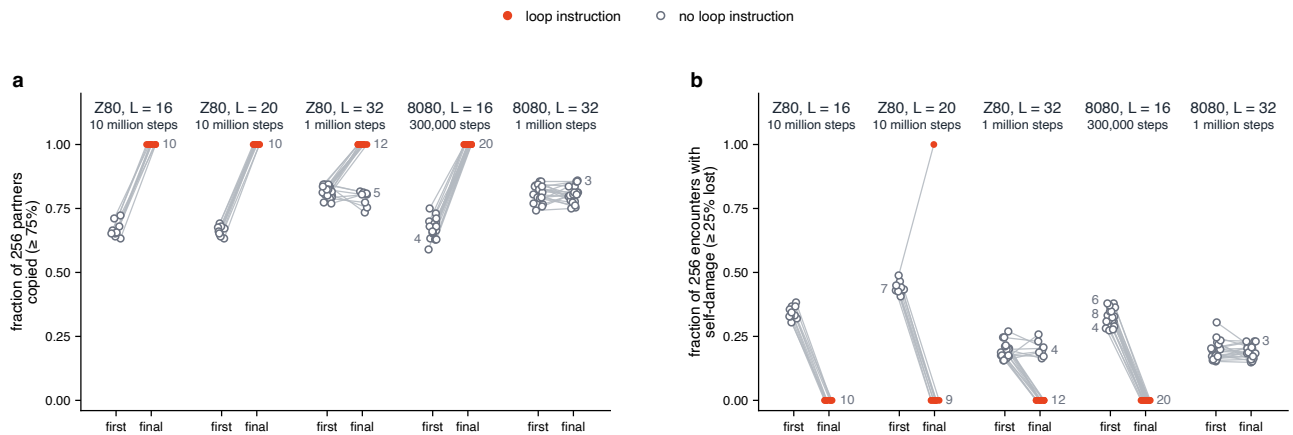
L	closing instruction	worlds	most common final tape in the class (worlds)	partners copied	self-damage
16	conditional return (RET NZ)	17	ad e3 21 e3 21 c0 ad c0 ... (17)	1.00	0.00
16	block copy (LDIR)	2	1e 50 f9 c3 8e 4e f6 b5 ... (1)	1.00	0.00
16	absolute jump (JP nn)	1	50 5e c3 eb 12 ee 8a 51 ... (1)	1.00	0.00
20	block copy (LDIR)	13	1e 04 ed b0 1e 04 ed b0 ... (4)	1.00	0.00
20	relative jump (DJNZ)	3	21 e5 21 e5 21 4e 10 e5 ... (2)	1.00	0.00
20	block copy (LDDR)	2	1e 9c 10 f6 6b c2 03 33 ... (1)	1.00	0.00
20	none: the open pusher	1	21 e5 21 e5 21 e5 21 e5 ... (1)	0.69	0.36
20	relative jump (JR)	1	21 e5 21 e5 21 fd 20 e5 ... (1)	1.00	0.00
50	relative jump (JR)	20	01 c5 01 c5 01 c5 01 c5 ... (14)	1.00	0.00
64	none: the open pusher	12	11 d5 11 d5 11 d5 11 d5 ... (5)	0.78	0.06
64	block copy (LDIR)	5	84 5e ed b0 84 5e ed b0 ... (3)	1.00	0.00
64	block copy (LDDR)	2	b8 4b 2c cf ef ec ae 83 ... (1)	1.00	0.00
64	absolute jump (JP (HL))	1	d5 d5 11 73 e9 d5 11 d5 ... (1)	1.00	0.00



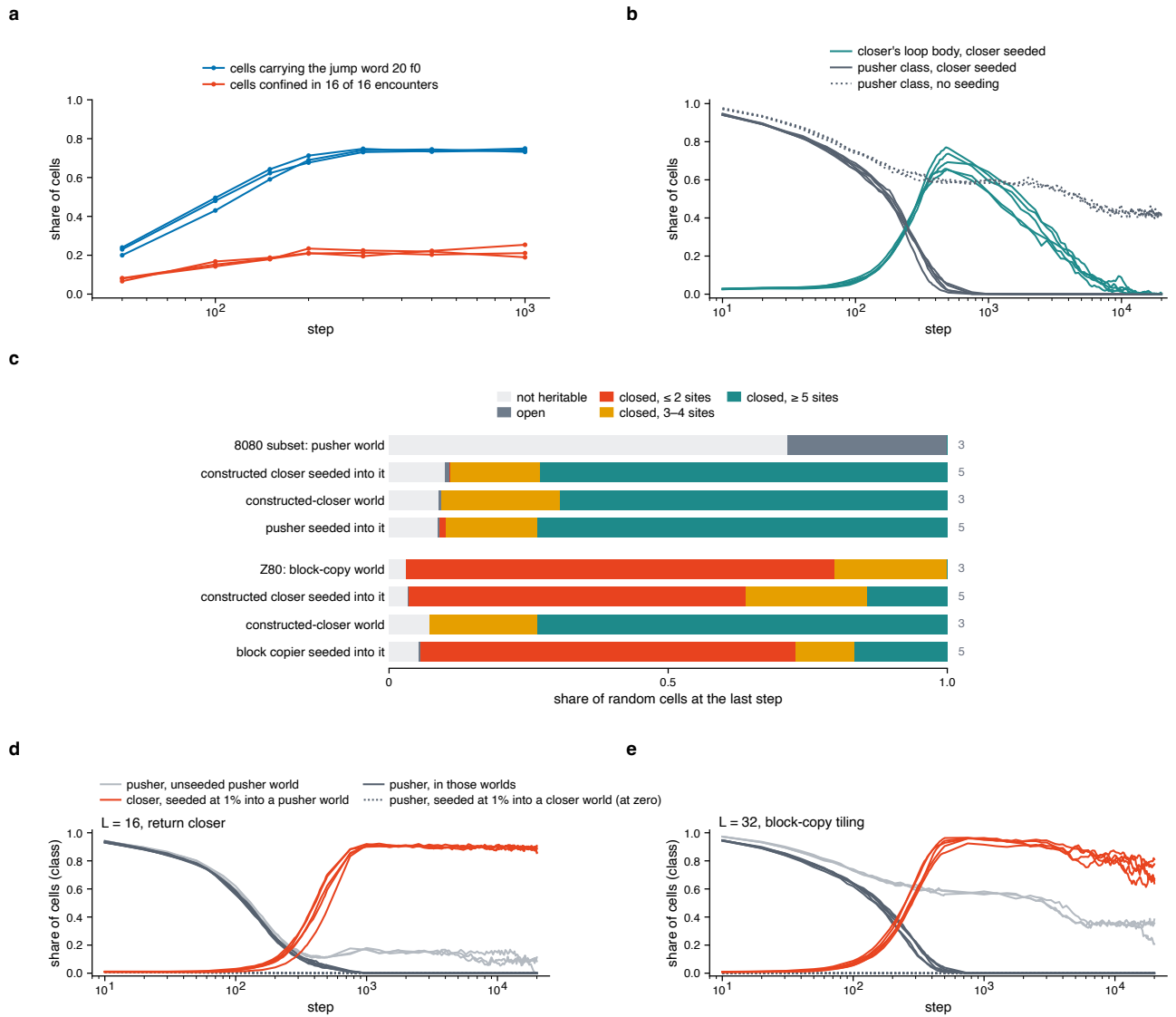
Extended Data Fig. 2 | What the instruction set must provide. **a**, Delay of emergence relative to the unablated soup, as the ratio of Kaplan–Meier medians (128 instructions per encounter, mutation 1/16, seeds 101–110), for each of the 15 ablations of Stage C; filled, all ten worlds alive (holding a heritable replicator); open, fewer; arrows, no ratio because fewer than half the worlds produced a replicator; right margin, worlds alive by 300,000 steps (black, all ten; grey, fewer), or by 1,000,000 steps for the three arms marked † (all-ld, rmw-only and no-copy), which were run and counted to one million steps; arms (a, d) as in Supplementary Table 4. Without immediate loads 2 of 10 worlds produce a replicator by 300,000 steps (1 of 10 without any load); without the stack writers, or without PUSH and POP alone, the first replicator arrives 160- to 992-fold later, as a block copier; without any copying route none arises in a million steps. Among the arms that keep the immediate loads and PUSH, the largest delays are 6.5-fold without the CB page and 3.5-fold without EX (SP),HL (seed-paired sign tests, $P = 0.75$ and 0.18). **b**, Fraction of worlds alive by 300,000 steps against tape length (Wilson 95% intervals; ten worlds per length) and the heredity (gen2) of the pusher tiling tested in isolation at each length (dashed; dotted line, the 0.3 threshold); the two diverge at several lengths below 16 bytes, where block copiers come first (at 5, 6, 9 and 12 bytes; the pusher still comes first at 8) and the stack and instruction pointers collide on the short ring. **c**, Fraction of worlds alive against the ring length P of the pair memory at $L = 8, 10$ and 12 (Stage F; circled, the native ring $P = 2L$, from Stage E; Wilson 95% intervals; at $P = 28$, the one ring two lengths share, $L = 10$ is drawn 0.45 byte left and $L = 12$ 0.45 byte right); the outcome depends on the ring as well as on the organism, a 28-byte ring raising it to 10 of 10 at $L = 10$ and 12 and a 32-byte ring lowering it to 2 of 10 at $L = 8$. Of the rings shown only $P = 16$ and 32 are aligned (a ring P is aligned if P divides 65,536, so that the 16-bit program counter and stack pointer fold onto it consistently; Methods). **d**, Fraction of 20 worlds alive at $L = 9$ by ablation at 128 and 512 instructions per encounter (Stage D): removing the stack writers makes a replicator more likely, not less.



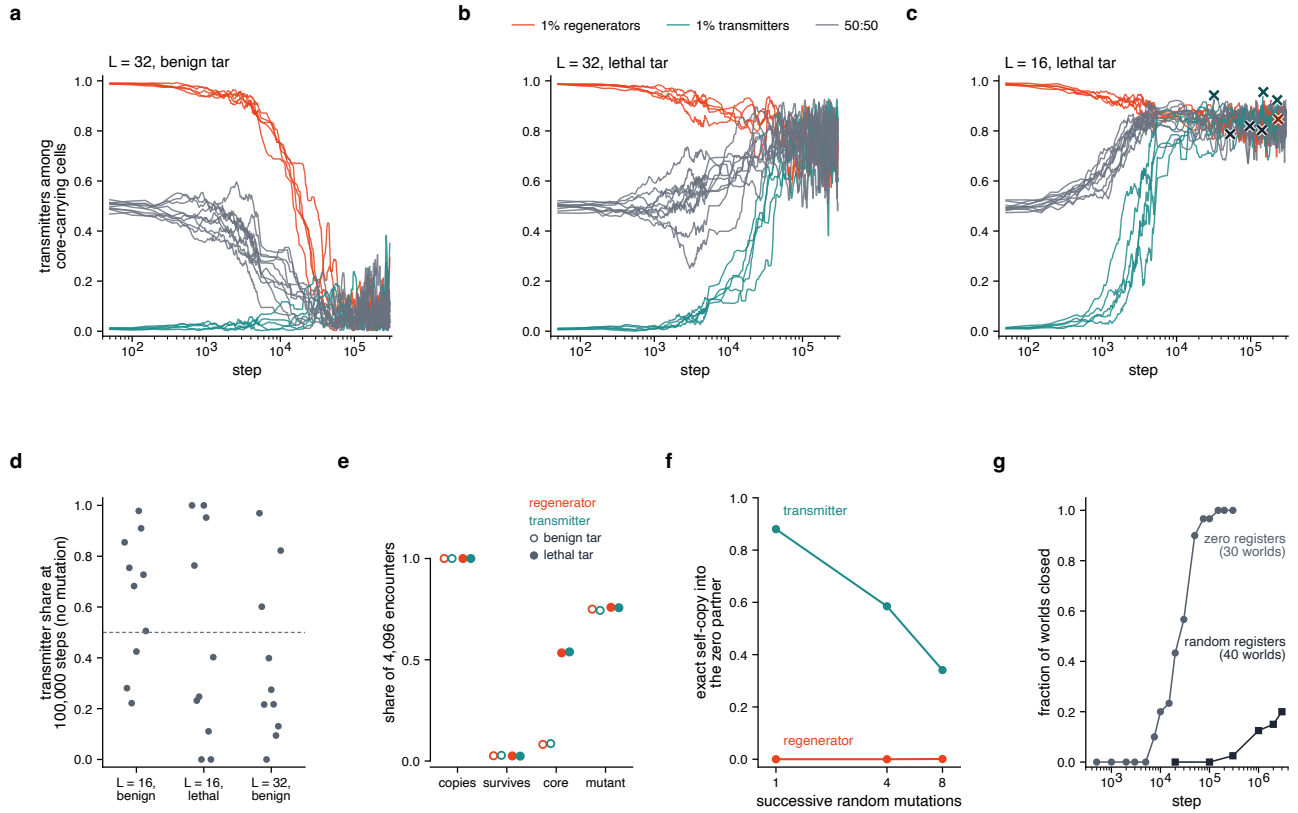
Extended Data Fig. 3 | Pointer confinement, information inflow and the executed part of the tape. Every first replicator (circles) and final dominant (squares) of Stages G, K and I, 110 of each, run against 256 seeded random partners on a traced copy of the executor (filled vermillion, loop instruction; open, none). **a**, Information the offspring carry about the partner, $H(Y|X=x)$, for replicators whose pointer fetches a partner byte in no encounter (0) or in every encounter (1), the only values that occur; first replicators and final dominants in separate columns, spread sideways only; numbers, coincident points at 0 bits and points at 7.9–8 bits; dotted line, the 8-bit ceiling of 256 partners. The 100 first replicators born under benign tar enter the partner in every encounter and carry 3.4–8 bits; the ten born under lethal tar and all 89 loop-bearing final dominants never enter it, and 85 of the 89 carry no inflow, the other four keeping one to three partner bytes (7–8 bits); the 21 final dominants without a loop instruction are open pushers. **b**, Fraction of the tape fetched as instruction stream in any of the 256 encounters, first replicator against final dominant per world (grey lines join the same world; points spread sideways within each column; numbers, coincident points). The pusher executes its whole tape; the closers execute part of it, and the evolved ones regenerate the rest from it (Fig. 4a,c).



Extended Data Fig. 4 | Closure across lengths, horizons and instruction sets. First heritable replicator (first) and final dominant (final) of every world of Stages K, L and M, partner test over 256 random partners (grey lines join the same world; filled, loop instruction; points spread sideways within each column; numbers, coincident points). **a**, Fraction of partners copied ($\geq 75\%$ copies). **b**, Fraction of encounters with self-damage ($\geq 25\%$ of the organism's bytes lost). Z80 at the aligned length $L = 32$ (Stage K, one million steps): 12 of 20 worlds close, all by block copy. Z80 at $L = 16$ and 20 for ten million steps (Stage L): 20 of 20 close (10 at each length). The 8080 subset (Stage M): at $L = 16$ all 20 worlds close on the return design ad e3 21 e3 21 c0 ad c0; at $L = 32$ none of 20 does within a million steps, although the subset can express a closer that, seeded, takes over (Supplementary Methods; Extended Data Fig. 5b,c). One $L = 20$ final dominant (seed 6003, a block copier at 0.3% of the soup) damages itself in every encounter while copying every partner.



Extended Data Fig. 5 | Five invasion tests. **a**, The $L = 50$ marker against confinement: the closed form with three jumps seeded at 1% into a world of its ancestor (three soups): share of cells carrying the jump word 20 f0 (blue) and share of 512 random cells whose pointer stays out of the partner in 16 of 16 encounters (vermilion), from step 50 (both start at 0.01 at seeding). The jump reaches about 0.74 of the cells; the confined share rises to about 0.2 and levels off. **b**, The constructed 8080 closer (20-byte core and a 12-byte random payload) seeded at 1% into a world of the 8080 pusher at $L = 32$ under the 8080 subset (five soups): share of cells carrying the closer's loop body 2b 46 2b 4e c5 15 c2 (teal; defined after the first run) and share within Hamming distance 8 of the pusher (slate, solid), with unseeded pusher worlds (dotted, three soups). The exact loop body then drifts by mutation, so its share falls while the closer's descendants hold the soup (c). **c**, Classes of 64 random cells at the last step (8080 subset, 20,000 steps; Z80, 50,000), mean over soups (numbers at right; classes as in Fig. 4d; the split of closed cells by transmissible sites rests on up to 16 scanned heritable cells per soup). The 8080 pusher world stays open; the constructed closer takes it over, and its descendants carry bytes they never execute; under the full Z80 the evolved block-copy tiling 04 5e ed b0 $\times 8$ holds its world against the constructed closer and, seeded at 1%, displaces it. **d**, **e**, Invasions at aligned lengths, under the Stage G dynamics at $L = 16$ (d) and the Stage K dynamics at $L = 32$ (e): the evolved return closer ad e3 21 e3 21 c0 ad c0 $\times 2$ and the evolved block-copy tiling 04 5e ed b0 $\times 8$ seeded at 1% into a world filled with the pusher (closer class, vermilion; pusher class in the same soups, slate; five soups each), the pusher seeded at 1% into a world filled with the closer (dotted, five soups; at zero within 140–270 steps at $L = 16$ and 20–30 at $L = 32$), and unseeded pusher worlds (grey, three each; at $L = 16$ one of them closed by itself). A class is the set of cells within Hamming distance $L/4$ of the design at any cyclic shift. At 20,000 steps every sampled heritable cell of the closer-seeded soups is confined. Colours as in Fig. 4: vermilion, closers that regenerate (d, e); teal, the transmitting constructed closer (b).



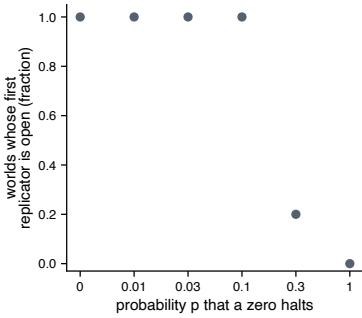
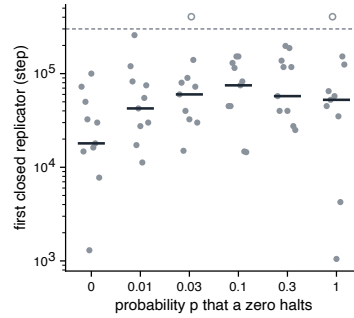
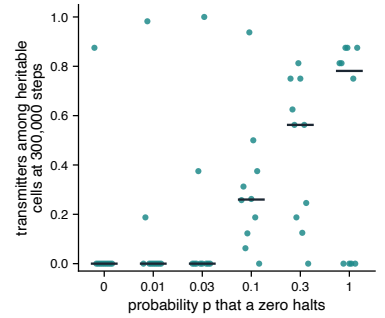
Extended Data Fig. 6 | The copy-offset switch and closure without supplied registers. **a–c**, A regenerator R and a transmitter T with the same core, $XX\ 5e\ ed\ b0$, differing only in the offset byte and in what follows the core ($L = 32$: $R = 04\ 5e\ ed\ b0 \times 8$, $T = a0\ 5e\ ed\ b0 + 28$ random bytes; $L = 16$: $R = 44\ 5e\ ed\ b0 \times 4$, $T = b0\ 5e\ ed\ b0 + 12$ random bytes, drawn for every cell), with the standard mutation: transmitters among the cells carrying the core against step, from 1% regenerators (vermillion), 1% transmitters (teal) or a 50:50 mixture (grey); one line per world (five, five and ten); after step 5,000 each line is a running median of nine samples, and samples with fewer than 1,000 core-carrying cells are then omitted; $\times$, in a darker shade of the line colour, the end of the line of a world that lost the core (its last sample joined to the one before; these worlds' lines are drawn above the others; in **c** the crosses of two 50:50 worlds ending within a marker of each other are drawn side by side). At $L = 32$ every start converges to a low transmitter share with benign tar (**a**) and to a high one with lethal tar (**b**); at $L = 16$ with lethal tar (**c**) to a high one, although the core was later lost in 7 of 20 worlds (1 of 5 from 1% regenerators, 3 of 5 from 1% transmitters, 3 of 10 mixtures). At $L = 16$ with benign tar and mutation the core fell below 1,000 cells in all 20 worlds, and that condition is not shown; 7 of the 20 were above 1,000 again at 300,000 steps, with transmitter shares of 0.14–0.63. **d**, Without mutation the share drifts: transmitter share at 100,000 steps in each of ten 50:50 worlds per condition (points spread sideways in world order; dashed, the start). **e**, Encounter components of the pristine pair at $L = 16$ (4,096 encounters each; partners and executors drawn from the final soups of the mixed worlds of each tar, under benign tar soups in which the core had collapsed): exact copy into a partner (copies), exact survival as the partner of a soup executor (survives), core bytes 1–3 intact as a partner (core), and exact copying by a random single mutant (mutant); hollow markers, benign tar; filled, lethal tar. **f**, Post hoc, ref. 23's robustness criterion for the $L = 32$ pair (exact self-copy into the all-zero partner, 512 instructions, after k successive random mutations; 1,000 trials; median over 20 transmitter tails): the transmitter scores high because it keeps mutations, the regenerator near zero because it erases them. **g**, Fraction of worlds at $L = 16$ in which most heritable cells among 64 random cells are confined, at the recorded snapshots, with zero registers (slate: Stages G and L, 30 worlds; 20 at 7,500 steps) and with every register drawn at random before each encounter (black: 40 worlds, seeds 8101–8120 and 8201–8220).



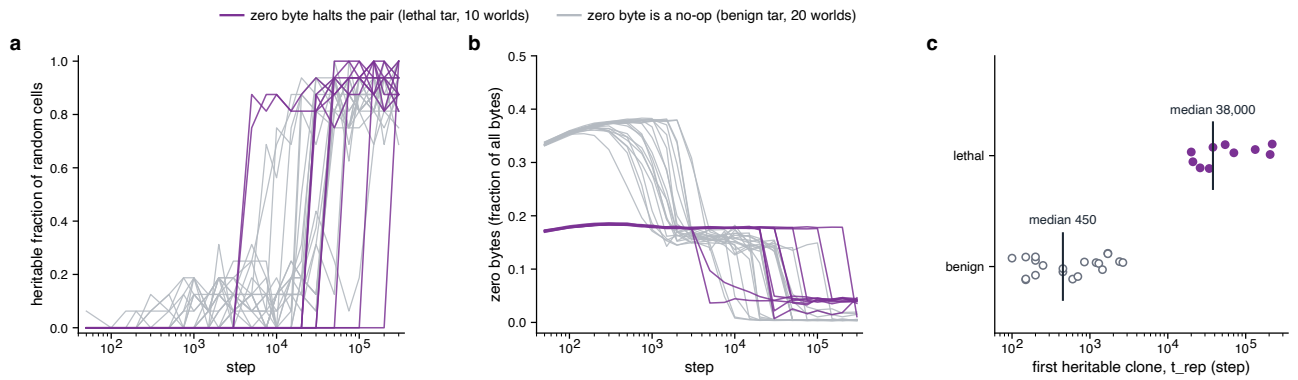
Extended Data Fig. 7 | The first confined copiers are not point mutants. Every founding event is a rewrite (16) or a copy of the partner (1). **a, b**, Lines of descent from the newest open copier older than the founder (chain start) to the founder, one world per executor kind, nearest the median chain (14 records, 79 steps). Bytes in hex, coloured by instruction class (key); filled if the byte is an instruction start in at least half of 64 runs with random partners. **a**, World 31010 (15 records, 85 steps): the executor rewrites its own half into the founder. **b**, World 31019 (11 records, 79 steps): a point mutation three steps earlier makes a confined tape that writes the founder into every partner. **c**, Share of 512 soup partners with which each first founder's executor makes it (17 worlds); filled, its pointer entered the partner in the founding encounter (10 of 17). All 17 executors are whole-tape non-copiers (84–89% of the soups; $P = 0.076$ by abundance). Twelve copy neither tape nor executed code (median 0.002 without the open producer of 31005, which makes it with every partner; none in 5; at most 0.46); five, confined, copy their executed code (at least 75% of it in at least half of the runs), always make the founder and arose one to three steps earlier. Point mutations made three writers (31005, 31006, 31019). In 31007 the founder is a copy of an open-copier partner. **d**, Bytes changed between each founder and the tape before it (best cyclic shift; median 8, range 4–10); dotted, the reach of one point mutation; text, the mutational-reach null. **e**, Steps from the newest open copier to the founder (median 79, range 1–360); only non-copiers between them in 16 of 17. In the soups before the founders, 21 e3 was carried by a median 0.24% of cells, typical of the 255 second-byte variants of 21 e5, and 21 e0 by 1.8%, 15 times their median (post hoc). Event kinds and copier tests are defined in Methods; a point mutation changes one, rarely two, bytes. Exploratory analysis of the first 17 worlds (Methods); the registered replication on 20 new worlds is in Supplementary Table 3.

a

	random partners, benign	random partners, lethal	closed benign world	lethal world	open phase (step 5,000)	random registers
pusher	○	○	○	○	○	○
return closer	●	○	●	○	●	○
block-copy R	●	●	●	●	●	○
block-copy T	●	●	●	●	●	○
lethal-tar closer 4009	●	●	●	●	●	○
self-initialising closer	●	●	●	●	●	●

b**c****d**

Extended Data Fig. 8 | Parent, environment and lethality. **a**, Six parents at $L = 16$ copied through eight serial transfers in six environments, in the order of the columns (every single-byte mutant in 16 lineages, 256 unmutated lineages): benign or lethal rule with uniform random partners; partners drawn from the closed final soup of Stage G world 2001 (benign rule), from the lethal final soup of Stage I world 4001 (lethal rule) or from the open phase of world 2001 at step 5,000 (benign rule); and random registers with random partners. Filled, unmutated lineages alive after eight transfers (1.00 in every filled cell); hollow, lost (at most 0.04 alive, except the pusher among partners from the closed world, 0.36, drawn hollow because fewer than half survive); number, transmissible sites after eight transfers. Colours: open pusher, slate; regenerators, vermilion; transmitters, teal. The mode belongs to the parent; the environment decides which parents survive. **b–d**, The lethality dial: a zero byte fetched as an opcode halts the pair with probability p , ten worlds per p at $L = 16$ (the same seeds at every p), 300,000 steps. **b**, Fraction of worlds with a replicator whose first replicator is open (its pointer enters the partner in at least half of 16 traced encounters); at $p = 1$ one world (seed 9004) has no heritable common tape and the fraction is of 9. **c**, Step of the first closed replicator (a top-3 tape holding at least 0.5% of cells, heritable and confined in 16 of 16 encounters, the registered endpoint) per world; hollow circles, not closed by 300,000 steps (dashed); bars, Kaplan–Meier medians (the fifth of ten ordered times). The endpoint misses closed populations of transmitters, whose genomes all differ so that no tape holds 0.5% of cells: both hollow worlds ($p = 0.03$, seed 9007; $p = 1$, seed 9004) are closed and transmitter-dominated at 300,000 steps (d), which biases the timing test against closure at high p . **d**, Share of heritable random cells that are transmitters (confined, at least five transmissible sites) at 300,000 steps per world; bars, medians (the mean of the fifth and sixth of ten).



Extended Data Fig. 9 | Lethal tar in the first machine. Ten pre-registered worlds at $L = 16$ in which a zero byte fetched as an opcode halts the pair (Stage I, purple) against the twenty benign-tar worlds of Stage G at the same length (grey). **a**, Heritable fraction of 16 random cells against step. **b**, Zero-byte fraction against step. **c**, Step of the first heritable clone, t_{rep} , in each world (jittered vertically), with Kaplan–Meier medians (38,000 against 450). Every lethal-tar first replicator is a block-copy replicator with a hardware loop, contains no zero byte, and copies every partner without self-damage from its first appearance. The first clone can lag the first heritable material: in two lethal-tar worlds 75–88% of random cells are heritable by step 5,000 (a), while the first tape among the three most common to pass the culture test is dated 21,000 and 206,000 steps; dated by heritable material, the delay is 32-fold (plain medians, 40,000 against 1,250 steps).

References

1. Maturana, H. R. & Varela, F. J. *Autopoiesis and Cognition* (Reidel, 1980).
2. Rosen, R. *Life Itself* (Columbia Univ. Press, 1991).
3. Fontana, W. & Buss, L. W. “The arrival of the fittest”: toward a theory of biological organization. *Bull. Math. Biol.* **56**, 1–64 (1994).
4. Krakauer, D., Bertschinger, N., Olbrich, E., Flack, J. C. & Ay, N. The information theory of individuality. *Theory Biosci.* **139**, 209–223 (2020).
5. Dyson, F. *Origins of Life* 2nd edn (Cambridge Univ. Press, 1999).
6. Szostak, J. W. The eightfold path to non-enzymatic RNA replication. *J. Syst. Chem.* **3**, 2 (2012).
7. Kauffman, S. A. Autocatalytic sets of proteins. *J. Theor. Biol.* **119**, 1–24 (1986).
8. Maynard Smith, J. & Szathmáry, E. *The Major Transitions in Evolution* (W. H. Freeman, 1995).
9. Szathmáry, E. The evolution of replicators. *Phil. Trans. R. Soc. Lond. B* **355**, 1669–1676 (2000).
10. Szostak, J. W., Bartel, D. P. & Luisi, P. L. Synthesizing life. *Nature* **409**, 387–390 (2001).
11. Moreno, A. & Mossio, M. *Biological Autonomy: A Philosophical and Theoretical Enquiry* (Springer, 2015).
12. Walker, S. I. & Davies, P. C. W. The algorithmic origins of life. *J. R. Soc. Interface* **10**, 20120869 (2013).
13. Woese, C. R. On the evolution of cells. *Proc. Natl Acad. Sci. USA* **99**, 8742–8747 (2002).
14. Nowak, M. A. & Ohtsuki, H. Prevolutionary dynamics and the origin of evolution. *Proc. Natl Acad. Sci. USA* **105**, 14924–14927 (2008).
15. Ray, T. S. An approach to the synthesis of life. In *Artificial Life II* (eds Langton, C. G., Taylor, C., Farmer, J. D. & Rasmussen, S.) 371–408 (Addison-Wesley, 1991).
16. Ofria, C. & Wilke, C. O. Avida: a software platform for research in computational evolutionary biology. *Artif. Life* **10**, 191–229 (2004).
17. Agüera y Arcas, B. et al. Computational life: how well-formed, self-replicating programs emerge from simple interaction. Preprint at <https://arxiv.org/abs/2406.19108> (2024).
18. Rasmussen, S., Knudsen, C., Feldberg, R. & Hindsholm, M. The Coreworld: emergence and evolution of cooperative structures in a computational chemistry. *Physica D* **42**, 111–134 (1990).
19. Pargellis, A. N. The spontaneous generation of digital “Life”. *Physica D* **91**, 86–96 (1996).
20. Koza, J. R. Spontaneous emergence of self-replicating and evolutionarily self-improving computer programs. In *Artificial Life III* (ed. Langton, C. G.) 225–262 (Addison-Wesley, 1994).
21. Hutton, T. J. Evolvable self-replicating molecules in an artificial chemistry. *Artif. Life* **8**, 341–356 (2002).
22. Dittrich, P., Ziegler, J. & Banzhaf, W. Artificial chemistries — a review. *Artif. Life* **7**, 225–275 (2001).
23. Cicala, F. et al. Coevolution of self-replication and function in a digital primordial soup. Preprint at <https://arxiv.org/abs/2607.09211v2> (2026).
24. Jha, K. et al. Tapes together strong: the co-evolution of computation and cooperation. Preprint at <https://arxiv.org/abs/2609.10817> (2026).
25. Knierim, C., Versari, L., Obryk, R., Agüera y Arcas, B. & Sauro, R. A. BFF: simple explanations for complex phenomena. Preprint at <https://arxiv.org/abs/2607.01483> (2026).
26. Benner, S. A., Kim, H.-J. & Carrigan, M. A. Asphalt, water, and the prebiotic synthesis of ribose, ribonucleosides, and RNA. *Acc. Chem. Res.* **45**, 2025–2034 (2012).
27. Sharma, A. et al. Assembly theory explains and quantifies selection and evolution. *Nature* **622**, 321–328 (2023).
28. Marshall, S. M. et al. Identifying molecules as biosignatures with assembly theory and mass spectrometry. *Nat. Commun.* **12**, 3033 (2021).
29. Boerlijst, M. C. & Hogeweg, P. Spiral wave structure in pre-biotic evolution: hypercycles stable against parasites. *Physica D* **48**, 17–28 (1991).
30. Langton, C. G. Self-reproduction in cellular automata. *Physica D* **10**, 135–144 (1984).
31. von Neumann, J. *Theory of Self-Reproducing Automata* (ed. Burks, A. W.) (Univ. Illinois Press, 1966).
32. Waddington, C. H. Canalization of development and the inheritance of acquired characters. *Nature* **150**, 563–565 (1942).
33. Wagner, A. *Robustness and Evolvability in Living Systems* (Princeton Univ. Press, 2005).
34. Wagner, A. Robustness and evolvability: a paradox resolved. *Proc. R. Soc. B* **275**, 91–100 (2008).
35. Adami, C. & LaBar, T. From entropy to information: biased typewriters and the origin of life. In *From Matter to Life: Information and Causality* (eds Walker, S. I., Davies, P. C. W. & Ellis, G. F. R.) 130–154 (Cambridge Univ. Press, 2017).

36. C G, N., LaBar, T., Hintze, A. & Adami, C. Origin of life in a digital microcosm. *Phil. Trans. R. Soc. A* **375**, 20160350 (2017).
37. Vasas, V., Szathmáry, E. & Santos, M. Lack of evolvability in self-sustaining autocatalytic networks constraints metabolism-first scenarios for the origin of life. *Proc. Natl Acad. Sci. USA* **107**, 1470–1475 (2010).
38. von Kiedrowski, G. A self-replicating hexadeoxynucleotide. *Angew. Chem. Int. Ed. Engl.* **25**, 932–935 (1986).
39. Cojocaru, R. & Unrau, P. J. Processive RNA polymerization and promoter recognition in an RNA World. *Science* **371**, 1225–1232 (2021).
40. Sumper, M. & Luce, R. Evidence for de novo production of self-replicating and environmentally adapted RNA structures by bacteriophage Q β replicase. *Proc. Natl Acad. Sci. USA* **72**, 162–166 (1975).
41. Godfrey-Smith, P. *Darwinian Populations and Natural Selection* (Oxford Univ. Press, 2009).
42. Bertschinger, N., Olbrich, E., Ay, N. & Jost, J. Autonomy: an information theoretic perspective. *BioSystems* **91**, 331–345 (2008).
43. Eigen, M. Selforganization of matter and the evolution of biological macromolecules. *Naturwissenschaften* **58**, 465–523 (1971).
44. Larsson, N. J. & Moffat, A. Off-line dictionary-based compression. *Proc. IEEE* **88**, 1722–1732 (2000).
45. Montévil, M. & Mossio, M. Biological organisation as closure of constraints. *J. Theor. Biol.* **372**, 179–191 (2015).
46. Dewdney, A. K. Computer recreations: in the game called Core War hostile programs engage in a battle of bits. *Sci. Am.* **250**(5), 14–22 (1984).

Supplementary Methods

Each paragraph below extends the Methods paragraph of the same heading; paragraphs whose heading does not appear in Methods are given here in full.

Soup. In the padding experiment of Stage F, $P \geq 2L$ and the padding bytes are never written back. Background mutation ($8,192/2^k$ random bytes per step) uses $k \in \{2, 4, 6\}$ in the mutation arms of Stages A, C and D and $k \in \{1, 2, 3, 5, 8\}$ in the Stage E mutation sweep at $L = 100$; the per-byte-constant arm of Stage E replaces $32 \cdot L$ bytes per step, the $L = 16$ rate per byte, at every length.

Recording. Stages C–M write a sample at steps 1, 2, 3, 5, 8, 13, 21 and 34, then every 50 steps to step 5,000, then every 500 steps (300,000-step runs), 1,000 (one-million-step runs) or 5,000 (Stage L), with the ten most common exact tapes and their shares, quasispecies⁴³ occupancy (fraction of cells within Hamming distance $\lceil L/4 \rceil$ of the dominant tape), byte histogram, zero-byte fraction and byte entropy, brotli-compressed bits per byte and the high-order entropy of ref. 17, species entropy and unique-tape fraction, per-interaction statistics (bytes written into the partner, copy events, destroyed programs), 16 random tapes for the functional census, and full soup snapshots at fixed steps and at emergence. Stages A and B sample every 500 steps from step 500, with the three most common tapes and without the byte histogram, the per-interaction statistics or the random tapes.

Culture test and partner tests. A tape is heritable if its copies copy in turn, and $\text{gen2} \geq 0.3$ tests this only approximately: gen2 is a normalised gain in best-shift similarity to the original tape, not a count of copies, so offspring that are not copies contribute about zero unless, like the offspring of a word that smears its return addresses into its partners, they raise their own partners' similarity to the word without copying it. Such words pass the test without being heritable: 246 two-byte census words reach $\text{gen2} \geq 0.3$ without producing a single 75% copy (Supplementary Fig. 3). Parent, offspring and granddaughter sequences of three lineages of each kind of replicator are tabulated in `results/review_r2/LINEAGES.md(lineage_examples.py)`. Replicators are therefore also scored by the copies they make (faithfulness and the partner tests), and dating the first replicator by the faithful criterion (`t_faith`, Methods) instead of by heredity alone gives the same tape at the same step in all 80 Stage G worlds.

Events and statistics. The final dominant is the most common tape in 57 of the 60 Stage G worlds at $L = 16, 20$ and 64 ; at $L = 50$ it is the second most common in all 20, behind the plain pusher tiling (5.5–6.6% of cells), while the tapes that carry its jump word `20 f0` hold 70–76% of cells. Its offspring is then the parent with one more `20 f0` inserted, the same string in every encounter; that offspring enters the partner in every encounter, the next generation (six jumps) is confined but fails the culture test ($\text{gen2} 0.19$), and the one after (ten jumps) no longer copies (`150_lineage.py`), so at $L = 50$ the final dominant is a stage of a lineage rather than a lineage. Between-variant transition rates in BFF use two-sided Fisher tests and Mann–Whitney tests on event times with non-transitioning runs censored at the horizon. Across the 233 unablated worlds of the exploratory stages, 92 gained a control-flow instruction between the first replicator and the final dominant and 3 lost one. In BFF, where replicator populations are quasispecies with small exact-class shares, the first replicator is the most common tape class (a tape and its reverse count as one class) at the first sample at which it is heritable, and the population event is a heritable fraction ≥ 0.5 of 32 random tapes. Re-dating the first replicator from the recorded runs under heredity thresholds from 0.2 to 0.4 leaves the load–push word first in 170 of 170 benign-tar worlds of 16–64 bytes; at 0.5 and 0.6, near the pusher's own gen2 , it is first in 155 of 169 and 107 of 164.

Definitions and theorems. A *self-writer* is a tape that, executed against the all-zero partner, leaves the partner holding a copy of itself. *Closure* (confinement): the organism's pointer never fetches a partner address as part of an instruction in any encounter with a partner of the sampled class. In the tests a tape is *confined* if no partner byte is fetched in any of a stated number of traced encounters with uniformly random partners, and *open* if one is fetched, for a first replicator in at least half of them and for a sampled cell in any; each use below states the number of encounters and any other criterion. Theorem 1: copying a 64-byte BFF tape into a partner that differs from it in every cell takes at least 128 instructions, 64 that change a cell and 64 head moves, so a program whose pointer stays in its own 64 cells executes some address twice in that encounter, through a backward bracket jump: a loop, a return of control into code already executed, not a repeat of the machine's state. Theorem 2: a pointer that stays in the organism's L cells executes some address twice, a return of control into code already executed (not a repeat of the machine's state), provided a copy costs more instruction executions than the organism has addresses (BFF) or execution cannot stop inside the organism before the budget is spent (the Z80 soups with benign tar, budget $128 > L$). By convention a `HALT`, which holds the processor on its own address for the rest of the budget, counts as executing that address again, so a confined program that halts inside the organism meets the conclusion by the convention alone, without returning control into code already executed. The second proviso fails wherever a zero byte can halt the pair, and the first is not established for the Z80, so the theorem does not apply in Stage I, in the lethality dial at $p > 0$ or in any other run under the lethal rule (the lethal arms of the copy-offset switch and of

the parent $\times$ environment matrix, the encounter censuses, and the serial transfers and scans of lethal-tar tapes), nor in the Stage E arms with budgets 32 and 64 at $L = 100$, where the budget ends before a pass over the tape. Theorem 6: the tiling of a self-consistent literal-write instruction is a straight-line, hence open, program the size of the literal that writes its own content at an advancing write pointer, and it is a self-writer wherever the encounter lets those writes cover the partner's region (in the Z80, by Proposition 3); without such an instruction, under the theorem's copy-cost hypothesis, which we do not prove for BFF, and in a machine whose execution ends when the pointer leaves the pair, every self-writer executes some address twice. A literal write is sufficient for an open first replicator, not necessary: open first replicators also arose without one, in BFF with harmless brackets and in the Z80 with a random stack pointer (Supplementary Information). Lemma 7: with the organism x fixed, $I(E; Y | X = x) > 0$ implies imperfect fidelity across partners, and perfect fidelity implies $I(E; Y | X = x) = 0$; the converse fails. Proposition 8, an elementary consequence of the overlapping block move: in the block-copy core `XX 5e ed b0`, if the first instruction passes control to the second, writes no memory below the copy offset and leaves HL, D and BC as stated in the Supplementary Information, and the copy offset $d = XX \bmod 2L$ satisfies $4 \leq d \leq L$ and divides L , every encounter leaves organism and partner holding the d -periodic tiling of the organism's first d bytes, so exactly the $d - 4$ bytes between the core and the offset are transmitted. Proposition 3 (computer-assisted, `census_period.py`): every word of period two, three and four, tiled to 16 bytes and executed against the zero partner; a self-writer is closed if its pointer fetches no partner byte against the zero partner and 16 random partners. Period two: exactly six self-writers, the trivial `00 00` and the five load-push words, all straight-line and open. Period three: one, `b0 5e ed` tiled, closed, a block copier whose copy offset equals the tape length. Period four: 641 of primitive period four, 558 of them closed, all block copies (455 `LDIR`, 103 `LDDR`), and 83 open. Proposition 4 states what the count of the initial soup does and does not show. Full statements and proofs are in the Supplementary Information.

Harness conventions. Draws are seeded by the step's batch seed and the pair index. Registered predictions (pre-registration document, sections R3-C and R4) and outcomes: the first replicator is a load-push word under random registers in at least 18 of 20 worlds (met, 20 of 20) and open under a random stack pointer in at least 15 of 20 (met, 18 of 20; the first replicator is a load-push word in 5, the stack move `e1 e3` in 13 and a closed block copier in 2); closure in at most 5 of 20 random-register worlds by 3,000,000 steps (not met: 6 of 20), and in 2–12 of 20 in the replication (met: 2, both even seeds). Closers of the closed random-register worlds (most heritable cells among 64 random cells confined in 16 traced encounters) were characterised post hoc (32 heritable cells each: whole-tape exact copy over 16 encounters, never-executed bytes by tracing). Invasions under random registers at $L = 16$: the 13-byte closer `21 00 00 11 10 00 01 10 00 ed b0 18 fe` with a random 3-byte tail per cell, seeded at 1% into a pusher-filled world, and the reverse, five seeds each and three unseeded controls, 50,000 steps; under the standard convention, the same closer against the 4-byte regenerator `44 5e ed b0` $\times 4$, 100,000 steps. The registered motif-share endpoints failed on definition (neutral operand bytes drift in a transmitter); classes of 64 random cells at the end (culture test, and confinement in 16 traced encounters, under the world's convention) are reported (`r4_analyse.py`). Under random registers the seeded closer's descendants held the soup closed in 5 of 5 pusher-filled worlds: at 50,000 steps 0.81–0.91 of random cells were heritable and 0.93–0.98 of those confined, against 0.03–0.08 heritable and none confined in the three unseeded pusher worlds; with the pusher seeded at 1% into the closer's world, the soup stayed 0.84–0.89 heritable and 0.93–0.98 confined (5 of 5). Under the standard convention neither design persisted as such, and every soup stayed closed (0.88–0.97 of random cells heritable, 0.98–1.00 of those confined), held by a diverse block-copy family.

Invasions at aligned lengths. Stage G dynamics at $L = 16$: the return closer `ad e3 21 e3 21 c0 ad c0` $\times 2$ against the pusher `01 c5` $\times 8$; Stage K dynamics at $L = 32$: the block-copy tiling `04 5e ed b0` $\times 8$ against `01 c5` $\times 16$. Each seeded at 1% into a world filled with the other, five seeds each way, three unseeded controls per resident, 20,000 steps; class share (within Hamming distance $L/4$ of the design at any shift) every 10 steps to 500 and every 250 after, and classes of 64 random cells at 1,000, 5,000 and 20,000 steps. Registered: the closer holds most cells and most heritable cells are confined (no partner byte fetched in 16 traced encounters) at 20,000 steps in at least 4 of 5 soups at each length (met, 5 of 5 at both), and the pusher stays below 5% in a closer world (met, 5 of 5).

The copy-offset switch. The matched pair (registered in the revision as test E, unrelated to Stage E): at $L = 16$, $R = 44$ `5e ed b0` $\times 4$ and $T = b0 5e ed b0$ followed by 12 random bytes drawn for every cell; at $L = 32$, $R = 04 5e ed b0 $\times 8$ and $T = a0 5e ed b0$ + 28 random bytes. Conditions $L = 16$ with benign tar, $L = 16$ with lethal tar and $L = 32$ with benign tar; starts 50:50 (ten worlds without mutation for 100,000 steps and ten with the standard mutation for 300,000 steps) and 1% each way with mutation (five each); every cell classified by its first four bytes every 50 steps to 5,000 and every 500 after (`offset_switch.py`). $L = 32$ with lethal tar was registered separately, after these results, to separate the tar from the length (E-L32L): ten 50:50 worlds and five 1% starts each way, standard mutation, 300,000 steps; its prediction, a transmitter share among core-carrying cells above 0.5 at 300,000 steps in at least 7 of 10 mixtures, was met (10 of 10; median 0.77, range 0.55–0.87). The registered predictions of a mutation-bias equilibrium (the switch's E1 and E2) were not met; without mutation the share drifts. Encounter components of the pristine pair were measured with 4,096$

encounters each against cells of the final soups (`offset_encounter.py`). Mutational flows (`offset_flow.py`, registered as E-mech): every single-byte mutant of R and T, four random partners each, classified by whether its offspring copies itself exactly and with what period; the registered prediction (EM1) failed and the flows of isolated mutants are the same under benign and lethal tar, so the environment dependence is reported without a mechanism.

Encounter censuses (exploratory). Lattice-neighbour pairs (200,000 per soup set) from the final soups of the ten 50:50 $L = 32$ worlds of each tar, executed under each rule (composition crossed with rule), with every cell classed by its first four bytes before and after (`offset_census_enc.py`); and core-free executors from the same soups against three partners, R, T with a random body and T with a tiled body (`a0 5e ed b0 × 8`), traced to see whether the executor's pointer entered the partner (`offset_intruder.py`, 16,384 encounters per combination). Both were run after the matched-pair results and are reported as descriptive. The in-situ demography (`n2_demography.py`, `n2_background.py`) summed per-capita copying between classes (`n2_mutual.py`), destruction by core-free executors, conversion of intruders and mutational terms. The inherited-zero test (N4) replaced a transmitter's zeros by random non-halting bytes (`n4_scars.py`); the outcomes of N2 and N4 are given under Pre-registration and stages. Payload zeros (`payload_toxin.py`, N1) rise towards the end of the payload, are about as common under benign as under lethal tar at $L = 32$ (median 0.067 against 0.051; at $L = 16$, 0.038 against 0.060) and are absent in most mutation-free worlds.

Genealogies (exploratory, then replicated). Same-seed runs diverge because the pair draw resolves collisions by atomic claims whose order varies. Records from which no living cell descends are freed. Validation: before any registered world, replaying every active pair of the first 200 steps of a validation world on the traced executor reproduced the recorded memories byte for byte (795,934 encounters), cells changed outside active pairs never exceeded the mutation count, and in a world seeded with the return closer at 1% among pushers all 64 sampled confined copiers traced to a seeded closer; the replay check held in all 40 recorded worlds. Lines, founders and first founders are computed by `lod_chain.py` and `lod_founders2.py`, the first copier on each line by `lod_openfirst.py`; every first founder and confined writer entered the partner in none of 64 encounters. Word shares are measured in the last soup snapshot (every 2,000 steps) before each founder against a random-tape baseline of about 2.4×10^{-4} per two-byte word (`lod_parts_first.py`) and, post hoc, against the shares of all 255 second-byte variants of `21 e5` in the same snapshot (`lod_neighbours.py`). The founding encounter of each first founder was replayed from the record to find which tape executed and on which side the founder was written; the executor and every earlier record on the chain were then run against 64 random partners to measure how often each writes the founder into the partner or turns itself into it, and whether its pointer enters the partner (`lod_founding.py`). Because the copier test asks for a copy of the whole tape, each founding executor was also tested by its executed bytes only (a tape copies its executed code if, in at least half of 64 random encounters, at least 75% of the organism positions it fetches reappear in the partner), and its production of the founder was measured with 512 partners drawn from the last soup snapshot before the founder (the executed-bytes test and this production test are also in `lod_founding.py`). Every single-byte mutant of every distinct tape on the chains was classed against 64 random partners to measure the reach of point mutation, and the composition of the soup before each founder gives the abundance null for the writers' kind (`lod_reach.py`). The registered descent prediction was met but uninformative, since by then nearly every cell present had an open copier among its ancestors of the previous 2,000 steps. Separating first founders from later conversions by existing closers identifies 15 later founders in 10 worlds, 11 of them with bytes from a confined copier, and 14 in 10 worlds of the replication, 11 of them with such bytes.

Parent × environment. Six parents at $L = 16$ (the pusher; the evolved return closer; R and T of the copy-offset switch, T with a fixed random tail; the evolved lethal-tar closer of world 4009; the self-initialising closer `2e 00 1e 10 ed b0` with a fixed random tail) were followed through eight serial transfers, with every single-byte mutant in 16 lineages and 256 unmutated lineages, as in the serial-retention test, in six environments: benign or lethal rule with uniform random partners; partners drawn from the final soup of Stage G world 2001 (closed: every sampled heritable cell confined) or from its snapshot at 5,000 steps (open phase, before closure), benign rule; partners from the final soup of Stage I world 4001, lethal rule; random registers with random partners (`pe_matrix.py`; registered as PE). Registered: wherever a parent's lineages survive, regenerators carry at most 2 sites and transmitters at least 8 after eight transfers (met, 24 of 24); the pusher's lineages are lost everywhere (not met: 0.36 alive with partners from the closed world, with 0 sites).

Lethality dial. Derived soup and executor shaders (`gen_dial_shader.py`) in which a zero byte fetched as the first byte of an instruction halts the pair with probability p , a fresh draw from a per-encounter generator for every such fetch; $p \in \{0, 0.01, 0.03, 0.1, 0.3, 1\}$. Before any soup ran, the $p = 0$ executor reproduced the standard executor and the $p = 1$ executor the Stage I executor exactly on 65,536 random pairs with 25% zero bytes. Ten worlds per p at $L = 16$ under Stage G conditions, the same seeds (9001–9010) at every p , 300,000 steps; top-3 tapes every 50 steps to 2,000, every 250 to 20,000 and every 2,500 after; every assay under the world's own p (`dial_soups.py`; registered as DZ). Registered: the first replicator is open (its pointer in the partner in at least half of 16 traced encounters) in at least 8 of 10 worlds at $p = 0$ and at most 2 at $p = 1$, with a falling trend (met: 10 of 10, and 0 of the 9 worlds with a heritable common tape; one-sided Cochran–Armitage

$z = -6.1$); first closure (a top-3 tape holding at least 0.5% of cells, heritable and confined in 16 of 16 traced encounters) later at higher p , monotone up to one reversal and Spearman-positive (Spearman met, $\rho = 0.25$, one-sided permutation $P = 0.023$; monotone part not met: Kaplan–Meier medians of 18,000, 42,500, 60,000, 75,000, 57,500 and 52,500 steps from $p = 0$ to $p = 1$, so that closure came later at $p = 0.1$, where the first replicator was open in 10 of 10 worlds, than at $p = 1$, where there was no open phase, which contradicts the monotone window reading of Model 5); the transmitter share of heritable cells rising with p (met, $\rho = 0.50$, one-sided permutation $P < 5 \times 10^{-5}$, no value as large in 20,000 permutations) with transmitter-majority worlds at most 3 of 10 at $p = 0$ and at least 7 at $p = 1$ (1 and 6: the second not met). The registered closure endpoint (a top-3 tape holding 0.5% of cells) misses closed transmitter populations, whose genomes all differ: two worlds counted as unclosed ($p = 0.03$ and 1) are closed at the population level (most heritable cells confined) and transmitter-dominated at 300,000 steps, which biases the timing test against closure at high p .

Ref. 23's robustness criterion. Every Stage K first replicator and final dominant ($L = 32$): 1,000 trials each of 1, 4 and 8 successive random single-byte mutations; a trial counts if the mutant, run for 512 instructions against the all-zero partner, leaves the partner equal to the mutant. Registered: block-copy finals more robust than pushers at every k (not met: medians 0.000–0.002 for both). The matched pair of the copy-offset switch was scored the same way post hoc (`robust_ref22.py`).

Mutational scan. For each Stage G first replicator and final dominant, every single-byte mutant (all 255 alternative values per position at $L = 16$ and 20; 32 seeded values per position at $L = 50$ and 64) was run as organism A against 32 shared random partners for 128 instructions and its offspring against 32 fresh partners, with the culture test's thresholds. The single-mutant variation score is $\sum_i \log_2(1 + 255 f_i)$ over positions, f_i being the fraction of transmissible values at position i , a summary of how many single mutants are heritable and reach first-generation copies, not a channel capacity (it ignores epistasis and later generations). Three predictions were written before the run (the successor more robust to mutation than the first replicator; the closed form with more transmissible sites; the matched pusher-with-jump of $L = 50$ with more sites than its pusher); all three failed and are reported as such.

Pair invasion. Soups of 20,000 cells at $L = 50$ under the Stage G dynamics, every cell set to one tape of the matched pair (the pusher tiling, or the same tiling with 20 $\epsilon 0$ inserted three times, the final dominant of 14 Stage G worlds, confined in all 256 traced encounters although its offspring reopen) and 1% of cells to the other at step 0; five seeds per direction, three seeds per resident with no invader; 20,000 steps. The pre-registered outcome, the invader's nearest-class share passing 50%, could not be met at this length because the quasispecies cloud splits between the two classes (a closed tape that has lost two jumps is nearer the pusher); the share of cells carrying the jump word, reported instead, was defined after the data were seen. In single-tape invasion assays the pusher tiling 01 c5 $\times 8$ seeded into 1% of the cells of a random soup at $L = 16$ (128 instructions, mutation 1/16, five soups) peaked at a median occupancy of 0.32 at step 250. The jump-word marker was checked against traced confinement before use (registered): in three re-runs of the closed-into-ancestor invasion, 512 random cells at each of eight steps were traced against 16 random partners. The marker share reached 0.73–0.75 while the share of cells confined in all 16 encounters stayed at 0.19–0.25, and only 26–33% of marker-carrying cells were confined; the registered agreement failed, so the $L = 50$ invasion is reported as the spread of the jump, not of closure (Extended Data Fig. 5a).

Serial allele retention. The fourteen genomes are four open pushers, the most common closed final dominant (confined in all 256 traced encounters) at each of $L = 16$, 20 and 32, the $L = 50$ final dominant, the lethal-tar final dominant with the most first-generation sites, the most common tape of three Stage L worlds at the first snapshot at which a genome-bearing block copier was most common, and the 20-byte 8080 closer with two random 12-byte payloads. Lineage r met the same partners for every mutant. The window is sought cyclically, which needs no alignment, so inserted jumps and shifts do not hide it. Every lineage is counted as lost (not alive), erased (alive, allele absent) or carrying the allele. Mutants whose window occurs in the unmutated tape would be excluded (none did). Design and predictions were registered before the run: the pusher's sites were predicted to persist (failed: its alleles reach first-generation copies only, and fade as fast when its lineages branch and persist, X1); the evolved closers to erase, the transient genomes to keep at least five sites and the constructed closer its payload (met). A post hoc variant drew partners from cells of the organism's own world (Stage G, seed 2001) instead of random bytes.

Invasions of a constructed closer. Stage M dynamics at $L = 32$. Against the pusher (registered as A1 in the revision, unrelated to Stage A): the 20-byte 8080 closer with payload 1 seeded at 1% into a world filled with the 8080 pusher, and the reverse, under the 8080 subset, five seeds each and three unseeded controls per resident, 20,000 steps. Against a block copier (A2): the same closer against the evolved block-copy tiling 04 5e ed b0 $\times 8$ under the full Z80, 50,000 steps. The registered outcome, the share of cells carrying the closer's exact 20-byte core, failed on definition: the core mutates within a few hundred steps while its descendants hold the soup. We report the share carrying the closer's loop body (post hoc) and the classes of 64 random cells per soup at the last step (Extended Data Fig. 5).

Address mapping and aligned lengths. The conventions, addresses taken modulo the pair length, the stack pointer at the end of the pair and every register empty at the start of an encounter, follow ref. 17, whose tapes were 16 bytes long, an aligned length. Of the 67 final dominants of Stage G that are closed (loop-bearing, and fetching no partner byte in any of 256 traced encounters), 25 carry a relative jump (JR or DJNZ) at a length where the 16-bit program counter does not fold consistently onto the ring: 20 at L = 50 and 5 at L = 20, one of which (seed 2003) is classed by its block repeat in Extended Data Table 1 (we did not check that every one of these jumps crosses address 0); the pair-invasion and mutational-scan results at L = 50 are results about the machine so defined. Stage K was pre-registered: first replicator a load–push word in ≥ 18 of 20, a closed successor in ≥ 15 of 20, kill below 10. Outcomes: the first replicator was a load–push word in 20 of 20 worlds, and the final dominant was closed in 12 of 20, all block copies.

The 8080 subset. Three deviations from a real 8080 are kept: the IX and IY prefixes DD and FD still select the indexed form of an 8080 instruction, where the 8080 executes them as CALL; the 8080’s other alternate opcodes are no-ops here (CB, JMP on the 8080, and ED, CALL, become two-byte no-ops with the byte that follows, and D9, RET, a one-byte no-op); and the core keeps Z80 flag behaviour, so P/V holds overflow after arithmetic where the 8080 sets parity, and conditional jumps, calls and returns on PE or PO can differ. The registered reading that the subset expresses no closer on a 32-byte ring was wrong and is withdrawn: a reviewer’s 20-byte closer of 8080 instructions (21 20 00 31 40 00 16 10 2b 46 2b 4e c5 15 c2 08 00 c3 11 00, a counted push loop closed by a conditional jump, then a jump to itself) copies itself and any 12 following bytes exactly into 256 of 256 random partners without entering them, in our executor under the subset (check_8080_closer.py).

Ten-million-step runs. Stage L samples every 5,000 steps, with full snapshots at the Stage G steps and at 2, 3, 5, 7.5 and 10 million. At every snapshot of the Stage L worlds and of the Stage G, K and I worlds, the most common exact tape was scanned (32 values per position, 32 partners) and traced (64 partners), giving its transmissible sites, single-mutant variation score, executed set and pointer confinement against step (a modal-tape analysis; population classes are in Fig. 4d,e). Predictions registered before the run: L1, in at least 5 of 10 worlds at each length the dominant at ten million steps has more transmissible sites than the closed dominant at 300,000 (L = 16) or one million (L = 20) steps; L2, every such recovery is in a block-copy lineage and its sites are copied-but-unexecuted positions; L3, no return-based or push-based dominant with more than one transmissible site at any snapshot. Outcomes: L1 not met (1 of 10 at L = 16, 0 of 10 at L = 20); L2 not met at L = 16 (the one recovery, from 0 to 4 sites, is a block-copy lineage with one of its four sites unexecuted) and not applicable at L = 20; L3 not met as written, at L = 16 (the open pusher of seed 6003 holds two transmissible sites at eight snapshots from 2,000 to 30,000 steps) and at L = 20 (the open pusher holds two to three sites and the jump-based closer two); the kill criterion fired (final dominant with at most one site in 9 of 10 and 8 of 10 worlds). Every world closed (the most common tape heritable and entering the partner in fewer than half of 64 traced encounters), first at 10,000–75,000 steps (L = 16) and 20,000–500,000 (L = 20); the heritable fraction of random cells at ten million steps is a median 0.89 and 0.95.

Controls. Pre-registered: H1, the open phase less heritable under mixing (killed: the median heritable fraction of 16 random cells over steps 2,000–20,000 is 0.125 in both, one-sided $P = 0.78$); H2, closure sooner under mixing, closure here meaning $t_{\text{close}70}$, the first sampled step at which the heritable fraction of 16 random cells reaches 0.7, not pointer confinement (killed: median 50,000 against 30,000 steps, reached by 300,000 steps in 7 of 10 worlds against 20 of 20, one-sided $P = 0.77$). Post hoc, after H2 was killed, the share of worlds reaching $t_{\text{close}70}$ by 300,000 steps was compared by a two-sided Fisher exact test (7 of 10 against 20 of 20, $P = 0.030$): closure by this heritability proxy was later and rarer under mixing, the direction opposite to the registered prediction. Also post hoc, the trend of population closure (most sampled heritable cells confined) with tape length at the aligned lengths L = 16, 32 and 64 (20 of 20 by 300,000 steps; 12 of 20 and 8 of 20 by one million) was tested by a two-sided Cochran–Armitage test on $\log_2 L$ ($z = -4.02$, $P = 5.7 \times 10^{-5}$); the shorter horizon at L = 16 makes the test conservative, and the per-byte mutation rate also falls with length. Descriptive: the first replicator met its own kind (Hamming distance at most 2) in 57% of its open-phase encounters on the lattice and 11% when mixed (kin_census.py). Under lethal tar the heritable fraction of random cells rose from 0 at 20,000 steps to 0.41 at 30,000 and 0.88 at 50,000 (medians over worlds). *Heritable material* is dated by the first sampled step at which at least 10% of the 16 random cells of the census are heritable (post hoc, timing_heritable.py): a median of 40,000 steps under lethal tar against 1,250 under benign tar (Stage G, L = 16), a 32-fold delay, against Kaplan–Meier medians of 38,000 and 450 steps for the first clone, t_{rep} .

BFF. 24 seeds each for the published and wrapping variants, 12 for each literal variant, 12 for the published rule with harmless brackets (run after review; pre-registered: every first replicator loop-bearing, met 7 of 7; a transition in at least 9 of 12, not met at 7; pointer-closed first replicators (entering the partner in at most 5% of encounters) in every world, not met in two, one intermediate (23% of encounters) and one open (every encounter)), 6 for each of $p = 0.01, 0.03, 0.1$ and 0.3 under wrap + literal push, and 6 for each of $r = 2$ and 3 under the literal push without a wrapping pointer; events every 64 epochs; high-order entropy with brotli quality 2 as in ref. 17. With the literal and halting brackets no closed class evolved:

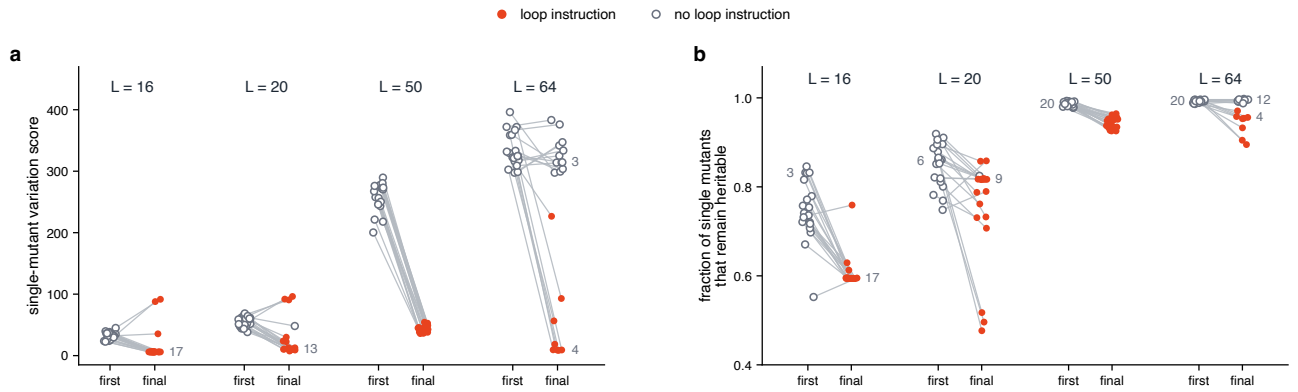
a closed top class (entering the partner in at most 5% of encounters and copying at least 95% of partners) appeared in none of the 12 soups with a wrapping pointer, where at least 6 had been registered (prediction (e3) of the theory note), and in none of the 12 without one (`bff_analysis.py`). Dial predictions, registered before the run as D1 and D2 of the revision (unrelated to Stage D): D1-1, the open population persists (share $\geq 10\%$ at 16,384 epochs) in at least 4 of 6 soups at $p = 0.01$ (met, 6 of 6) and in at most 1 of 6 at $p = 0.3$ (not met: the share falls below 1% at 832–896 epochs in 6 of 6 and recovers to 0.10–0.11 in 5 of 6, with 0.31–0.44 of random tapes heritable); D1-2, the epoch of collapse falls monotonically with p within a factor of three of $1/p$ (not met: no collapse at $0 < p \leq 0.1$, 896 at $p = 0.3$, 192 at $p = 1$); D1-3, no closure by control flow at any p (met); D2-1, inflow at most 1 bit at $r \geq 2$ (met: 0.00 bits in 12 of 12 soups at $r = 2$ and 3, against 8.00 in 12 of 12 at $r = 1$); D2-2, the first replicator still the loop-free, pointer-open tiling (met, 12 of 12); D2-3, collapse slower or absent at $r \geq 2$ (met: absent, with an all- P share at epoch 1,024 of 22–31% at $r = 2$ and of 3.6–7.3% in 5 of 6 soups and 0 in one at $r = 3$, against 0–0.24% at $r = 1$, and every random tape heritable at 16,384 epochs).

Searches. Two-byte census: all 65,536 words tiled to 16 bytes, culture test (32 partners) and partner test (256), plus the deterministic fixed-point test against the zero partner. BFF straight-line search: 16,447,860 programs (prefix ≤ 3 instructions + tiling unit of period 2–5 with a copy instruction), 2 random partners, wrapping pointer, alphabet prefilter then exact similarity, full culture test for survivors. Closed-replicator search: all 1,309,528 periodic tilings of $\{P, [,], \text{no-op}\}$ up to period 10 under the literal, wrapping, no-halt rules, three partners each; closed means that the pointer entered none of the three, and no closed tiling copied at least 90% into every partner. The search covers these four symbols only, not the full language.

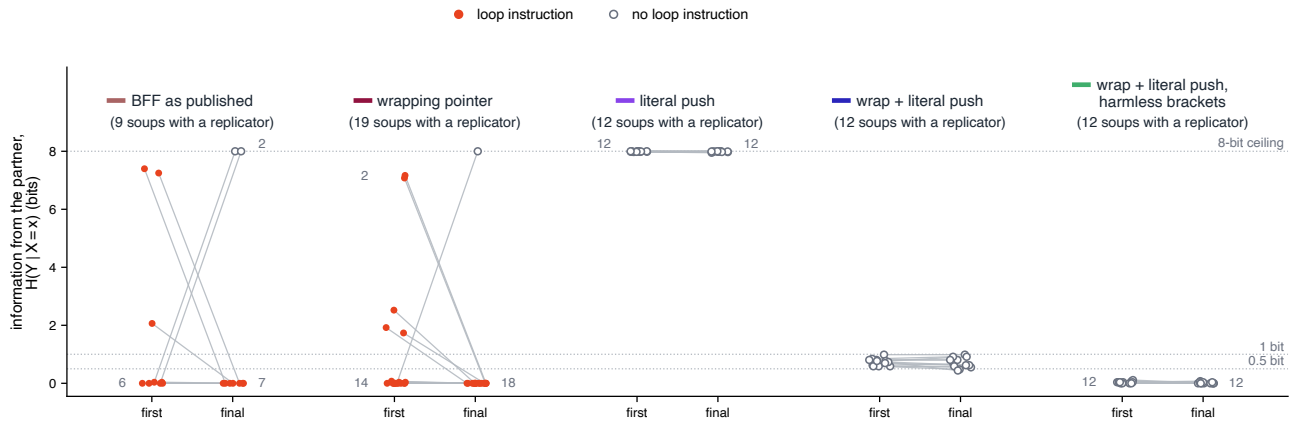
Pre-registration and stages. Predictions were fixed before the runs they concern in four documents, each with a dated change log: the plan of the instruction-set atlas (Stages A–M), the theory note (the BFF cells), the pre-registration of the two biology analyses (information inflow and assembly) and the revision pre-registration (the tests of the second and third revisions). Labels below are those of the documents; where a label could be read as a stage letter, it is given with the name of its test. Stages A (630 runs, $L = 16$: the unablated soup and six ablations $\times$ budgets $\{32, 128, 512\} \times$ mutation $\{1/4, 1/16, 1/64\}$, ten seeds, 300,000 steps), B (640: the unablated soup and three ablations, block-copy, stack-writes and no-copy, $\times$ budgets $\{128, 512\} \times$ eight lengths (4, 9, 25, 36, 49, 64, 81 and 100) $\times$ ten seeds), C (490: the unablated soup and 15 ablations at $L = 16$, seeds 101–110, 450 runs, the all-ld, rmw-only and no-copy arms run to one million steps; and the unablated soup and the stack-writes ablation at $L = 36$ and 100, 40 runs) and E (1,230: the size axis, 21 lengths from 3 to 100, under four control arms, budget and mutation sweeps at $L = 100$) are exploratory. Stages A and B stopped a run early once quasispecies occupancy reached 50% (112 of 630 and 144 of 640 runs stopped early); later stages have no early stop. Stages A and B were registered, with an occupancy event as their emergence outcome; after 19 of their runs had been read, emergence was dated instead by the assay-based t_{rep} , and both stages were then reclassified as exploratory. By the letter of the registered outcome, the hypothesis that the no-copy ablation leaves no replicator was refuted (zero floods crossed the occupancy event in 10 no-copy runs of Stage B, none of them a replicator); the acceptance region for the block-copy ablation could not be resolved at 500-step sampling; no mutation rate minimising the time to emergence could be established from the three rates run; and the clause that emergence slows with tape length was not supported. Stages C and E had predictions fixed before their runs but are used here only descriptively; their misses were C1b (the all-ld ablation replicating in more than 0.3 of runs: 2 of 10), the factor-of-two bound of C3a (2.4-fold), C3c (half met: removing PUSH and POP alone delayed emergence more than removing all 22 stack writers), C5 (met for first replicators only: the final dominants at $L = 100$ have periods of 9 and 10, not at most 8), the length floor of Stage E (not supported: the pusher works at $L = 8$ and from $L = 16$ on, but not at $L = 9$ –12) and the second clause of E4 (no whole-tape first replicator at any budget). Misses of the later stages: in Stage D, the second clause of D3 (the read-only arm within 0.1 of the unablated soup: 10 of 20 against 5 of 20 at 512 instructions), D4 by its letter (one first replicator has period 6, not 3 or 9) and D5 (not supported: 4 of 20 at 32 instructions against 6 of 20 at 128); in Stage F, F1 (the divisor law on padded rings: falsified, and the ring-arithmetic reading withdrawn), the second clause of F2 (tiled block-copy units did appear on prime rings), F3 (falsified: odd rings slow the pusher) and F4b (half met: emergence at $L = 10$ was rescued on a 28-byte ring, with heritability at the threshold); in Stage G, the heritable-fraction thresholds of G1(c) at $L = 50$, and G2, whose syntactic loop criterion omitted hardware loops (6 of 20 and 3 of 20 syntactic against 19 of 20 and 8 of 20 behavioural), so that by the registered criterion, which decides, $L = 20$ fell between prediction and alternative and $L = 64$ in the stated alternative (fewer than 5 of 20); in Stage K, the expectation of 15 of 20 closures at $L = 32$ (12, closure meaning a loop-bearing final dominant that copies at least 95% of 256 random partners without self-damage); Stage H's H1 and H2 were killed (Controls); Stage I's I2 (the open phase cut short, “open, then extinct” in at least 5 of 10 worlds) was not met (0 of 10) and I3 (closure rarer under lethal tar) was killed (10 of 10 worlds closed, most sampled heritable cells confined); the misses of Stage L, of the mutational scan, of the pair invasion and of the 8080 subset are given in their paragraphs above. In BFF (theory note), the strict form of prediction (a), every first replicator of the published rule closed (entering the partner in at most 5% of encounters and copying at least 95% of partners) and loop-bearing, held in 7 of 9 worlds (two intermediate), and the

chunk-copier phase it predicted before the transition was not seen; the mechanism first claimed for an open BFF replicator under a wrapping pointer was falsified by the harness test before any soup ran; (e3), a closed dominant in at least 6 of the wrap + literal worlds, was not met (0 of 12); (12), the collapse of the literal without a wrapping pointer in at least 9 of 12, held by its letter in only 5, although no world kept an open population; and with harmless brackets B5-2 and B5-3 were not met (BFF). In the biology analyses, whose inflow thresholds of 1 bit and 0.5 bit are those of Supplementary Fig. 2, the inflow prediction Q4a (the first replicators of published BFF below 0.5 bit in at least 90% of soups) fell between prediction and kill (6 of 9), Q4b (the literal-push first replicators above 1 bit in at least 90%) was killed (12 of 36: a copier that floods the ring enters the partner yet carries no inflow), the assembly analysis's P3 (assembly no better than an AUC of 0.65 as a detector of heredity) fell in its registered grey zone (0.72), and its P1 was met but could not have failed, every first replicator having the minimal assembly index for its length. In the second revision the registered tests were serial allele retention, the marker check, population sampling, the constructed-closer invasions, the harness conventions, the aligned invasions, first-closure timing, the census by period, ref. 23's criterion, the copy-offset switch with its mutational flows, closure under random registers and its replication, the parent $\times$ environment matrix, the matched pair at $L = 32$ under lethal tar, and the lethality dial (revision pre-registration, sections R2–R4, E, E-mech, C4a-rep, PE, E-L32L and DZ). Their misses: S1 of serial retention (its alleles reach first-generation copies only, and fade even when its lineages branch and persist, X1), V1 of the marker check (the jump spreads at $L = 50$ but confinement does not), Q1 of population sampling (in worlds whose modal tape is confined, at least 90% of heritable cells confined: 105 of 109 worlds), P1 of the census by period (a closed self-writer of period three exists), F1 of the robustness test (ref. 23's criterion scores pushers and block copiers alike), E1 and E2 of the copy-offset switch (no mutation-bias equilibrium), EM1 of its mutational flows (isolated flows do not explain the environment dependence), C4a of the random-register test (6 of 20 worlds closed, most heritable cells confined, against at most 5 predicted), PE2 (0.36 of the pusher's lineages survived among partners drawn from a closed world), the monotone part of DZ2 (closure is slowest at intermediate lethality) and the endpoint count of DZ3 (6 of 10 transmitter-majority worlds at $p = 1$, against at least 7); the motif endpoints of the constructed-closer invasion (A1), of the copy-offset switch and of the random-register invasion (C4b) failed on definition, since the motifs mutate while their descendants hold the soup, and functional classes of random cells are reported instead. In the third revision the registered tests were N1 (the unexecuted bytes of transmitters as a selected defence), N2 (in-situ demography, with the backup-core hypothesis), N3 (the line of descent of the first confined replicators), N4 (inherited zeros as the carriers of the lethal rule's protection), N5 (the stepwise path from the pusher to a closer) and N6 (the replication of the genealogies with the frozen pipeline; N6-1, N6-2, N6-3 and N6-5' met, N6-4 missed, Supplementary Table 3) and X1–X3 (branching transfer, inflow among the soup's own cells, open first along lines of descent; X1-2 failed and its registered reading was applied, the rest met) (revision pre-registration, sections N1–N6 and X1–X3). Their misses: N1 was rejected (N1-1 to N1-4 failed at their registered endpoints, and removing the halting bytes changes a host's survival by about 0.001 per encounter against a registered 0.02); N2-1 and N2-2 failed (a regenerator with a damaged first core never copies itself, 0 of 4,096); N3's validation criterion on per-record confinement flags (N3-V1, unrelated to the marker check's V1) was not met (0.922 and 0.799 against 0.95; a genotype-level check was added as a dated amendment); N3-1 was met but uninformative; N3-2 was registered on the earliest record flagged confined on each line, which proved a transient that mostly does not copy itself, and is scored instead on the founder, a definition fixed after inspection: its writer part failed, since no founding executor is an open copier (all 17 are non-copiers by the whole-tape test; by their executed bytes, 5 are confined tapes that copy the code they run and 12 copy neither), and its point-mutation part cannot fail under the record definitions and is not counted; N4 was killed (88–94% of the protection remains without the zeros); N5 was uninformative, and a stepwise path read from a lenient copier criterion was withdrawn after the culture test. Fixed after the N3 data were inspected, and frozen for N6: the copy-gain rule, the genotype-defined founder, functional parents, the chain start at the newest open copier older than the founder (corrected after the first analysis started chains too early), the separation of first founders from later conversions by existing closers, the executed-bytes copier test, and the reach and abundance nulls.

Supplementary Figures and Tables

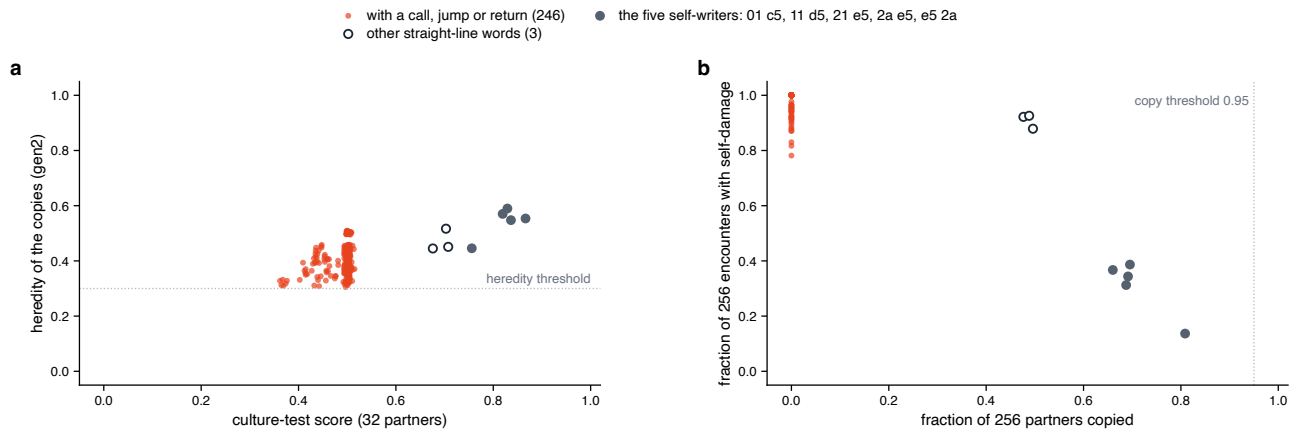


Supplementary Fig. 1 | Single-mutant scan of first and final dominants. Single-byte mutational scan of the first heritable replicator (first) and the final dominant (final) of every Stage G world, grouped by tape length (grey lines join the same world; filled, loop instruction; open, none; points spread sideways within each column; numbers, coincident points); transmission is scored in first-generation copies; serial transfers are in Fig. 4a–c. **a**, Single-mutant variation score, $\sum \log_2(1 + 255 \times \text{fraction of transmissible values})$ over positions, a summary score, without unit. **b**, Robustness: fraction of single mutants that remain heritable ($\text{gen2} \geq 0.3$). All 255 alternative bytes per position at $L = 16$ and 20, 32 seeded values at $L = 50$ and 64; 32 partners per generation.



Supplementary Fig. 2 | Information inflow in the second machine. $H(Y | X=x)$ over 256 random partners for the first replicator and the final dominant of every BFF soup with a replicator, by variant (grey lines join the same soup; filled, both brackets present, a loop; numbers, coincident points; the bar before each name, the variant's colour in Fig. 5), with the 8-bit ceiling of 256 partners and the 1-bit and 0.5-bit thresholds registered for the inflow analysis (Supplementary Methods, Pre-registration and stages). Without a literal channel (as published, and with a wrapping pointer) every first replicator carries a loop and most carry no inflow; the literal push without a wrapping pointer sits at the ceiling; under a wrapping pointer the literal tiling floods the ring and carries less than 1 bit, with harmless brackets at most 0.11, although its pointer enters the partner in every encounter. The loop-free finals at the 8-bit ceiling are the literal pushers without a wrapping pointer, open copiers (11 of 12 copy 89–94% of partners, the twelfth none) that lie outside Theorem 1, which concerns published BFF programs whose pointer stays in their own cells, and three one-symbol fills (two as published, one with a wrapping pointer) that had displaced the replicator and copy 0.00–0.02 of partners.

Supplementary Table 1 | Uncertainty across worlds and threshold sensitivity. Medians over worlds with 95% bootstrap intervals (10,000 resamples of worlds; Kaplan–Meier medians of t_{rep} with 2,000 resamples, censored at the horizon). Counts with Wilson 95% intervals: first replicator a load–push word, benign tar, $L = 16$ –64: 170 of 170 (0.98–1.00); worlds closed at the population level (most heritable cells among 256 random cells confined in 16 traced encounters), Stage G, $L = 16$: 20 of 20 (0.84–1.00); $L = 20$: 19 of 20 (0.76–0.99); $L = 32$ (Stage K): 12 of 20 (0.39–0.78); $L = 50$: 0 of 20 (0.00–0.16); $L = 64$: 8 of 20 (0.22–0.61); 8080 subset, $L = 32$: 0 of 20 (0.00–0.16). Threshold grid: the first replicator of every benign-tar world of 16–64 bytes (Stages G, H, K, L and M) re-dated from the recorded runs under gen2 thresholds of 0.2 to 0.6 (the paper uses 0.3) and informative-partner thresholds of 0.65 to 0.85 (the paper uses 0.75; it changes nothing); open means that the pointer fetches a partner byte in at least half of 16 traced encounters: gen2 threshold 0.2: load–push first in 170 of 170 worlds with a replicator (0.98–1.00); open first in 170; gen2 threshold 0.3: load–push first in 170 of 170 worlds with a replicator (0.98–1.00); open first in 170; gen2 threshold 0.4: load–push first in 170 of 170 worlds with a replicator (0.98–1.00); open first in 170; gen2 threshold 0.5: load–push first in 155 of 169 worlds with a



Supplementary Fig. 3 | Every two-byte word. All 65,536 two-byte words were tiled to 16 bytes and tested; shown are the 254 that pass the heredity threshold ($\text{gen2} \geq 0.3$). **a**, The heredity of the copies (gen2) against the culture-test score over 32 random partners. The 246 words that contain a call, jump or return (vermillion) pass the threshold without being heritable, by smearing return addresses into the partner: gen2 is a normalised gain in best-shift similarity to the original tape, so smeared return addresses raise the partners' similarity to the word without making copies, and heredity is read together with the copies of the partner test (b). The five self-writers (slate), the four load–push words 01 c5, 11 d5, 21 e5 and 2a e5 plus e5 2a, the push-first rotation of 2a e5, are exactly the words, besides the trivial 00 00, that write themselves into an all-zero partner (Proposition 3); the three other straight-line words (black), c5 01, d5 11 and e5 21, are the push-first rotations of the first three. **b**, The partner test over 256 random partners: the fraction of encounters in which the word loses a quarter or more of its own bytes against the fraction of partners that become a $\geq 75\%$ copy; dotted line, 0.95 of partners copied, the copy threshold of the partner-test criterion for a closed copier (at least 95% of partners copied with at most 5% self-damage; Methods), applied here without tracing the pointer. No word reaches it, and none of the 246 produces a single 75% copy. Coincident points are spread by a small jitter upwards and to the right, except that the words that copy no partner are spread upwards only and stay at 0, so that no point is drawn below its value or the threshold.

replicator (0.87–0.95); open first in 155; gen2 threshold 0.6: load–push first in 107 of 164 worlds with a replicator (0.58–0.72); open first in 99. At thresholds of 0.5 and above the pusher's own gen2 (0.46–0.62) falls near the threshold, so later closers are dated first in some worlds.

quantity	condition	median	95% bootstrap interval	worlds
first replicator: share of 256 random partners copied	Z80, L = 16	0.68	0.65–0.70	20
first replicator: share of encounters with self-damage	Z80, L = 16	0.33	0.31–0.35	20
first replicator: gen2	Z80, L = 16	0.58	0.58–0.58	20
first replicator: share of partners holding an exact copy	Z80, L = 16	0.58	0.55–0.59	20
first heritable clone t_{rep} , Kaplan–Meier median (steps)	Z80, L = 16	450	200–1,200	20
first: share of single mutants heritable	Z80, L = 16	0.74	0.72–0.78	20
final: share of single mutants heritable	Z80, L = 16	0.59	0.59–0.60	20
first replicator: share of 256 random partners copied	Z80, L = 20	0.66	0.64–0.67	20
first replicator: share of encounters with self-damage	Z80, L = 20	0.43	0.42–0.45	20
first replicator: gen2	Z80, L = 20	0.46	0.44–0.46	20
first replicator: share of partners holding an exact copy	Z80, L = 20	0.32	0.31–0.33	20
first heritable clone t_{rep} , Kaplan–Meier median (steps)	Z80, L = 20	700	550–1,000	20
first: share of single mutants heritable	Z80, L = 20	0.86	0.82–0.88	20
final: share of single mutants heritable	Z80, L = 20	0.82	0.75–0.82	20
first replicator: share of 256 random partners copied	Z80, L = 50	0.68	0.66–0.70	20
first replicator: share of encounters with self-damage	Z80, L = 50	0.11	0.10–0.12	20
first replicator: gen2	Z80, L = 50	0.58	0.58–0.63	20
first replicator: share of partners holding an exact copy	Z80, L = 50	0.05	0.04–0.05	20
first heritable clone t_{rep} , Kaplan–Meier median (steps)	Z80, L = 50	600	500–650	20
first: share of single mutants heritable	Z80, L = 50	0.99	0.98–0.99	20
final: share of single mutants heritable	Z80, L = 50	0.95	0.94–0.95	20
first replicator: share of 256 random partners copied	Z80, L = 64	0.73	0.72–0.75	20
first replicator: share of encounters with self-damage	Z80, L = 64	0.06	0.06–0.07	20
first replicator: gen2	Z80, L = 64	0.62	0.62–0.62	20
first replicator: share of partners holding an exact copy	Z80, L = 64	0.00	0.00–0.00	20
first heritable clone t_{rep} , Kaplan–Meier median (steps)	Z80, L = 64	750	600–1,000	20
first: share of single mutants heritable	Z80, L = 64	0.99	0.99–0.99	20
final: share of single mutants heritable	Z80, L = 64	0.99	0.96–0.99	20
first heritable clone t_{rep} , Kaplan–Meier median (steps)	Z80, L = 16, lethal tar	38,000	21,000–132,500	10
last snapshot: share of heritable random cells confined	Stage G, L = 16	1.00	1.00–1.00	20
last snapshot: share of random cells heritable	Stage G, L = 16	0.86	0.85–0.88	20

last snapshot: share of heritable random cells confined	Stage G, L = 20	1.00	0.99–1.00	20
last snapshot: share of random cells heritable	Stage G, L = 20	0.94	0.92–0.95	20
last snapshot: share of heritable random cells confined	Stage G, L = 50	0.29	0.27–0.30	20
last snapshot: share of random cells heritable	Stage G, L = 50	0.55	0.54–0.57	20
last snapshot: share of heritable random cells confined	Stage G, L = 64	0.18	0.09–1.00	20
last snapshot: share of random cells heritable	Stage G, L = 64	0.28	0.23–0.97	20
last snapshot: share of heritable random cells confined	Stage I, L = 16	1.00	1.00–1.00	10
last snapshot: share of random cells heritable	Stage I, L = 16	0.93	0.92–0.95	10
last snapshot: share of heritable random cells confined	Stage K, L = 32	1.00	0.00–1.00	20
last snapshot: share of random cells heritable	Stage K, L = 32	0.94	0.35–0.96	20
last snapshot: share of heritable random cells confined	Stage L, L = 16	1.00	1.00–1.00	10
last snapshot: share of random cells heritable	Stage L, L = 16	0.86	0.85–0.88	10
last snapshot: share of heritable random cells confined	Stage L, L = 20	1.00	1.00–1.00	10
last snapshot: share of random cells heritable	Stage L, L = 20	0.95	0.94–0.96	10
last snapshot: share of heritable random cells confined	Stage M, L = 16	1.00	1.00–1.00	20
last snapshot: share of random cells heritable	Stage M, L = 16	0.86	0.85–0.88	20
last snapshot: share of heritable random cells confined	Stage M, L = 32	0.00	0.00–0.00	20
last snapshot: share of random cells heritable	Stage M, L = 32	0.49	0.36–0.54	20

Supplementary Table 2 | Branching transfer (registered, X1). Unmutated survival is given for reference only: it varies between runs, and with soup partners it is inflated by pool cells that already copy the parent (13% at L = 16, 24% at L = 64). Populations founded by every single-byte mutant (32 seeded values per position, four populations each) and 64 by the unmutated genome, each member run against two fresh partners per generation, copies ($\geq 75\%$ at the best cyclic shift) carried forward, populations capped at 32; partners uniform random bytes or cells of the open-phase soups (steps 5,000 and 10,000 of the 20 Stage G worlds of the same length). Return closer and block-copy tiling, regenerators; transmitter, world 6003. Sites as in the serial-retention test; mutants carrying, mutants with a population carrying their allele.

genotype	partners	unmutated alive, g = 1 / 4 / 8	sites, g = 1 / 2 / 4 / 8	mutants carrying, g = 8
pusher, L = 16	uniform random	0.94 / 0.81 / 0.75	1 / 0 / 0 / 0	0 of 512
pusher, L = 16	open-phase soup	0.98 / 0.88 / 0.75	0 / 0 / 0 / 0	0 of 512
pusher, L = 64	uniform random	0.94 / 0.42 / 0.12	29 / 21 / 0 / 0	5 of 2048
pusher, L = 64	open-phase soup	1.00 / 0.70 / 0.44	32 / 14 / 0 / 0	18 of 2048
return closer, L = 16	uniform random	1.00 / 1.00 / 1.00	0 / 0 / 0 / 0	0 of 512
block-copy tiling, L = 32	uniform random	1.00 / 1.00 / 1.00	0 / 0 / 0 / 0	3 of 1024
transmitter, L = 20	uniform random	1.00 / 1.00 / 1.00	9 / 9 / 9 / 9	301 of 640

Supplementary Table 3 | Registered replication of the genealogy analysis (N6, and X3 on the same worlds). Twenty new benign-tar worlds at L = 16 (seeds 32001–32020), recorded and analysed with the pipeline frozen before they were run; n = number of closed worlds with a first founder. Kill rule (point mutations complete at least a quarter of first founders): not triggered (0 of 20). Registered decision: with N6-1 and N6-2 met, the founder result is stated as confirmed; N6-4 is reported as failed.

prediction	registered criterion	observed	verdict
N6-1	rewrite or copy of the partner ≥ 0.8 n and point mutation ≤ 0.1 n	18 and 0 of 20	met
N6-2	only non-copiers between chain start and founder ≥ 0.75 n	20 of 20	met
N6-3	median share of executed words $\geq 5 \times 0.00024$	0.00190 (7.9 $\times$)	met
N6-4	executor whole-tape non-copier ≥ 0.8 n; executed-bytes non-copier ≥ 0.6 n (with the abundance null)	20 and 11 of 20; abundance P = 0.062	not met
N6-5'	expected founders from recorded point mutations < 0.05	0.0064	met
X3-1	first copier on every sampled line open in ≥ 0.9 n worlds	20 of 20	met

Supplementary Table 4 | Instruction-set ablations. Each ablation replaces the listed instruction families or mnemonics by no-ops through a prefix-aware fetch hook; the count is of opcodes on the base, CB and ED pages (indexed DD/FD forms follow their base opcode); a leading minus removes a family from the set. i8080 is the 8080 subset of Stage M.

ablation	opcodes made no-ops	patterns (instruction families or mnemonics)
none	0	(none)
block-copy	4	block-copy
stack-writes	46	stack, ex, call-ret, rst
stack-write-only	22	PUSH, EX (SP),HL, CALL, RST N
stack-read-only	24	POP, RET, EX DE,HL, EXX, EX AF,AF'
push-only	8	stack
push	4	PUSH
ex-sp-only	1	EX (SP),HL
call-rst	34	call-ret, rst
call-rst-write	17	CALL, RST N
ld-mem	31	ld8-mem, ld16-mem
ld-imm	11	ld8-imm, ld16-imm
ld-reg	54	ld8, ld-special
all-ld	96	ld8, ld8-imm, ld8-mem, ld16-imm, ld16-mem, ld-special
ed-loads	8	8 ED-page opcodes
cb-page	256	rotate, rotate-mem, bit, bit-set, bit-set-mem, -RLD, -RRD
no-copy	112	ld8, ld8-imm, ld8-mem, ld16-imm, ld16-mem, ld-special, stack, ex, block-copy
rmw-only	46	writes-mem, -incdec-mem, -rotate-mem, -bit-set-mem
i8080	342	bit, bit-set, bit-set-mem, rotate, rotate-mem, EX AF,AF', DJNZ, JR, EXX, 78 ED-page opcodes

Supplementary Table 5 | Every Z80 soup run of the numbered stages. From the run summaries: arms (their number, or the arm itself: none, the unablated soup; mixed, well-mixed pairing; lethal, lethal tar; i8080, the 8080-like subset; the ablations are listed in Supplementary Table 4), tape lengths L, budgets (instructions per encounter), background mutations per step, seeds, horizons (k, thousand steps; M, million) and the runs stopped early by the occupancy rule of Stages A and B; 3,660 runs in all. The revision-2 tests (harness conventions, invasions, the copy-offset switch, the lethality dial) are listed in Supplementary Methods.

stage	runs	arms	L	budgets	mutations / step	seeds	horizon	early stops
A	630	7	16	32, 128, 512	128, 512, 2,048	1–10	300k	112
B	640	4	4, 9, 25, 36, 49, 64, 81, 100	128, 512	512	1–10	300k	144
C	490	16	16, 36, 100	32, 128, 512	128, 512, 2,048	101–110	300k, 1M	0
D	280	6	9	32, 128, 512	128, 512	1001–1020	300k	0
E	1,230	4	3–100 (21)	32–2,048 (13)	32–4,096 (20)	1–1020 (31)	300k	0
F	210	none, stack-write-only	8, 10, 12, 16, 36	128	512	101–110	300k	0
G	80	none	16, 20, 50, 64	128	512	2001–2020	300k, 1M	0
H	10	mixed	16	128	512	3001–3010	300k	0
I	10	lethal	16	128	512	4001–4010	300k	0
K	20	none	32	128	512	5001–5020	1M	0
L	20	none	16, 20	128	512	6001–6010	10M	0
M	40	i8080	16, 32	128	512	7001–7020	300k, 1M	0

Supplementary Information: theorems and proofs

What kind of statement each item is: a **Theorem** has a pen-and-paper proof, and where a part of one is applied to a machine through a computed fact, as the Z80 case of Theorem 6 (ii) is through Proposition 3, this is marked; a **Proposition** is an exact statement that is either elementary (Propositions 4 and 8) or decided by finite computation, with the computation described (Proposition 3); a **Lemma** is a short general statement used to read a measurement (Lemma 7); a **Model** is a result about a stylised model with stated assumptions, not about the simulator. Empirical premises are marked as such, and every number quoted is from the generated tables behind the main text.

Notation. An encounter executes a program held in a shared memory. The organism A occupies addresses 0 to $L - 1$ and the partner B the next L addresses; in the Z80 soups the $2L$ addresses form a ring, in BFF the 128 addresses do not wrap. Execution starts at address 0 with a fixed initial machine state (registers and flags zero; the Z80 stack pointer at the 16-bit address that aliases the last byte of B) and lasts at most T instruction executions ($T = 128$ in the Z80 soups, 2^{13} in BFF). A *control transfer* is a jump, call, return, repeat-in-place or halt; a *straight-line program* executes none, so that under default flow its pointer advances past each executed instruction. An address is *executed* when the first byte of an instruction is fetched from it; an instruction that repeats in place (LDIR) fetches its own address again at each repetition, and each repetition counts as one execution. An address is *repeated* if it is executed twice in one encounter, that is, if control returns into code that has already been executed. A repeated address is not a repeated machine state: registers, flags, data heads and memory may differ between the two visits. The *no-op context* is the partner all of whose bytes are no-ops (zero in the Z80, where byte 0 is NOP; zero in BFF, where every byte outside the ten instruction characters is a no-op). A tape x is a *self-writer* if its encounter with the no-op context ends with B holding a copy of x (up to a cyclic shift, as in the culture test), and a *full self-replicator in context e* if the encounter (x, e) ends with B holding a copy of x . It is *closed* (confined) if its pointer fetches no byte of B as part of an instruction, opcode or operand, in any context, and *open* if it does in some context. Confinement is a property of execution, measured by tracing the pointer; it is not closure in the sense of organisational theories of life, where closure is the mutual maintenance of a set of constraints⁴⁵, and a confined organism still writes into its partner and relies on the harness for its initial registers and its turn to run. A value is *prescribed* if it is determined by x alone, the same for every partner.

Theorem 1 (the cost of a copy in BFF). *Setting.* BFF as published: a 128-byte pair; the instruction pointer p and the two data heads $h0$ and $h1$ start at address 0; p advances by one after every instruction unless a bracket jumps; $< >$ and $\{ \}$ move $h0$ or $h1$ by one cell; $+ -$ change the byte under $h0$ by one; $.$ copies the byte under $h0$ into the cell under $h1$ and $,$ the byte under $h1$ into the cell under $h0$; every other byte is a no-op; execution ends when p leaves the pair, on an unmatched bracket, or after 2^{13} steps. *Statement.* Let A be the organism's 64-byte tape, at addresses 0–63, fix a cyclic shift s , and let B be a partner whose byte differs in every cell from the copy it is to receive (A rotated by s). Any execution of the pair that leaves B holding that copy runs at least 128 instructions: at least 64 that change a cell and at least 64 head moves. If p stays among A's 64 addresses throughout, some address is therefore executed at least twice, and the execution takes a backward bracket jump: control returns into code it has already executed. *Proof.* Every cell of B must change, and an instruction changes at most one cell ($+$, $-$ and $,$ the cell under $h0$, $.$ the cell under $h1$), so at least 64 changing instructions run, and each cell of B is under a head when it changes. Both heads start at address 0, which lies in A, and each move changes one head's position by one, so after v moves the heads have occupied at most v positions besides 0; occupying all 64 cells of B needs $v \geq 64$. Moves change no cell, so the two sets of instructions are disjoint, and at least 128 instructions run. An execution of at least 128 instructions at 64 addresses executes some address twice. Without a bracket jump p only increases, and $[$ jumps forward, so p can return to an executed address only through a taken backward jump at $]$. ■ *Corollary 1.1.* A BFF program that copies its tape, at some shift, into a partner differing in every cell from that copy while its pointer stays in its own cells takes a backward bracket jump in that encounter, that is, runs a loop. The loop is a repeated address, not necessarily a repeated state: between two visits to an address the heads and the tape may differ. *Scope.* The count is per encounter: it needs neither the same instruction sequence in every encounter nor a no-op partner, and it says nothing about a program whose pointer enters the partner, which then fetches the partner's bytes. An earlier version stated two further parts, a bound uniform over partners for programs whose pointer stays out of B and the claim that no straight-line execution writes a copy of a tape with more than 32 non-zero bytes into a zero partner. Both proofs were wrong, the first because brackets test the byte under $h0$ and $,$ copies B's bytes into A, so that a confined program's control flow can depend on B, the second because bytes written into B ahead of the pointer are executed when the pointer arrives; both parts are withdrawn. That every observed BFF first replicator carries a loop (35 of 35 under the published rule, the wrapping pointer and harmless brackets) is an observation explained, not forced, by the count; whether the loop also keeps the pointer out of B is a separate, measured fact (22 of the 28 first replicators under the published rule and the wrapping pointer never entered the partner in 256 encounters).

Theorem 2 (a confined execution repeats an address). *Statement.* Consider one encounter of an organism of L bytes

whose pointer executes only addresses of A. Some address of A is executed at least twice, so that control returns into code already executed, provided either (i) the encounter completes a copy that costs more than L instruction executions, or (ii) execution cannot end inside A before a budget of $T > L$ executions is spent. *Convention.* In the Z80 a HALT holds the processor on its own address for the rest of the budget; the executor stops there, which leaves memory as holding would, and a held HALT counts as executing its address again, so that execution does not end at a HALT in the sense of (ii). A confined program that halts inside A therefore meets the conclusion by this convention alone, without returning control into code already executed. A closed full self-replicator is confined in every encounter, so the conclusion holds in each of its encounters that meets (i) or (ii). *Proof.* Every executed address lies among the L addresses of A. If none is executed twice, the encounter comprises at most L executions. Under (i) these are too few to complete the copy; under (ii) the budget requires more than L. In either case two executions share an address. ■ *What the theorem does not say.* It gives a repeated address, not a repeated state. At the two visits the registers, flags, heads and memory may differ, and in a copy loop they do, since each pass moves a different byte; nothing here says that the machine's state is periodic. A periodic orbit would follow from the pigeonhole principle only over the whole state, every register and flag and all 2L bytes of memory, which takes far more values than the budget has steps. *Hypothesis (H1), default flow.* In the absence of an executed control transfer the pointer advances past each instruction, so that a straight-line program runs from the organism's region into the partner's. The theorem does not use (H1); Corollary 2.1 and Theorem 6 do. Under (H1), a confined execution that repeats an address has executed a control transfer, since without one its address rises from 0 until it leaves A. *The machines.* BFF satisfies (i) in every encounter with a partner that differs in every cell from the copy it is to receive: by Theorem 1 the copy costs at least 128 executions, more than the organism's 64 addresses. The Z80 soups with benign tar satisfy (ii): the budget is 128 executions for tapes of at most 100 bytes, and no instruction ends execution (a HALT is held, under the convention above). *Where the provisos fail.* Under lethal tar a zero byte fetched as an opcode ends execution, possibly inside the organism: in Stage I, in the lethality dial at every $p > 0$ (with probability p) and in every other run under the lethal rule (the lethal arms of the copy-offset switch and of the parent $\times$ environment matrix, the encounter censuses, and the serial transfers and scans of lethal-tar tapes). In the Stage E arms with budgets of 32 and 64 instructions at $L = 100$ the budget ends before a pass over the tape. In all of these (ii) does not hold, (i) is not established for the Z80, and the theorem does not apply. *The write-ratio reading.* A variant of (i) uses the write ratio. Suppose the partner differs from the copy in every cell, so that a copy writes at least L bytes, and the organism writes at most $\beta < 1$ bytes per byte of code it fetches. An execution confined to A in which no byte of A is fetched twice as part of an instruction fetches at most L bytes of code and so writes at most $\beta L < L$ bytes, too few; hence some byte of A is fetched twice. This counts fetched bytes rather than instruction addresses, because an instruction can begin inside an earlier one. The ratio describes the replicators observed: the first replicator writes two bytes per four bytes of code (the executor trace shows eight bytes written when the pointer leaves its sixteen cells), BFF's copy instructions move one byte per instruction and need a head move per cell, and the literal push of modified BFF writes two bytes per three. It is not a machine-wide property of the Z80, in which a chain of pushes writes two bytes per byte of code; for the Z80 the theorem rests on (ii). *Corollary 2.1.* Under (H1) every straight-line program is open: with $T > L$ its pointer reaches the partner's region. A closed self-writer therefore contains an executed control transfer, which may be a repeating instruction such as LDIR, besides the instructions that write its content. For the Z80 the census of Proposition 3 makes this exact at small periods: the self-writers of period two are five straight-line load-push words, all open, and the smallest closed self-writer has period three (b0 5e ed tiled, whose tiling contains the block copy b0 5e ed b0). The corollary does not make closed self-writers rare: at period four they outnumber the open ones.

Proposition 3 (the short self-writers of the Z80; computer-assisted). Tile a word of period k to 16 bytes and execute it for 128 instructions as A against the all-zero partner; it is a self-writer if B ends holding the tiling at some cyclic shift. By exhaustive execution of every word: at period two, of 65,536 words exactly six leave B holding a tiling of the word, the trivial 00 00, which B already holds, and the five load-push words 01 c5, 11 d5, 21 e5 (LD rr, nn ; PUSH rr) and 2a e5, e5 2a (LD HL, (nn) ; PUSH HL); none contains a control transfer, so each runs on into the partner, and every two-byte self-writer of this machine is open. At period three, of 16,777,216 words exactly one is a self-writer, b0 5e ed, which is closed: tiled, it contains the block copy b0 5e ed b0 (OR B ; LD E, (HL) ; LDIR), whose copy offset is the tape length. At period four, of 4,294,967,296 words, 641 are self-writers of primitive period four: 558 closed, every one of them a block copy (455 with LDIR, 103 with LDDR), and 83 open. Closure is tested by tracing the self-writer against the zero partner and 16 random partners (closed if the pointer fetches no partner byte in any). The stochastic census of period two agrees: the same five words copy into 0.66–0.81 of 256 random partners, none into 0.95 or more, and the 246 words that pass the heredity threshold by smearing return addresses produce no 75% copy. *What it shows.* The smallest self-writers are open, and a closed self-writer needs only one period more; but closed self-writers are the majority at periods three and four, the largest the census covers, because a single repeating block move closes a copy. Counting self-writers does not, therefore, explain why the open word comes first.

Proposition 4 (a count on the initial soup). In a soup of N tapes of L uniformly random bytes, the expected number

of occurrences of a given k -byte word, at any of the $L - k + 1$ positions of a tape, is $N(L - k + 1) / 256^k$: for $N = 20,000$ and $L = 16$, 4.6 for a given two-byte word, 0.017 for a given three-byte word and 6.1×10^{-5} for a given four-byte word. *What the count does and does not show.* The load–push word works only in phase, as the four bytes 01 c5 01 c5 (LD BC, nn reading c5 01, then PUSH BC), the same length as the shortest evolved closers (1e a4 ed b0, 44 5e ed b0), so a given open and a given closed working unit have the same expected count at step 0, 6.1×10^{-5} . Proposition 3 counts more closed than open four-byte self-writers. The count therefore gives the open form no head start; the head start is measured (Kaplan–Meier medians over the 20 Stage G worlds at $L = 16$ of 450 steps to the first replicator and 21,000 to the first closed replicator, a top-3 tape heritable and confined in 16 of 16 traced encounters; 38,000 to the first closed replicator in the 10 Stage I worlds under lethal tar, where the first replicator is closed). An earlier version compared a two-byte word with a four-byte closer and concluded that the open form is exponentially likelier; that comparison was wrong and is withdrawn. Why the open word is found first, perhaps because partial load–push words write useful bytes from any position along an executed path while a closer works only from the start of execution, is not modelled.

Model 5 (the closure window). *Assumptions.* An open replicator population of $n(t)$ tapes; closed functional variants arise at a rate q per open tape and step, by point mutation or written by programs that carry parts of the open population (for point mutation alone, which must place the right m bytes at the right phase, q is of order sites $\times \mu \times 256^{-m}$; writing from parts already present can make q much larger); a closed variant, once present, grows faster than the open form and fixes with probability of order one. *Statement.* The expected number of closed variants produced while the open population lasts is $q \int n dt$, and closure occurs with probability $\approx 1 - \exp(-q \int n dt)$; the integral is the window. If the open form’s own copies are viable in the soup it creates, n stays near the carrying capacity, the window grows without bound and closure is a matter of waiting; if the soup becomes lethal to the open form before $q \int n dt$ reaches order one, the open form dies without closing. The rate q need not be constant: where the parts are spread by open copiers, q may itself grow with n , and $q \int n dt$ is then to be read as $\int q n dt$, which rises faster than linearly with the open population. *What the experiments found.* How closers arise: in exact genealogies at $L = 16$, settled on 17 closed worlds and replicated as registered on 20 more (Methods, Genealogies), the founder (the first confined copier after the newest open copier: a $\geq 75\%$ copy in at least half of 64 random encounters, the pointer entering the partner in fewer than half; every one entered in none) was out of reach of a single point mutation of the tape before it and was written in one encounter, by a program that copies neither its tape nor the code it runs in 23 of 37 worlds and by a confined tape that copies only the code it runs in 14, and in 34 of 37 worlds its parts were one-byte variants of the pusher’s own word. Z80, benign zeros: closure of the population (most sampled heritable cells confined) in 20 of 20 worlds by 300,000 steps at $L = 16$, in 19 of 20 by one million at $L = 20$ and in 8 of 20 at $L = 64$, where the open form is least damaged and the motif must land in the longest tape (smallest q); at $L = 50$ a confined final dominant (no partner byte fetched in 256 traced encounters) arises in 20 of 20 worlds but its offspring reopens, so the model’s assumption that a closed variant breeds true fails there. BFF with the literal push and lethal brackets: under a wrapping pointer the open form held 0.58–0.63 of $N = 2^{17}$ tapes at epoch 64 and $2\text{--}3 \times 10^{-4} N$ at epoch 256, an integrated window $\int n dt$ of $8.2\text{--}8.7 \times 10^6$ tape-epochs over 16,384 epochs, $5.0\text{--}5.5 \times 10^6$ of them before epoch 256; without a wrapping pointer the window was $2.5\text{--}4.1 \times 10^5$; a closed top class (entering the partner in at most 5% of encounters and copying at least 95% of partners) in 0 of 12 soups in each. BFF with the literal push and benign brackets: the open form persists for 16,384 epochs, holding a median 0.98 of the soup at the end, yet there is no closure by control flow in 12 of 12. The model does not explain this by $q = 0$. The search that found no closed heritable tiling up to period 10 covered only the four symbols {P, [,], no-op} (Supplementary Methods), and the full language makes closed, loop-bearing replicators under related rules: they are most of the first replicators of BFF without the literal, 23 of 28 under halting brackets and 5 of 7 with harmless ones closed (entering the partner in at most 5% of encounters). In the model’s terms $q \int n dt$ stayed below order one over a long window, so q was small there, for reasons the model does not give. Z80 with lethal zeros (Stage I): the open form never establishes, so the window is empty from the start, and the first replicator is closed (no partner byte fetched in 256 traced encounters) and late (Kaplan–Meier median t_{rep} of 38,000 steps in the 10 Stage I worlds against 450 in the 20 Stage G worlds at $L = 16$; 10 of 10 worlds). The registered lethality dial (zeros halt with probability p ; ten worlds per p , the same seeds at every p) tested the model’s monotone prediction and refuted it: the first closed replicator (a top-3 tape heritable and confined in 16 of 16 traced encounters) comes at Kaplan–Meier medians of 18,000, 42,500, 60,000, 75,000, 57,500 and 52,500 steps at $p = 0, 0.01, 0.03, 0.1, 0.3$ and 1. Closure came later at $p = 0.1$, where the first replicator was open in 10 of 10 worlds, than at $p = 1$, where there was no open phase: the open phase was not needed for closure, and a damaged one delayed it. The dial therefore contradicts the window reading between its ends; the reading describes at most the contrast between benign and lethal tar, and that contrast changes lethality as well as the open phase. At the aligned length $L = 32$, 12 of 20 worlds closed by one million steps, and given ten million steps 20 of 20 closed at $L = 16$ and 20 (population closure, as above). In a world filled with the open form from step 0 at $L = 50$, n at carrying capacity throughout, the jump arose by mutation and spread to half the cells within 5,000–13,250 steps in 3 of 3, though its carriers are mostly not confined. The model’s qualitative prediction, that an open phase feeds closure, is consistent with the first closed replicator appearing at a Kaplan–Meier median of 22,500 steps under benign

tar (the 30 worlds of Stages G and L at $L = 16$) against 38,000 under lethal tar (the 10 Stage I worlds), where there is no open phase (log-rank $P = 0.007$), but that comparison changes lethality as well, the dial contradicts the reading between the extremes, and the open phase's part in closure is not settled; in exact genealogies the founders' lines run back to open copiers, as expected under any account (37 of 37 worlds; Methods, Genealogies). When an unmatched bracket halts with probability p , the open form held 98% of the soup at $0 < p \leq 0.1$ (median at 16,384 epochs) and fell below 1% at $p = 0.3$ (by epochs 832–896, in 6 of 6 soups) and at $p = 1$ (by epoch 192, median, in 12 of 12); at $p = 0.3$ it recovered to about a tenth of the soup by 16,384 epochs (at least 0.10 in 5 of 6), at $p = 1$ it did not (median 0.002). The registered expectation that the collapse time would scale as $1/p$ was not met, the lethality acting as a threshold between 0.1 and 0.3. When the literal writes two or three copies of its word, so that a copy completes in one pass, no partial copies are made, no lethal tar forms and the open form persists for 16,384 epochs in 12 of 12 soups, with zero inflow from birth. The model does not predict which regime a machine is in; that depends on how the open replicator's own by-products act on it, which is what the experiments measure.

Theorem 6 (a literal write and the loop-free self-writer). *Hypotheses.* (H1) Default flow, as in Theorem 2, with execution ending when the pointer leaves the pair (as in BFF) or continuing around a ring (as in the Z80 soups). (H2) *Copy cost:* in the no-op context, an execution that runs each address at most once cannot write a copy of the tape. (H2) is a hypothesis to be checked per machine. We do not prove it for BFF, where writes can land ahead of the pointer and be executed, and we have no direct test of it: the nearest evidence, a search of 16,447,860 straight-line BFF programs, was run under a wrapping pointer against two random partners, not in one pass against the no-op context, and found none that copies a tape of period two or more, only one-symbol fills (Supplementary Methods). (H3) *Self-consistent literal write:* an instruction that writes its own operand bytes verbatim at a write pointer that starts in, or advances into, the partner's region, and whose operand can be its own encoding in the phase executed (LD BC, nn ; PUSH BC with $nn = c5\ 01$; P a b with $a = b = P$). *Statement.* (i) Under either proviso of Theorem 2, every closed self-replicator executes some address at least twice in each encounter that meets the proviso. (ii) Under (H1) and (H3), the tiling of the literal instruction is a straight-line program, hence open, that writes its own content at the advancing write pointer; it is a self-writer wherever the encounter lets the written bytes cover the partner's region, which holds in the Z80 at $L = 16$ by the computer-assisted Proposition 3 (eight pushes over two passes of the 2L-byte ring); in BFF with P the tiling is observed to arise first (below), which the theorem does not decide. (iii) Under (H1) and (H2) without (H3), in a machine whose execution ends when the pointer leaves the pair, no straight-line program is a self-writer, and every self-writer executes some address at least twice in its encounter with the no-op context: born loop-bearing (whether its pointer also stays out of the partner is a separate, measured property). (iv) Under uniform random initialisation over an alphabet of size σ , the expected number of occurrences of a fixed k -symbol pattern among N regions of length L is $N(L - k + 1)\sigma^{-k}$, so the expected count of a minimal open pattern shorter than every closed one exceeds that of any closed pattern by a factor σ raised to the length difference. This is a statement about expected counts, not about the probability that a pattern is present, which saturates once patterns are common (a given two-byte word is expected 4.6 times in the initial soup at $L = 16$; Proposition 4); it concerns the presence of the pattern, not of a replicating tiling; and nucleation is not modelled. In the Z80 at $L = 16$ the minimal working units, open and closed, are both four bytes long, so (iv) gives the open form no advantage there (Proposition 4); this does not hold at every length (at $L = 9$ the replicator is the three-byte block copier `ld ed b0, DEC E; LDIR, tiled`). *Proof.* (i) is Theorem 2. (ii) The tiling executes no control transfer, so by (H1) its pointer runs into the partner's region: it is open; its writes are its operands, by (H3) the tiling itself; completion of the copy is the per-machine fact cited. (iii) In the no-op context a straight-line program runs each address at most once before the pointer leaves the pair, and by (H2) such an execution cannot complete a copy; so a self-writer executes some address twice. (iv) is linearity of expectation. ■ *What it shows.* In a machine that meets (H1) and (H2) and whose execution ends when the pointer leaves the pair, a self-consistent literal write is what allows a straight-line, and hence open, self-writer: without one, every self-writer repeats an address in its encounter with the no-op context. The theorem does not make a literal necessary for an open first replicator, and the data show that it is not: in BFF as published with harmless brackets, which has no literal, one of the 7 first replicators, loop-bearing, entered the partner in every encounter (and one more in 23%), and in the Z80 with the stack pointer drawn at random before each encounter the first replicator was open (its pointer in the partner in at least half of 16 traced encounters) in 18 of 20 worlds but a load-push word in only 5; in 13 it was `e1 e3 (POP HL; EX (SP),HL)`, a stack move that copies memory rather than writing a literal (Methods, Harness conventions). A literal channel is therefore sufficient for an open first replicator, not necessary. Whether the first replicator is open in practice is observed. The Z80 has a self-consistent literal write, and its first replicator is open (in the partner in at least half of 16 traced encounters) in 170 of 170 benign-tar worlds of 16–64 bytes (Stages G, H, K, L and M; Supplementary Table 1). BFF has none, since its instructions increment in place or copy from memory, and its first replicators are loop-bearing in 35 of 35 worlds with a replicator under the published rule, the wrapping pointer and harmless brackets: none of the 28 under the first two rules enters the partner in more than 34% of encounters (22 never), and of the 7 with harmless brackets five enter in at most 3%, one in 23% and one in every encounter. BFF with one literal-write instruction added is open first, loop-free and in the partner in every encounter, in

12 of 12 soups under each pointer rule. The theorem says nothing about how long the open phase lasts (Model 5). *Scope*. (H1) is a property of a von Neumann architecture in which a program's code lies in a shared memory and execution runs on past it: it holds in the Z80 soups, BFF, Core War⁴⁶ and Tierra¹⁵, and not in Avida¹⁶, whose genomes are circular, so that a straight-line Avida program returns to its own start. The theorem also inherits the provisos of Theorem 2, which fail under lethal tar and in budget arms shorter than the tape. (H2) is a property of the copy instructions and must be checked per machine; Theorem 1 does not imply it (one pass over 128 addresses meets Theorem 1's bound), and we do not prove it for BFF. In the ring Z80 the premise that execution ends when the pointer leaves the pair fails, so (iii) does not apply there: the two-byte self-writer completes its copy on a second pass of the ring. For the Z80 the open-first conclusion rests on observation; the census (Proposition 3) shows that the smallest self-writers are open, but not why they come first. For a chemistry the hypotheses are a reading, not a theorem: (H1) would be a copier that runs off its template into adjacent material by default, (H3) a template whose content specifies its own complement without an interpreter.

Lemma 7 (perfect fidelity requires zero information inflow). *Setting.* Let X be the organism's tape, E the partner, drawn from a context distribution independently of X , and $Y = F(X, E)$ the offspring, the partner's L bytes after the encounter. Fix an organism x . The *information inflow* into its offspring is the conditional mutual information $\lambda(x) = I(E; Y | X = x) = H(Y | X = x) - H(Y | X = x, E)$, the information the offspring carries about the partner when the organism is held fixed. Execution is deterministic given (x, E) , so $H(Y | X = x, E) = 0$ and $\lambda(x) = H(Y | X = x)$: the entropy of the offspring over partners, with the organism fixed, which the partner test measures, equals the information that flows from the partner into the offspring. *Statement.* (i) If $\lambda(x) > 0$ then $P(Y = x | X = x) < 1$: a replicator with inflow cannot be perfectly faithful across contexts. (ii) If $P(Y = x | X = x) = 1$ then $\lambda(x) = 0$: perfect fidelity across contexts requires zero inflow. (iii) $P(Y \neq x | X = x) \geq (H(Y | X = x) - 1) / \log_2 |\mathcal{O}| \geq (\lambda(x) - 1) / \log_2 |\mathcal{O}|$, where $\mathcal{O}$ is the set of possible offspring. (iv) The converse of (ii) is false: $\lambda(x) = 0$ says that the offspring does not depend on the partner, not that it is x . *Proof.* (ii) If $Y = x$ with probability one given $X = x$, then $H(Y | X = x) = 0$; a conditional mutual information is at most the corresponding conditional entropy, so $0 \leq \lambda(x) \leq H(Y | X = x) = 0$. (i) is the contrapositive of (ii). (iii) Fano's inequality, applied to the guess $Y = x$, gives $H(Y | X = x) \leq 1 + P(Y \neq x | X = x) \log_2 |\mathcal{O}|$, and $\lambda(x) \leq H(Y | X = x)$. (iv) A regenerator of Proposition 8 whose tape carries a mutation at a position $i \geq d$ has the same offspring in every context, so $\lambda = 0$, and that offspring lacks the mutation, so $Y \neq x$. ■ Parts (i)–(iii) do not use determinism; it is used only to identify $\lambda(x)$ with the measured entropy. *Measurement.* The partner test estimates $\lambda(x)$ by the plug-in entropy of the offspring over 256 uniformly random partners; the estimate cannot exceed $\log_2 256 = 8$ bits and is biased downwards when the offspring takes many values. *Context.* For one generation and a fixed organism, $\lambda(x) = 0$ is the condition that the environment adds no information about the system's next state given its present one, which Bertschinger et al.⁴² use, with related conditional mutual informations, to express autonomy, and which Krakauer et al.⁴ develop into an information theory of individuality. *Reading, with the measured qualification.* Confinement (execution never enters the partner) is neither necessary nor sufficient for $\lambda = 0$: it is not sufficient because offspring positions the organism never writes keep the partner's bytes (7 of the 22 BFF first replicators whose pointer never entered the partner have $H(Y | X = x) > 0$, three of them ≈ 7 bits from a single unwritten byte), and it is not necessary because a copier that overwrites everything it touches has $\lambda = 0$ although its pointer enters the partner (the BFF literal that writes two or three copies of its word, whose offspring carry 0 bits of the partner while its pointer enters the partner in every encounter; Fig. 5d). Inflow comes from partial copying; openness of control flow is the mechanism that makes the Z80 first replicator's copies partial. Measured: the first replicator has $\lambda > 0$ in 80 of 80 Stage G worlds (medians 3.9, 6.2, 7.8 and 8.0 bits at $L = 16, 20, 50$ and 64 , the last at the ceiling) and the loop-bearing successor $\lambda = 0$ in 63 of 67 (one offspring string in all 256 contexts), the four exceptions copying all but one to three bytes (Fig. 3c).

Proposition 8 (the copy period). The proposition is elementary. It rests on a property of the forward overlapping block move that programmers use on purpose: a move whose destination lies d bytes ahead of its source fills memory with its first d bytes repeated (with $d = 1$ this is the usual way to fill or clear memory with `LDIR`; the same overlap is why `memmove` and `memcpy` differ). Its interest is the consequence for heredity: in a recurring evolved core the displacement d is one byte of the genome, and it sets how many sites a lineage can transmit. The core is not new: the appendix of the second version of Cicala et al.²³ lists the `xx 5e ... ed b0` core with byte 0 annotated as the "partner offset"; what is added here is the count and its reading. *Overlap lemma.* Let M be a memory of P cells with addresses taken mod P , and let a forward move with displacement d , $1 \leq d < P$, perform $M[s + d + k] \leftarrow M[s + k]$ for $k = 0, 1, \dots, K - 1$ in this order. Then the byte written at step k is the initial byte at $s + (k \bmod d)$. If moreover d divides P and $K \geq P - d$, the move leaves every cell $s + j$ holding the initial byte at $s + (j \bmod d)$. *Proof.* By induction on k . For $k < d$, cell $s + k$ has not been written before step k , since the writes to it come at the steps congruent to $k - d \bmod P$, the first of which is $P - d + k > k$; it still holds its initial byte. For $k \geq d$, its last write before step k came at step $k - d$ (writes to one cell are P steps apart, and $d < P$) and carried, by the induction hypothesis, the initial byte at $s + ((k - d) \bmod d) = s + (k \bmod d)$. For the second part, the write at step k lands on the cell whose offset from s is congruent to $d + k \bmod P$, hence, when d divides P , to $k \bmod d$: every cell written receives the initial byte of its own residue class, cells s to $s + d - 1$ keep their bytes, and the steps $k < P - d$

reach every cell from $s + d$ to $s + P - 1$. ■ *Setting.* The Z80 soup at tape length $L \leq 64$: the pair memory is a ring of $2L$ bytes, addresses are taken mod $2L$, every register and flag except the stack pointer is zero at the start of an encounter (the stack pointer aliases the last byte of B), and an encounter executes 128 instructions, each repetition of `LDIR` counting as one. Let the organism's tape A begin with `XX 5e ed b0`, and let $d = XX \bmod 2L$ satisfy $4 \leq d \leq L$ and $d \mid L$ (hence $d \mid 2L$). Assume that, in every encounter, the one-byte instruction `XX` (a) passes control to address 1, (b) writes memory, if at all, only at addresses d to $2L - 1$, and (c) leaves $HL \equiv 0$ and $256 \cdot D \equiv 0 \pmod{2L}$ and BC either 0 or at least 126. Hypothesis (b) admits, for example, `f5` (`PUSH AF`), a working value at $L = 20$, which writes two bytes at the end of the partner. *Statement.* Whatever the partner holds, (1) the encounter leaves the whole ring, organism and partner, equal to the d -periodic tiling of A's first d bytes, with the partner in phase with the organism; (2) a mutation at a position i with $4 \leq i < d$, to any byte value, is carried by the offspring and, in every later encounter in which a descendant runs as the organism, by that descendant's offspring; (3) the byte at a position $i \geq d$ has no effect on either tape after the encounter, so a mutation there is erased from the offspring and from the parent; (4) outside the four-byte core, the positions at which every mutant value is transmitted are therefore exactly the $d - 4$ positions 4 to $d - 1$. *Proof.* By (a) and (b), `XX` hands control to address 1 and leaves addresses 0 to $d - 1$, the core among them, unchanged. `LD E, (HL)` reads address $HL \bmod 2L = 0$, which holds `XX`, so $E = XX$ and, by (c), $DE \equiv XX \equiv d \pmod{2L}$. Neither HL nor DE passes `0xFFFF` in the next 125 increments (after a one-byte `XX` that meets (a)–(c), HL is at most `0xFF00`, reached with `DEC H`, and DE at most `0xFF1B`, reached with `DEC DE`), so the k -th repetition of `LDIR` performs $M[(d + k) \bmod 2L] \leftarrow M[k \bmod 2L]$. By (c), BC is 0 (65,536 repetitions) or at least 126, so `LDIR` repeats for all 126 executions left, provided its own bytes at addresses 2 and 3 do not change. They do not: apply the overlap lemma with $s = 0$ and $P = 2L$ to the memory at the start of the move, which agrees with the encounter's initial memory at addresses 0 to $d - 1$ (a byte `XX` wrote at address d or above is overwritten by the move before it is read, since the move reads address $k \geq d$ only after writing it at step $k - d$); every write therefore gives its cell A's byte of the cell's residue class mod d , and addresses 0 to 3 lie below d , so they receive their own bytes. Execution therefore stays at addresses 0 to 3, and the lemma, with $K = 126 \geq 2L - d$ (because $L \leq 64$ and $d \geq 4$), leaves every address a holding A's original byte at $a \bmod d$. Because $d \mid L$, the partner's position j , at address $L + j$, holds A's byte at $j \bmod d$, which is (1). Bytes 4 to $d - 1$ are never executed, and they are read only by the block move, as data (`XX` reads at most addresses 0 and $2L - 1$, and `LD E, (HL)` address 0); so their values do not change the execution, and each is copied to every position congruent to it mod d . The offspring therefore begins with the same core and the same $d - 4$ bytes, meets (a)–(c), and the argument repeats in the next generation, which is (2). Positions d and beyond are overwritten in both tapes with copies of earlier bytes, whatever they held, which is (3); (4) follows from (2) and (3). ■ *Why the hypotheses are needed.* Without (b) the statement is false: at $L = 16$, `70` (`LD (HL),B`) passes control to address 1 and has $d = 16$, but it writes 0 over address 0, so that $E = 0$ and nothing is copied. Without (a) it is false too: with every flag clear, the returns `RET NZ`, `RET NC`, `RET PO` and `RET P` (`c0`, `d0`, `e0`, `f0`) are taken, write no memory and leave HL , D and BC at zero, yet send control to address $256 \cdot XX + b \pmod{2L}$, b being the partner's last byte, so that the outcome depends on the partner; their offsets divide L at $L = 16$ (`d0`, `f0`), 32 (`d0`, `e0`), 48 (`d0`, `f0`) and 64 (`c0`). For $d \leq 3$, one of the first two repetitions overwrites a byte of the block move with an earlier byte of the core, and the next fetch is another instruction. The bound $L \leq 64$ lets the 126 repetitions reach the end of the ring. *Offsets that do not divide L.* If d divides $2L$ but not L , the proof holds up to its last step: the ring holds the d -periodic tiling of A's first d bytes, but the partner holds it in phase $r = L \bmod d \neq 0$, so the offspring begins with A's bytes $r, r + 1, \dots \pmod{d}$, starts with the core only if the parent repeats the core at those positions, and the count $d - 4$ does not follow. If d does not divide $2L$, the tiling does not close around the ring: the writes that wrap past address $2L - 1$ bring bytes of other residue classes onto the core, in general before the budget is spent, and the proposition says nothing. *Remarks.* The variant `1e NN ed b0` (`LD E, n ; LDIR`) has offset $d = NN \bmod 2L$ and the same count; its block move has 127 repetitions. The 4 in $d - 4$ counts the core, the four bytes that are executed: `XX`, `LD E, (HL)` and the two bytes of `LDIR`. A change there alters what is executed: at position 0 another working value $XX' \equiv XX \pmod{2L}$ keeps the offset and is carried, `00` sets $d = 0$ and nothing is copied, and `HALT` (`76`) at position 1 or 2 stops execution before the copy; the single-mutant scan counts a site only if most of its values are carried, so the core contributes none. In the soups (Fig. 6c), 2,289 of the 2,343 confined block-copy cells whose offset divides L have exactly $d - 4$ transmissible sites in the single-mutant scan, the median equals $d - 4$ in all 15 (L, d) groups, and the three cells whose offset does not divide L have 0, 3 and 0 sites. A transmitter is the case $d = L$: it copies itself verbatim and carries $L - 4$ sites; every $d < L$ regenerates, rewriting the organism's own body from its first d bytes.